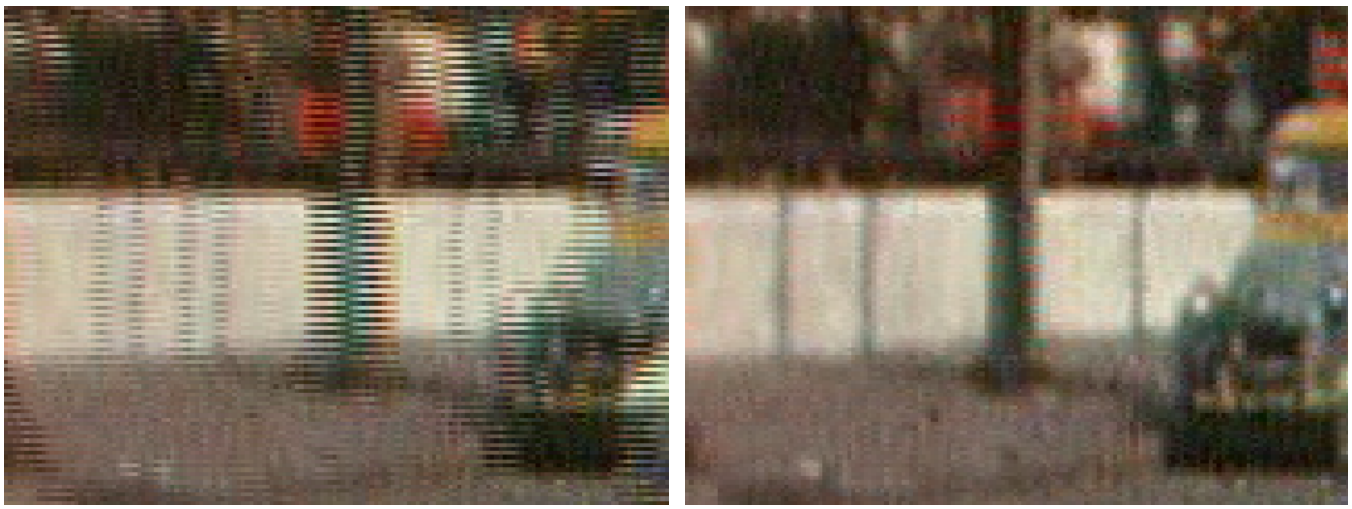
TECHNICAL REPORT · DVD RESTORATION

The Champagne Spy, Restored Pixel by Pixel

A technical report on turning a rare German PAL (European TV standard) DVD into a media-server-ready library at 50 smooth frames per second. Every half-picture (field) was decided individually; every restoration process is shown in its own colour. Calibrated H.265 compression, sub-millisecond audio sync and complete subtitles.

ORIGINAL DVD

RESTORED



Archive film, Paris street at 3:43, zoomed. Left: the disc joins halves of two different film frames, so the fence and the passing car shred into stripes. Right: each half re-paired with its true partner from the neighbouring disc frame. Nothing was invented or sharpened: these are the disc's own pixels in their correct order (Chapter 4).

One colour per restoration process, used in every change map and chart in this paper:

- **Slate** = untouched: the disc's own pixel, exactly as on the DVD
- **Blue** = invented: calculated by the deinterlacer (the tool that rebuilds a missing half-picture) where the disc has no picture
- **Gold** = lifted: moved pixel for pixel from the neighbouring disc frame
- **Green** = reworked: the disc's own lines cleaned of old line noise

The Champagne Spy (2007), documentary directed by Nadav Schirman, about Wolfgang Lotz (Ze'ev Gur-Arie), the Israeli Mossad agent who posed as a wealthy German horse breeder in 1960s Egypt, told largely through his son Oded Gur-Arie. Source: the German DVD release *Der Champagner Spion* (Neue Visionen, 2009, PAL). **Scope and rights.** The disc was purchased; it carries no copy protection, so nothing was circumvented; the restored files exist for personal viewing only and are not distributed. This report describes the method and its measurements. Frames from the film appear only as small crops for technical commentary; the film is © its producers and Neue Visionen.

Technical report · October 2026

Levi Horsley

Status: complete. The final encode, assembly and frame-by-frame verification finished on 14 October 2026; every size and score in this paper is a measured final result.

Contents

1	The Vision	3
2	The Source: A Rare Disc, Preserved Bit-Exactly	8
3	Two Moments in Every Frame: 25 Interlaced → 50 Progressive	11
4	Process 1, Gold: Deciding 271,174 Fields and Repairing Mis-paired Film Exactly	16
5	Process 2, Blue: True Video Motion, Reconstructed by QTGMC and Tuned for Fidelity	24
6	Process 3, Green: Where Does QTGMC Add Value? Cleaning Archive Line Noise	31
7	All Together: One Method per Frame, 50 Times a Second	38
8	Framing: Proving the Picture Was Already Right	44
9	Encoding: One Quality Target, Chosen from 60 Test Encodes	48
10	The Honest Size Story	54
11	The German Dub, Re-timed to Sub-Millisecond Accuracy	58
12	Audio Integrity: A Seamless Ending, an Untouched Original, Frame-Exact Intros	62
13	Subtitles I: From Disc Bitmaps to Correct Text	66
14	Subtitles II: Nothing on Screen Left Untranslated	71
15	The Extras: Trailers Reframed Subtitle-Safe, Interviews Treated Like the Film	78
16	Delivery: A Media-Server-Ready Package, Verified	83
17	Method and Effort: AI-Assisted Versus Conventional	87
A	Glossary	92
	Colophon	95

Chapter 1 gives the bottom line and the measured results. Chapters 2–16 each prove one improvement in the same order: the problem, the analysis, the decision, the impact, and an Effort ledger of what was built, measured and discarded. Chapter 17 estimates the same job without AI. Figures are numbered F1–F30 and charts C1–C4 in order of appearance.

CHAPTER 1

The Vision

Summary. A rare, barely findable German DVD of *The Champagne Spy* was preserved as a byte-for-byte copy and rebuilt as a media-server-ready set that plays at 50 smooth frames per second, with the German dub back in sync with the lips and every German title on screen translated into English. About 6 of every 7 pixels on screen are still the disc's own; the rest were calculated only where the disc's picture was missing (the other half of a moving frame) or carried stripes from an old video transfer. The work is complete: every frame of the finished film was checked against the restored master, and the whole set came out slightly smaller than the disc image. This chapter is the project in brief; Chapters 2–16 each prove one claim.

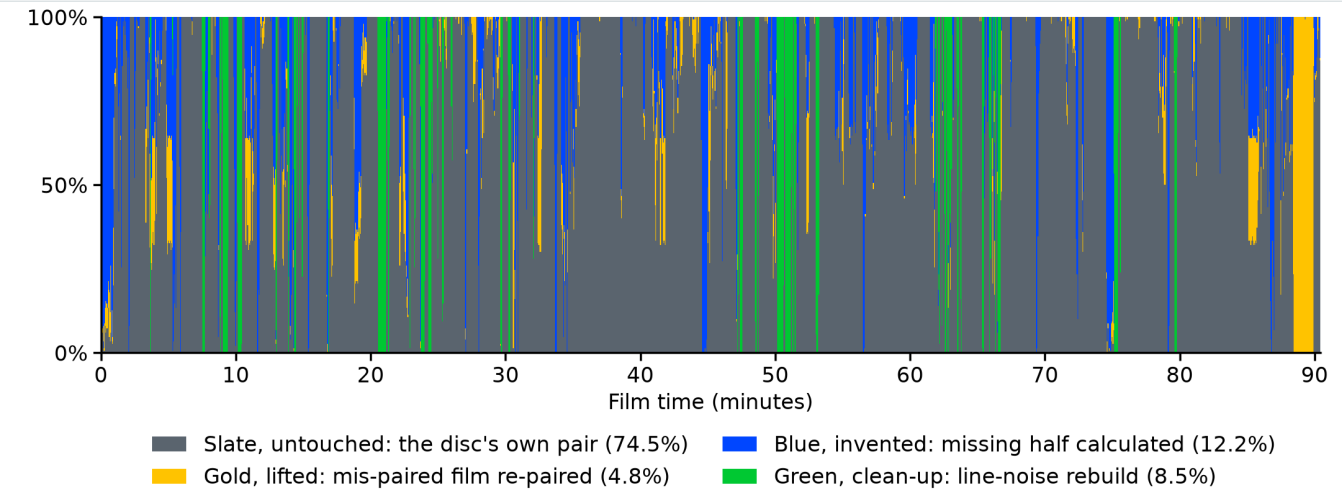


Figure F1. How every one of the film's 271,174 output frames was made, in 3-second columns from the first minute (left) to the last (right). Slate, untouched: the disc's own picture (74.5%). Gold, lifted: mis-paired film re-paired exactly (4.8%). Slate and gold together, 79.2%, are the disc's exact picture with no calculation at all. Blue, invented: the deinterlacer calculated the missing half of a true-motion frame (12.2%). Green, clean-up: a full rebuild of archive shots carrying old line noise (8.5%, 116 of 908 shots).

What to look for: the long slate stretches are the modern interviews, kept exactly as on the disc. The solid gold block at minutes 88–90 is the rolling end credits, almost all re-paired film. Blue at the very start is the opening archive TV footage. The green clusters, e.g. around minutes 21 and 50–52, are the noisy archive sequences that earned the clean-up.

1.1 By the numbers

What the numbers mean: every decision was measured across the whole film, not spot-checked, and for every change that shipped, several alternatives were built, measured and thrown away.

271,174
half-pictures (fields) decided one by one across 908 shots; each became its own output frame (from 135,587 disc frames)

85.4%
of the 112.5 billion output pixels are the disc's own (83.0% untouched + 2.4% lifted); 14.6% computed

0.66 ms
worst German lip-sync error after correction, from up to 83 ms; checked at 87 + 37 points

96.08
average VMAF (a 0–100 picture-quality score) of the finished film against the restored master, on all 271,174 frames; worst 1% of frames 91.10

7.41 GB
the finished set against the 7.92 GB disc image: 6.4% smaller, with twice the frames and a lossless German track

788 + 15
English subtitle lines read by OCR and corrected + new lines translating on-screen German, each timed to its title

8 + 1
whole-film interlace forensic passes + one complete second render of the film to test where the deinterlacer helps (542,348 frame renders)

60 → 1
test encodes (30,000 frames, each scored by 2 quality meters) → 1 setting shipped

≈ 6.5 h + 8 h
analysis and engineering, then unattended encoding (film: 7 h 59 min) and 48 min of automatic verification, on one laptop

1.2 The vision in seven moves

The brief: near-zero quality loss first, size second, with no limit on processing time. The vision that followed: invent nothing that is not physically missing, and prove every choice with a measurement or a side-by-side check.

- **1.2.1 Preserve first.** The source is an obscure German DVD, *Der Champagner Spion* (Neue Visionen, 2009, PAL, the European TV standard), barely findable and finally obtained from an online retailer. Before anything else it was copied into a bit-exact ISO image (one file that is a byte-for-byte clone of the whole disc): 7.92 GB, 0 read errors. In plain terms: the original survives whatever happens to the physical disc. → *Ch 2*
- **1.2.2 Understand before touching.** The film was broken into its 908 shots and then into all 271,174 half-pictures (fields), and each was decided by what it actually is, because the film mixes progressive film, interlaced video and still title cards, sometimes inside one shot. → *Ch 3–7* A field is a half-picture of every other line, captured at its own instant; an interlaced frame weaves two fields together, and when they come from different instants every moving edge tears into horizontal stripes (combing). Deinterlacing turns woven half-pictures into whole progressive frames, the way modern screens show them.
- **1.2.3 Four treatments, one colour each.** Each field has three candidate partners: the other field of its own disc frame, or a field of the previous or next frame. If one was captured at the same instant, joining them ("weaving") reproduces the original picture exactly; if none was, the missing half must be calculated. → *Ch 4–7*
  - *(a) Untouched, slate (74.5% of frames):* woven with its own partner, the disc's exact picture.
  - *(b) Lifted, gold (4.8%):* archive film the disc stores with its halves mis-paired, re-joined with the true partner from the neighbouring disc frame. Original pixels, moved, not made.

  - *(c) Invented, blue (12.2%):* true video motion, where no partner exists. QTGMC is a motion-tracking deinterlacer: it follows how things move across neighbouring frames to calculate the half-picture a field is missing, instead of just blurring or doubling lines. Here it keeps the disc's own half bit-exact, with no denoising and no sharpening.
  - *(d) Clean-up, blue + green (8.5%):* 116 of 908 shots (7.7 minutes) in which an old video transfer had printed fine horizontal stripes (line noise) into the picture. A full QTGMC rebuild invents the missing lines (blue) and recomputes the disc's own lines clean (green); the measured stripe share fell from a median 0.187 to 0.0001, essentially none. The rebuild costs some fine detail, so it was used only where a measurement and an eye check agreed the gain was visible, and never on text or credits.

  In plain terms: nothing is calculated where the disc already holds the real picture. Each process is proved in turn: gold (Ch 4), blue (Ch 5), green (Ch 6), then all together across eight consecutive frames (Ch 7).

- **1.2.4 Keep every moment of motion.** PAL carries 50 fields per second, each a separate moment; 25 whole frames would discard half of them. Every field became its own frame: 25 interlaced frames per second (25i) became 50 progressive frames (50p), 135,587 disc frames became 271,174 output frames, at a measured +21% file size on a worst-case motion clip. In plain terms: motion as smooth as the original broadcast, without the stripes. → *Ch 3*
- **1.2.5 Encode for quality, then size.** The picture is compressed with H.265 (also called HEVC), a modern video codec, in its 10-bit "Main 10" form, which stores brightness in four times finer steps than the DVD's 8-bit video. One quality target, CRF 14, was chosen: CRF (constant rate factor) is a quality setting, not a size setting, and lower numbers mean higher quality. It came from 60 test encodes scored frame by frame with two automatic picture-quality meters (VMAF and SSIMULACRA2), then checked by eye at TV size. The finished film scores a VMAF mean of 96.08 against the restored master over all 271,174 frames, and its file is 6.20 GB, about the size of the disc's own film streams (≈ 6.2 GB) with twice the frames. In plain terms: one setting for the whole film, under which every scene automatically receives the bits it needs. → *Ch 9–10*
- **1.2.6 Make the German dub match the picture.** The dub ran 22.16 parts per million fast, so it slid steadily out of sync until it was 83 ms ahead of the lips by the end, enough to see. Speed-corrected and stored as FLAC (a lossless audio format: every sample kept exactly), its worst measured error is now 0.66 ms, against a 20 ms frame. Its silent gap before the end credits became a seamless crossfade into the original music, and the Hebrew original is copied bit for bit; both were verified bit-identical in the finished file. In plain terms: voices match lips from the first minute to the last. → *Ch 11–12*
- **1.2.7 Complete subtitles, delivered ready for a media server.** The disc's picture subtitles were read by OCR (optical character recognition: software that reads text out of pictures) and corrected (788 English lines). 15 new lines translate German on-screen titles the disc never subtitled, each shown only while its German title is on screen and placed clear of on-screen text. The film, the trailers and both interviews sit in the folder layout a media server expects, with German audio and English subtitles on by default. In plain terms: an English viewer now reads every title a German viewer reads. → *Ch 13–16*

1.3 How much of the picture is original

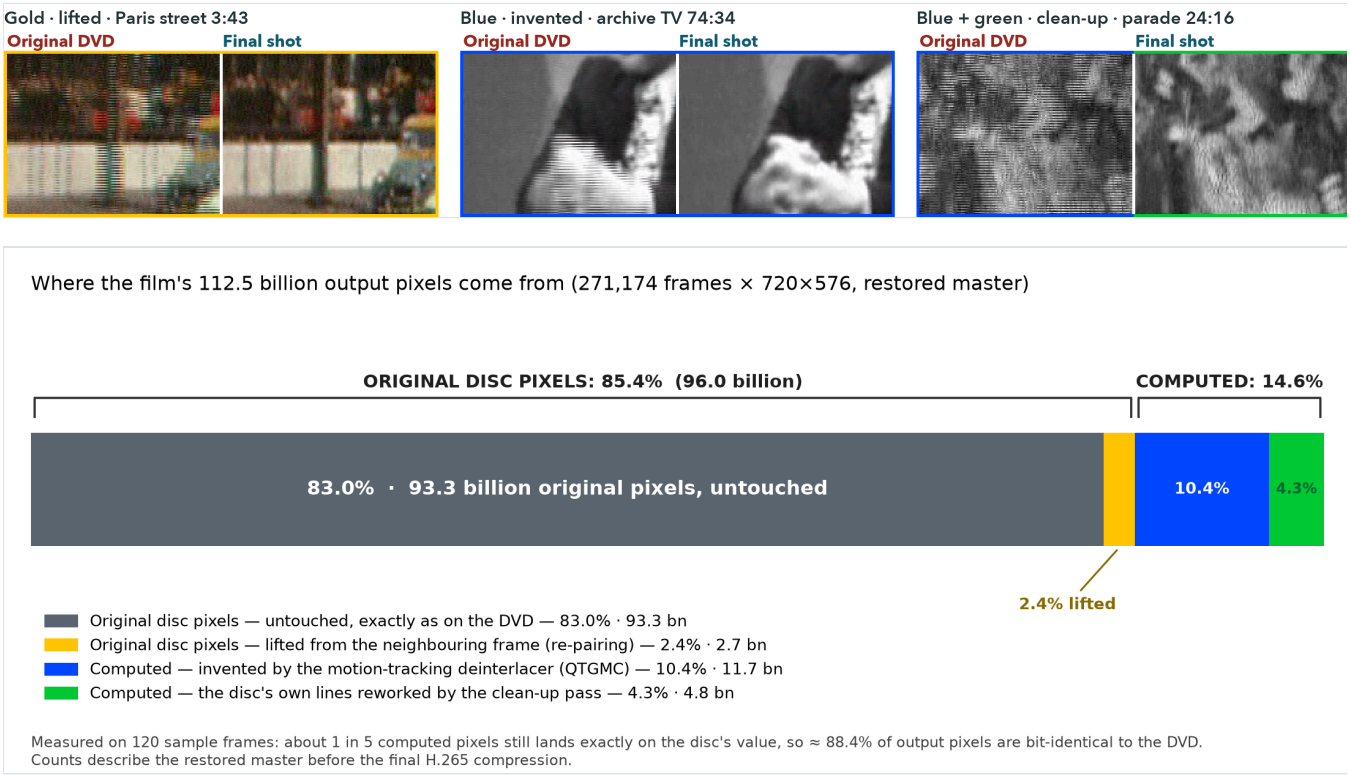


Figure F2. The three processes, one example each, and their share of the picture. Top, left to right, each pair zoomed, with the disc ("Original DVD", dark-red label) on the left and the restoration ("Final shot", teal label) on the right. Left pair, gold, lifted: Paris street 3:43, mis-paired film re-joined with its true partner (Ch 4). Middle pair, blue, invented: archive TV footage 74:34, the missing half of a moving hand calculated by QTGMC (Ch 5). Right pair, blue + green, clean-up: archive parade 24:16, old line noise removed (Ch 6). Bottom: where the film's 112.5 billion output pixels come from (restored master). Original disc pixels, 85.4%: 83.0% untouched, exactly as on the DVD (slate), and 2.4% lifted from the neighbouring frame (gold). Computed, 14.6%: 10.4% invented by QTGMC (blue) and 4.3% the disc's own lines reworked by the clean-up (green). Measured on 120 sample frames, about 1 in 5 computed pixels still lands exactly on the disc's value, so about 88.4% of all output pixels are bit-identical to the DVD. Counts describe the restored master before H.265 compression.

What to look for: top, the horizontal stripes on the left of each pair and the solid picture on the right; bottom, how long the slate segment is: about 6 of every 7 pixels are the disc's own.

- **1.3.1 Why frames and pixels give different numbers.** Every frame outside the clean-up keeps at least half of its lines exactly where the disc had them. So the 79.2% exact frames (all original pixels) plus the kept half of the 12.2% half-invented frames make 85.4% original pixels. The 10.4% invented pixels are the calculated half of those half-invented frames (6.1%) plus the calculated half of the clean-up frames (4.3%); the clean-up's other half is the 4.3% reworked. Worked through in Ch 7.
- **1.3.2 Caveat.** These maps and counts describe the restored master that feeds the encoder. The finished H.265 file, like the DVD's own MPEG-2 video, is lossy compression: to save space it approximates nearly every pixel by a small amount. At the chosen quality setting (CRF 14) that is not visible at TV size, but "bit-identical" refers to the master, not the compressed file (Ch 9).

1.4 The project's nine goals, checked

Goal	Result
1. Archive first	**Done.** Bit-exact ISO, 0 read errors (Ch 2)
2. Media-server-ready MKV set	**Done.** Film, intros and both interviews in the folder layout a media server recognises; every frame of every file scored against its restored master and every audio track verified bit-identical; placed in the library after a final visual review (Ch 16)
3. Quality first	**Done.** CRF 14 from 60 scored encodes; finished film scored on every frame, VMAF mean 96.08 (Ch 9–10)
4. Per-frame interlace analysis	**Done.** 908 shots and 271,174 fields decided individually (Ch 3–7)
5. Perfect audio sync	**Done.** German within 0.66 ms worst case (Ch 11)
6. Interlacing, bars, stretch	**Done.** 50 progressive frames/s, every shot's field order checked (0 reversed); film framing verified and untouched; two trailers reframed (Ch 3–8, 15)
7. Subtitles miss nothing, cover nothing	**Done.** 15 new lines, each timed to its German title and checked against on-screen text (Ch 13–14)
8. Both tracks, all extras, defaults	**Done.** German FLAC (default) + Hebrew original + English subtitles on; all 8 extras kept; tracks and flags verified in the finished files (Ch 12, 15, 16)
9. Plays everywhere	**Tested on the primary device.** H.265 Main 10, which Apple TV decodes on its built-in video chip (it cannot for the more efficient AV1); Apple's hardware decoder ran the whole film with 0 errors. A test clip in the final format played through the media server on an Apple TV (a stutter was traced to its Match Frame Rate setting, not the file). Fire TV, Roku and phones are covered by the codec choice rather than by a test (Ch 9, 16)

Done = finished and verified · Tested on the primary device = checked on an Apple TV; the other platforms are covered by the codec choice. The author made every judgement call from a short visual or listening check (screenshots, listening clips, a television test clip), and no long-running job was started without the author's explicit approval. Every later chapter ends with an Effort ledger of what was built, measured and discarded; Chapter 17 estimates the same job without AI.

CHAPTER 2

The Source: A Rare Disc, Preserved Bit-Exactly

Thesis. Every later result in this paper is only as trustworthy as the copy it starts from. So before a single field was analysed, three things were done. The disc was cloned byte for byte. Its contents were mapped. The software that reads it was checked against a second, independent decoder. The whole chapter comes down to four measurements:

7,922,694,144	0	None	3 / 3
bytes imaged from the disc	read errors	CSS copy protection present	frames pixel-identical in two decoders (0 · 67,793 · 135,586)

2.1 The problem

2.1.1 A rare disc is a single point of failure

- **An obscure release.** The source is the German DVD *Der Champagner Spion* (distributor Neue Visionen, 2009, PAL). This edition was barely findable and was finally obtained from an online retailer.
  - There is one physical copy. Discs scratch, age and eventually fail, and there may be no second copy to buy.
  - That set the project's first goal, which came before everything else: archive the disc perfectly before any conversion work touches it.

2.1.2 What the disc holds

The picture format is a legacy format. The film is interlaced and anamorphic, two broadcast-era conventions that modern screens must convert. The chapters that follow are about doing that conversion properly; this chapter only records what the disc holds.

Element	As stored on the disc
Main film video	MPEG-2 (the DVD's video compression format), 720×576, 16:9 anamorphic (pixel aspect 64:45, displayed as 1024×576)
Frame rate	25 interlaced frames/s (PAL), which is 50 fields (half-pictures) per second
Length	135,587 frames = 1:30:23.5, 24 chapters
Video stream	5.76 GB at about 8.5 Mb/s average
Audio 1	German dub, AC-3 (Dolby Digital) stereo, 320 kb/s
Audio 2	Hebrew original, AC-3 (Dolby Digital) stereo, 320 kb/s
Subtitles	VobSub bitmaps (pictures of text, not text) in German and English

In plain English: *Anamorphic* means the widescreen picture is stored squeezed into a 4:3-sized grid of 720×576 pixels and stretched back to 1024×576 on playback. Each stored pixel is shown about 1.42 times wider than it is tall. *Interlaced* means each of the 25 frames per second is really two half-pictures, one made of the odd lines and one of the even lines, and the two may have been captured at different moments. *VobSub* subtitles are small images, not text, so a computer can only edit, search or correct them after reading them with OCR (optical character recognition), which a later chapter covers.

2.2 The analysis

2.2.1 Imaging: a clone, not a re-encode

- **Sector-by-sector copy with `dd`.** The disc was read from its first byte to its last and written unchanged into a single ISO image.
  - Size: 7,922,694,144 bytes, which is exactly 3,868,503 DVD sectors of 2,048 bytes. That is dual-layer (DVD-9) capacity.
  - **0 read errors.** Every sector was read without error, so the image contains no gaps or substituted data.
  - **No CSS present.** The disc carries no Content Scramble System encryption. The bytes in the image are the bytes on the disc, with no decryption step in between.
  - Imaging took 16 minutes.
- **Why an image rather than a "rip".** A ripping tool that writes the film straight into a new container is convenient, but its output is a new file and not a copy of the disc. Menus, navigation data and any stream it was not asked for are left behind. The image keeps everything, so any later question about the disc can still be answered from it.

2.2.2 Title sets mapped and extracted from the image, never from the drive

Title set	Content	Length	Destination in the library
VTS_01	Copyright notice	16 s	Featurettes / Intros (own chapter)
VTS_02	Neue Visionen logo	24 s	Featurettes / Intros (own chapter)
VTS_03–06	Four trailers for other films	–	Featurettes / Intros (one chapter each)
VTS_07	Interview with Oded Gur-Arie	7:53	Interviews
VTS_08	Interview with director Nadav Schirman	14:16	Interviews (4 chapters)
VTS_09	The film	1:30:23.5	Main film file, 24 chapters

- **Seconds per title set.** Each title set was extracted from the ISO file in seconds. The physical disc was never read a second time.
  - Every re-run of every step therefore reads identical input. Nothing depends on how well a drive happens to read a disc on a given day.

2.2.3 Decoder trust test

- **Two independent decoders compared.** VapourSynth's BestSource, the frame-accurate source filter that feeds the whole restoration pipeline, was checked against ffmpeg. At frames 0, 67,793 and 135,586 (the first, middle and last frame of the film) the two decoders matched pixel for pixel.
  - A single dropped or duplicated frame shifts every frame after it. Agreement at the very last frame is therefore a strong end-to-end check of the frame numbering.
- **Why it matters.** Later chapters decide each of the film's 271,174 fields individually, and they verify those decisions by comparing specific frames. If a decoder were off by one frame, the analysis would be pairing the wrong half-pictures. Every later measurement would then be invalid, and nothing would visibly break to show it.

2.3 The decision

- **The ISO is the master of record.** All work derives from it. Nothing is ever re-ripped, and the original always remains available.
  - Every analysis, test encode and comparison in this paper reads the image, not the disc.
  - The image itself is never altered. Every result is a new file beside it.
  - Streams that need no improvement are carried forward straight from the image. The Hebrew original audio is copied bit for bit, and the intro audio keeps every original AC-3 frame unchanged (verified by per-frame MD5 checksums: digital fingerprints that change if a single bit changes).

2.4 The impact

- **Preserved.** The rare original now exists independently of the physical disc. If the disc is lost or fails, nothing is lost.
- **Reproducible.** Every analysis reads the same bytes, so any measurement in this paper can be re-run on identical input, and every frame-level result (the 908 shots, the 271,174 field decisions) points to a specific disc frame.
- **Comparable.** Because the exact original stays available, every "Original DVD" panel in later figures, and the count of output pixels copied unchanged from the disc, is measured against the real disc picture rather than against a second-generation copy.
- **An honest yardstick.** The 7.92 GB image is also the benchmark for the size story. The finished library is 7.41 GB, 6.4% smaller than the image: not a drastic cut, because quality was put first (Chapter 10).

> **In plain English:** The rare disc has been turned into a permanent digital original. Everything else in this project is measured against it, so any claim in this paper about what the disc shows can be checked against the disc itself.

EFFORT LEDGER	CHAPTER 2
BUILT AND DISCARDED	nothing. This is the one step designed to be done exactly once and then verified.
MEASURED	the byte count (7,922,694,144); 0 read errors; no copy protection present; two decoders agreeing pixel for pixel at 3 frames spread across the film.
TIME	16 minutes of imaging; seconds per title set.
KEPT	one bit-exact 7.92 GB ISO and its 9 title sets (the film plus 8 extras).

CHAPTER 3

Two Moments in Every Frame: 25 Interlaced → 50 Progressive

Thesis. A PAL DVD does not hold 25 pictures a second. It holds 50 half-pictures, each its own moment in time, woven together in pairs. Shown raw on a modern screen, every moving edge tears. Converted to 25 fps, half the motion is thrown away. This restoration keeps all 50 moments as 50 clean frames per second (Figure F3), at a measured cost of +21% in file size on a worst-case motion clip.

50	20 ms	135,587 → 271,174	+21%
fields (half-pictures) per second on a PAL disc	between two fields, each captured at its own instant	disc frames → output frames	measured size cost of 50 fps with H.265

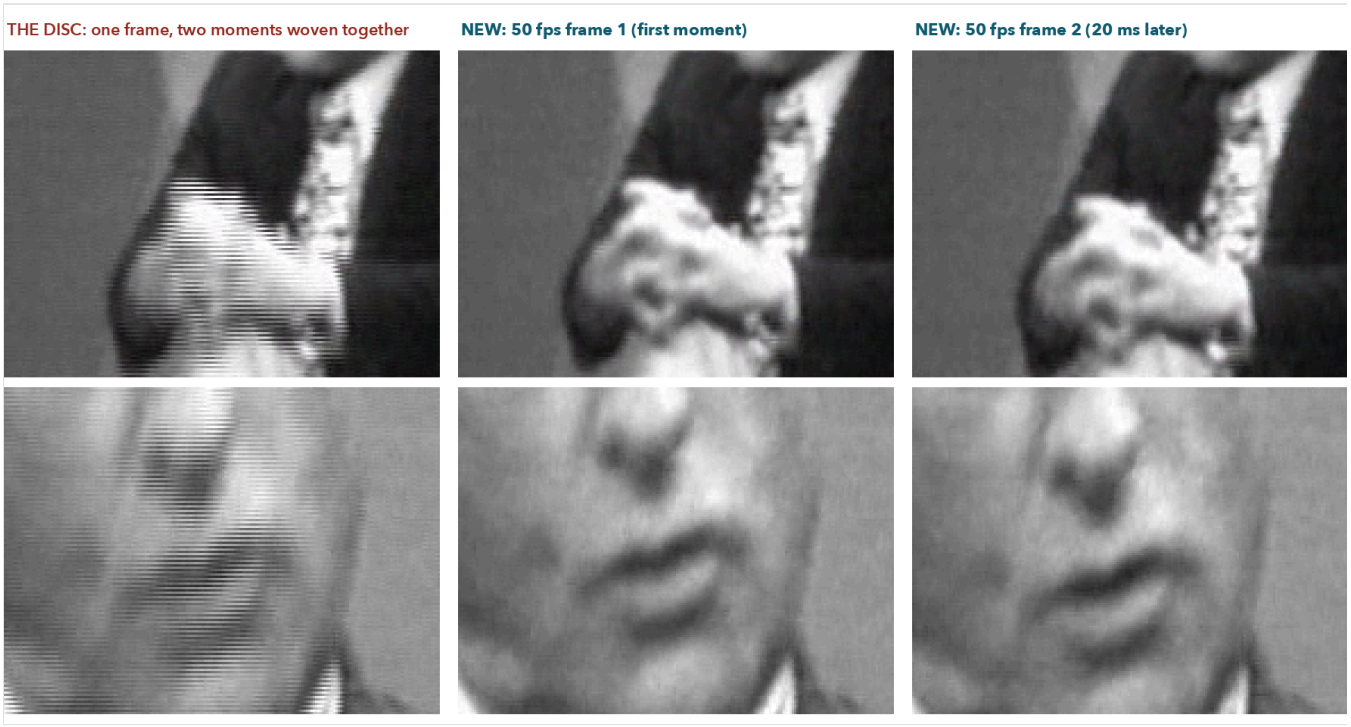


Figure F3. One disc frame becomes two clean frames (3× zoom, archive TV footage: a hand at 74:34 on top, a speaking mouth at 0:44 below). Left: the disc frame, where two moments 20 ms apart are woven together, so the hand and the lips tear into horizontal stripes. Middle: the first new 50 fps frame, a clean picture of the first moment. Right: the second new 50 fps frame, 20 ms later. This is the change from 25 interlaced to 50 progressive frames per second, made visible.

What to look for: one striped picture on the Left becomes two clean, slightly different pictures. The fingers and lips move between Middle and Right; that is the motion a 25 fps version would have thrown away.

3.1 The problem: every disc frame is two pictures

3.1.1 PAL video carries 50 fields per second

- **Fields, not frames.** An interlaced PAL camera records 50 *fields* a second. Each field holds every other line of the picture: the top field has the odd lines, the bottom field the even lines. In a 720×576 frame, each field is 720×288.

  - Each field is exposed at its own instant, 1/50 s = 20 ms after the one before it.
  - The DVD stores the fields woven together in pairs, interleaved line by line, as 25 frames per second. The film's 135,587 frames therefore hold 271,174 fields.
- **Why broadcast TV did this.** It gave 50 picture updates a second for the bandwidth of 25. A tube TV drew the two fields one after the other, so the viewer never saw both at once.

3.1.2 A modern screen shows both moments at once

- **Combing.** Flat-panel TVs, computers and phones are progressive: they show every line of a frame at the same time. A woven frame shown raw puts two instants 20 ms apart on screen together. Every moving edge then splits into horizontal "comb teeth" (left of Figure F3; left panel of Figure F5).

3.1.3 Two ways a woven pair goes wrong

- **The halves differ in brightness (5:49).** Alternating lines are light and dark even where nothing moves, so the picture looks like a venetian blind.
- **The halves come from different moments (66:34).** A moving arm is in one place in the first field and somewhere else in the second. Woven together, the figures appear doubled and zig-zagged.
- **The key insight: each field on its own is clean.** The defects live in the *pairing*, not in the fields. Chapters 4–7 build on this. For each of the 271,174 fields, the restoration asks whether the disc holds a partner captured at the same instant. If it does, the two are joined exactly. If not, the missing lines are calculated.

Figure F4. What a "field" is. Every PAL frame is two half-pictures interleaved line by line. Left: the whole frame as the disc stores it. Middle: the top field alone. Right: the bottom field alone (each field stretched to full height). Top row (5:49): the two halves differ in brightness, so the woven frame turns into venetian-blind stripes. Bottom row (66:34): the halves were captured at different moments, so the woven frame shows doubled, zig-zag figures. Each half on its own is clean, which is why this project works one field at a time.

What to look for: compare Left with Middle and Right in each row. The stripes and the doubled arms exist only in the woven frame. The small tile labels (S1/S2) are internal review labels; in the top row, "1st/2nd moment" just means top/bottom field. The difference to look for there is brightness.

> **In plain English:** Each DVD frame is two photos taken a fiftieth of a second apart, sliced into strips and shuffled together. A modern screen shows both at once, so anything that moves looks torn. Each photo is fine on its own, so the fix is to treat each photo separately.

3.2 The analysis: four treatments, a measured price, an honesty check

3.2.1 A four-way demo, so the choice could be made by eye

- **One shot, four treatments.** A moving street shot (a man walking past a pedestrian, a bus and a red car) was rendered four ways. It was shown as a still (Figure F5) and as a moving 2×2 clip (1920×1080, 50 fps, 8.8 s). In the clip, the 25 fps versions show each frame twice so that all four play in real time and their smoothness can be compared directly.
- **The two deinterlacers tested.**
  - *bwdif* is ffmpeg's standard motion-adaptive deinterlacer. It is fast and fills each missing line from nearby lines and fields, but it does not follow objects as they move.
  - *QTGMC* is a motion-tracking deinterlacer. It follows how things move across neighbouring frames to calculate the half-picture a field is missing, rather than blurring or doubling lines. The demo ran it at its most thorough preset (Placebo), with no denoising, in its common "Lossless=2" mode. Later measurement showed that mode alters the original lines it claims to keep, so the final setting differs (Chapter 5); the frame-rate decision does not depend on it.

#	Treatment	Output	Combing	Moments kept per second	Verdict
1	Disc as stored	25 interlaced	Yes	50, woven in pairs	Rejected
2	bwdif	25 progressive	No	25	Rejected: half the motion lost
3	QTGMC	25 progressive	No	25	Rejected: steadier edges, still half the motion
4	QTGMC	50 progressive	No	50	**Chosen**

Figure F5. The choice as presented: the same moving street shot four ways, read left to right. Left: the disc as stored, with comb teeth on the walking pedestrian, the red car and the front of the bus. Second: a simple deinterlacer (bwdif) at 25 fps; no combing, but half the moments are thrown away. Third: QTGMC at 25 fps; steadier edges, still half the moments. Right: QTGMC at 50 fps, every field its own frame; this one was chosen. Each panel shows the same left two-thirds of the demo frame. (The demo used an earlier QTGMC setting; the final one is described in Chapter 5.)

What to look for: the comb teeth on the walking pedestrian and the front of the bus, in the Left panel only. The advantage of 50 fps is smoother motion, which a still cannot show, so the two right-hand panels look alike here.

3.2.2 Why 25 fps loses information

- **One frame per disc frame means one moment per pair.** A 25 fps output keeps one of the two moments in each woven pair and throws the other away. That is half of the 50 motion samples per second the camera recorded, so motion becomes twice as choppy as the original broadcast. The loss shows most on gestures, walking figures and pans.

3.2.3 The cost, measured before choosing

- **The test.** The demo shot was chosen as worst-case motion. Its QTGMC output was encoded at 25 fps and at 50 fps in H.265 and in AV1.

- **The result.** 50 fps cost **+21%** file size with H.265 and **+63%** with AV1. H.265, the codec finally chosen (Chapter 9), paid the smaller penalty.
- **Why double the frames is not double the size.** A modern codec predicts each frame from its neighbours and stores mostly the difference. A frame 20 ms after the last one is nearly identical to it, and an exact repeat costs almost nothing. That is why the worst case was the one measured.

3.2.4 The honesty check: 50 fps fakes no motion

- **No in-between frames are invented.** This is not a TV's "motion smoothing". Every output frame comes from exactly one field the original camera or film transfer recorded.
- **Film and progressive material.** Here both fields come from one instant, so that one original picture is simply shown in two consecutive 20 ms frames. Before the clean-up step, per-field matching (Chapter 4) found such a same-instant partner on the disc for about 87% of fields (80.3% in their own frame, 6.5% in the neighbouring frame). After Chapter 6 rebuilt 116 noisy archive shots, 79.2% of output frames remain the exact disc picture.
- **True video motion.** The other 13.2% of fields (before the clean-up step) had no same-instant partner. Only there does 50 fps add genuinely new moments, the ones a 25 fps version would drop.
- **Net effect.** 50 fps is a fidelity choice, not an enhancement trick. It delivers exactly the motion information that is on the disc, no more and no less.

> **In plain English:** The same shot was made four ways and the best was picked by eye. The winner shows 50 pictures a second instead of 25. It costs about a fifth more space, not double, because each picture is mostly a copy of the one before. It invents nothing. Where the original had only 25 different pictures a second, each is shown twice. Where it had 50, all 50 are kept.

3.3 The decision

- **50 progressive frames per second, constant, for the whole film.** Every field becomes its own full frame, so 135,587 disc frames become 271,174 output frames.
  - Treatment 4, QTGMC at 50 fps, was chosen. Treatments 1–3 were rejected.
  - Film and progressive shots stay at 50 fps by showing each picture twice, so player and TV run at one steady rate throughout.
- **What this sets up.** Chapters 4–6 explain how each field is turned into a full frame: joined exactly with its true partner, reconstructed by QTGMC, or cleaned up by QTGMC. The final mix:

79.2%	12.2%	8.5%
of output frames: the exact disc picture (74.5% untouched + 4.8% lifted by repairing)	QTGMC with the disc's own lines kept bit-exact (true video motion)	QTGMC clean rebuild of archive line noise (116 shots, 7.7 min)

3.4 The impact: two clean moments where there was one torn frame

- **One disc frame becomes two clean frames 20 ms apart (Figure F3, top of this chapter).** On the disc, a hand at 74:34 and a speaking mouth at 0:44 are torn into stripes because two moments are woven to-

gether. In the 50 fps version each moment is a whole, solid picture. Motion is as smooth as the original broadcast, and nothing the camera captured is thrown away.

- **Where to see it again.** The same two moments appear in context in Chapter 5 (Figure F11: whole frame, then zoom). The lines QTGMC calculated for the hand at 74:34 are coloured **blue** in Figure F13.
- **Twice the frames, about the same space.** The finished film file is 6.20 GB, about the size of the disc's own main-film streams (5.76 GB of MPEG-2 video plus two AC-3 audio tracks, about 6.2 GB), while carrying twice the frames, 10-bit colour and a lossless German track. The full size story is in Chapter 10.

> **In plain English:** Where the disc showed one striped picture, the new file shows two sharp ones in a row, exactly as the camera recorded them.

EFFORT LEDGER	CHAPTER 3
BUILT AND DISCARDED	a four-way deinterlacing demo (a still plus a moving 2×2 clip), with 3 of its 4 treatments rejected; 25 vs 50 fps test encodes of a worst-case motion clip in H.265 and AV1.
MEASURED	+21% size for 50 fps with H.265 and +63% with AV1, before any decision was made.
TIME	not separately logged.
KEPT	50 progressive frames per second, with every field as its own frame (271,174 output frames).

CHAPTER 4

Process 1, Gold: Deciding 271,174 Fields and Repairing Mis-paired Film Exactly

Thesis. When the right partner for a half-picture already exists one frame away on the disc, the correct repair is to find it, not to invent new pixels. This chapter shows how that partner was found for every field of the film, one decision at a time. Where the disc had stapled two different film frames together, the film was repaired exactly: the right lines were moved to the right place and nothing was calculated.

271,174	8	4.8%	0
individual field decisions, one per output frame	whole-film interlace forensic passes over 135,587 frames	of output frames: mis-paired film re-paired exactly (final mix)	invented pixels in a re-paired frame

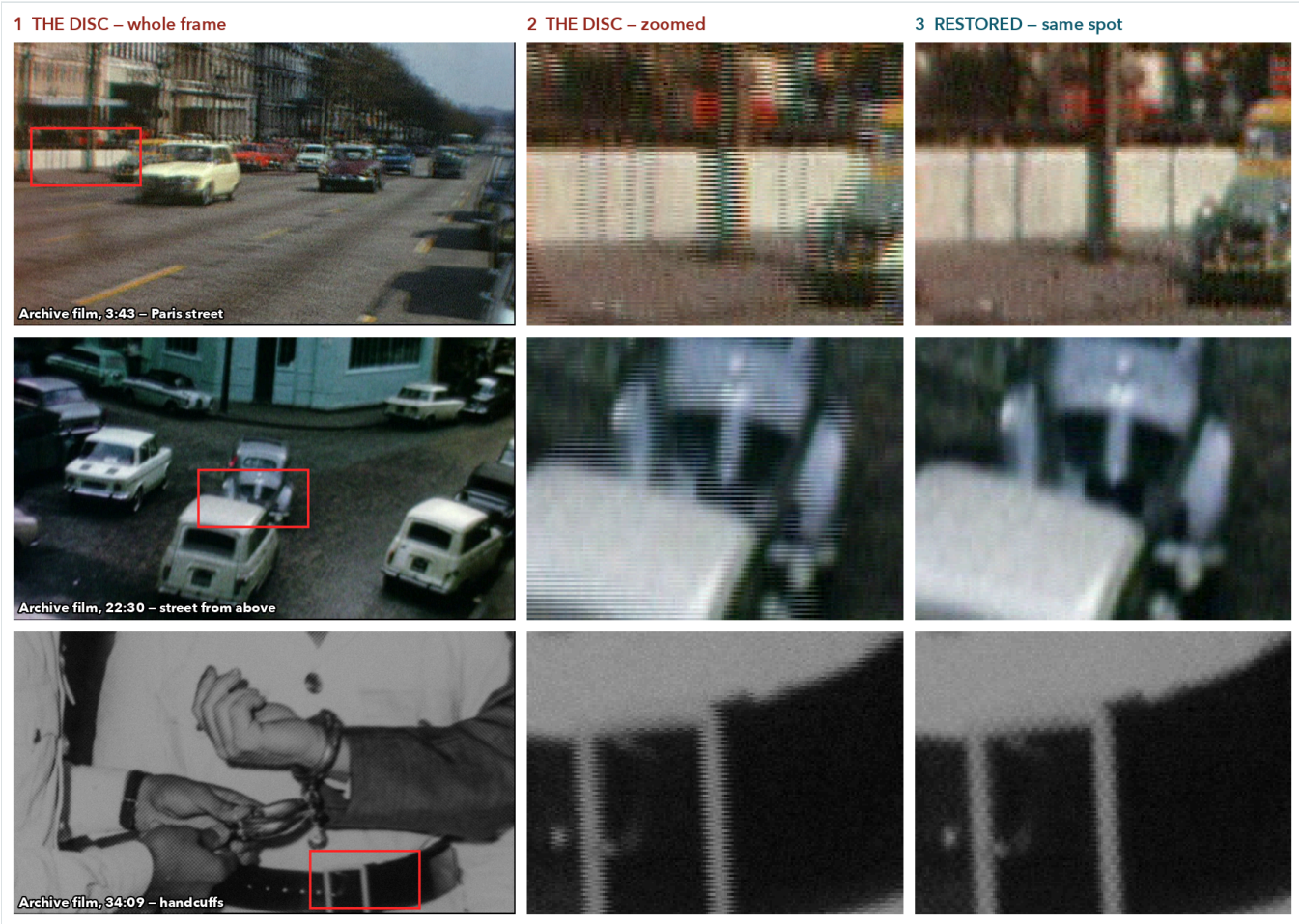


Figure F6. Mis-paired archive film, repaired exactly. Each row reads left to right. Left: the disc's whole frame with a red box (Paris street 3:43; a street from above 22:30; handcuffs 34:09). Middle: the box zoomed from the disc. The fence, the car roof and the belt are shredded into stripes because the disc pairs halves of two different film frames. Right: the same spot restored, each half re-paired with its true partner from the neighbouring disc frame: solid fence posts, a clean car roof, straight belt edges. Nothing was invented. These are the original film pictures, recovered bit-exact. Zooms show the disc's 720×576 pixels unstretched (not widescreen-corrected), so they look narrower than the boxed area; this keeps every line exactly as stored.

What to look for: Middle against Right in each row. Comb teeth become solid edges without any softening. The Right column is not sharpened or reconstructed; it is the disc's own pixels in their correct order. The faint colour streaks left in the crowd above the fence are not added by the repair: re-pairing moves the disc's lines and alters none of them.

4.1 The problem

4.1.1 One film, four kinds of picture

- **Mixed material.** *The Champagne Spy* cuts between very different sources: modern interviews shot progressive, archive film, archive TV video, and credits and title cards.
  - Each kind is stored differently inside the same 25-frames-per-second interlaced PAL stream, so each needs a different treatment.
  - The kind of picture sometimes changes *within* a single shot (106 shots, 4.2.3), so the decision had to be made per field, not per shot.

4.1.2 Discovery: film stored out of step

- **Phase-shifted film.** Some archive film is stored with its two fields mis-paired. On top of that, there are field-insertion cadences left over from a 24 → 25 frames-per-second conversion.
  - Stretching 24 film frames into 25 video frames per second means inserting extra half-pictures at intervals. After each insertion, the halves that follow are out of step with their frames.
  - The result is striped "combing" on what is really progressive film (Figure F7). The forensic grain-proof field difference scores such a frame like interlaced motion (8.0 in Figure F7), well above the ~5.3 ceiling of clean progressive film.
- **In plain terms:** each half is a perfect piece of a real film frame, but the disc staples together halves of two different frames.

Figure F7. Archive film stored out of step (27:37). Left: the whole frame as stored; the tree trunks have saw-tooth edges. Middle: the top field alone. Right: the bottom field alone. Each half is clean, but the trunks sit in slightly different places, because the disc joined halves of two different film frames. The fix is not to invent pixels. Each half is re-paired with its true partner in the neighbouring disc frame, which restores the original film frame exactly.

What to look for: the zig-zag trunk edges on the Left against the clean trunks in the Middle and Right. "diff8.0" is the grain-proof field-difference score: progressive film scores at most about 5.3, interlaced at least 6. The leading number is the disc frame index.

4.1.3 Why a generic deinterlacer gets this wrong

- **Two bad outcomes.** A standard deinterlacer takes each frame's two fields as a given pair. It sees the stripes as motion and does one of two things.
  - It invents new pixels for the "moving" half, which softens film that was never moving.
  - Or, if it decides the frame is progressive, it leaves the stripes in.
- **The missed fact.** The true partner field *is on the disc*, one frame over. Even QTGMC, the motion-tracking deinterlacer used for real video motion (Ch 5), calculates a missing half. It does not go looking for the original one.

> **In plain English:** picture a stack of photos that someone cut in half and taped back together with each top half on the wrong bottom half. One could repaint the bottom halves to fit, or find each one's real other half, which is lying in the next photo. This chapter does the second thing, for the whole film.

4.2 The analysis

4.2.1 A first approach, built and thrown away

- **A complete per-scene bwdif master.** The first plan deinterlaced the film scene by scene with bwdif ("Bob Weaver" deinterlacing filter, ffmpeg's adaptive deinterlacer) into a lossless FFV1 master: 55.8 GB, about 5 minutes to render. Lossless bwdif masters of all 8 extras were built the same way.
  - All of them were discarded once the per-field approach proved better. A per-scene decision cannot follow material that changes inside a shot, and bwdif never looks one frame over for the true partner.
  - **In plain terms:** a finished, working answer was thrown away because a measurably more exact one existed.

4.2.2 Eight whole-film forensic passes

Each pass ran over the whole film: 135,587 disc frames (1:30:23.5), decoded frame-exactly (BestSource and ffmpeg were checked to agree pixel for pixel, Ch 2). The custom passes are Python/NumPy measurements.

- **Pass 1, ffmpeg idet.** ffmpeg's built-in interlace detector. *Plainly:* a quick first vote, frame by frame, on whether the picture looks interlaced.
- **Pass 2, comb detector.** A comb-detection filter (CombMask), scored by its densest 16×16 block, as in TIVTC, a long-standing inverse-telecine tool. (Inverse telecine is the standard technique for undoing a film-to-video conversion by putting each film frame's two half-pictures back together.) *Plainly:* it counts "comb teeth" pixels. Scattered grain cannot fill a block, but real combing along a moving edge can. The first comb pass took about 2 minutes.
- **Pass 3, blurred grain-proof field difference.** Both fields are blurred so that film grain averages out, then compared. Calibrated: progressive ≤ ~5.3, interlaced ≥ 6. *Plainly:* it measures real movement between the two halves and ignores grain.
- **Pass 4, cadence analysis.** Each frame's own-field difference is compared with its difference to the next frame's field. *Plainly:* true video differs both ways. Out-of-step film is near zero against the neighbour, and that is the fingerprint of a mis-paired film frame.
- **Pass 5, cross-field difference.** The grain-proof difference between each field and the fields of the neighbouring frames. *Plainly:* the raw evidence of which neighbour shares its instant.
- **Pass 6, cross-weave comb test.** 271,174 test weaves: every field actually joined to its neighbour-frame candidate, then checked by the comb detector. *Plainly:* each possible repair was tried out, not assumed.
- **Pass 7, local field motion.** The grain-proof difference measured block by block, taking the worst 16×32 block. *Plainly:* a small moving object, such as a hand, cannot hide in an otherwise still frame.
- **Pass 8, stillness.** The same half (top with top, bottom with bottom) of consecutive frames compared after blurring (8 minutes for the whole film). *Plainly:* it detects frames in which nothing moved at all.

The core forensics (passes 3–7) took about half an hour in total, 2–6 minutes per pass. A separate whole-film OCR text audit ran alongside (Ch 14).

4.2.3 The shot map

908 shots (749 merged scenes), each classified by several independent detectors:

Shot class	Shots
Progressive	424
Interlaced	136
Mixed within the shot	106
Interlaced by motion, little visible combing	66
Line structure without field motion (grain or baked-in stripes)	175
Static	1
Total	**908**

Why it matters: 106 shots change kind partway through and 175 show stripes with no real motion, so no single setting per shot, let alone per film, could treat them correctly. That is why every field was decided on its own.

4.2.4 The detectors on trial

Every detector was checked against real crops before it was trusted. Each failure below was found by eye.

- **(a) The comb detector raises false alarms on texture.** Clean hair, skin and fabric scored up to 125 of 256 pixels in a block (Figure F8). A high score is therefore not proof of combing.
- **(b) The blurred measure is blind to fine combing.** But it reliably identifies *which neighbour shares the same instant*, and that is the question the repair needs answered.
- **(c) Still title cards fooled the field measures.** Sharp letter edges fall on one field's lines and not the other's, so a still card looked "different" between its fields. Yet a still card changes by only about 0.06 brightness levels from frame to frame, against 1–3 for grainy film. That gap led to the stillness rule.
- **In plain terms:** each detector is right about something and wrong about something else. The rule therefore uses each one only for what it is good at.

Figure F8. Why no single detector was trusted. Two clean interview close-ups, read left to right. Left pair: the whole frame with a red box, then a 3× zoom of jacket fabric that a standard comb detector scored at 125 of the 256 pixels in a 16×16 block, as if combed. Right pair: the whole frame, then a 3× zoom of an eyebrow scored at 110 of 256. Neither has any interlacing damage, so the rule pairs the comb detector with a grain-proof field-difference measure. Whole frames are shown at their 16:9 display shape; zooms show the disc's pixels unstretched, as in Figure F6.

What to look for: ordinary weave and hair texture with no stripes anywhere, despite scores that suggest heavy combing.

4.2.5 The rule, stated precisely

- **Candidates.** Each field has three possible partners: the other field of its own frame, the previous frame's field, or the next frame's field. If one of them was captured at the same instant, weaving the two reproduces the original picture *exactly*, with no interpolation.

- **A pairing is kept as exact when any one of four pieces of evidence is clear:**
  - **Evidence 1, a clear same-instant partner.** The local grain-proof difference (blurred fields, worst 16×32 block) must favour the best candidate by a ratio of at least 2 over the other candidate, and the best candidate must itself be a close match.
  - **Evidence 2, static and clean.** The comb detector is trusted in one direction only. A low score (at most 32 of 256 pixels in its worst block), together with a small local difference, confirms a static, clean pairing.
  - **Evidence 3, shot context.** In shots the shot map classed as progressive, static, or line structure without field motion, a frame's own two fields stay together when they agree closely.
  - **Evidence 4, stillness (overrides the rest).** If the matching half-pictures (top with top, bottom with bottom) of the previous, current and next frame are identical after blurring (differences of at most 1.0 brightness level), the frame is kept exactly as it is on the disc.
- **Otherwise.** When no evidence is clear, the field goes to QTGMC (Ch 5). The rule never guesses toward "exact".

4.2.6 Calibration by eye

- **More than 100 diagnostic crops** were rendered and inspected: borderline combing, clean texture, re-paired film, still cards. Figure F8 is one of these sheets.
  - None of them appears in the film. They exist only to make the thresholds right.
- **The rule was revised on evidence.** Before the stillness evidence was added, the rule's split was 77.4% own pair, 7.0% re-paired and 15.5% QTGMC. The stillness evidence moved 2.9% of all fields back to the exact own-frame picture, mostly from QTGMC: still frames no longer get a calculated half.
- **Lineage.** The principle is the one behind classic inverse-telecine field matching (re-pairing half-pictures to undo a film-to-video conversion). Here it is driven by evidence calibrated on this film and applied to every field individually.

> **In plain English:** before any frame was changed, the whole film was measured eight different ways, and every measuring tool was itself tested and caught making mistakes. The final rule asks one question per half-picture, "which neighbour was photographed at the same moment?", and answers it only when the evidence is clear.

4.3 The decision

The per-field rule was adopted for the film and all extras. Its result on the film's 271,174 fields, before the clean-up step of Ch 6:

80.3%	6.5%	13.2%	~87%
joined with their own frame's partner	re-paired with the neighbouring frame's field: repaired exactly, nothing rebuilt	sent to QTGMC (true video motion)	exact disc picture before the clean-up step

- **Both directions were needed.** The true partner was in the previous frame for 3.3% of fields and in the next frame for 3.2%. A repair that looked only one way would have missed about half of the mis-paired film.

- **Which evidence made each call.** All 271,174 decisions are logged with the evidence that decided them, for auditing (table below).

Deciding evidence (4.2.5)	Share of fields
Evidence 1 · clear same-instant partner (ratio ≥ 2)	54.2%
Evidence 2 · static and clean	4.8%
Evidence 3 · shot context	23.3%
Evidence 4 · stillness override	4.5%
No clear evidence → QTGMC	13.2%

What this shows: over half of all decisions rest on the strongest evidence, a direct same-instant match, and every unclear case went to calculation instead of being guessed exact.

- **How these numbers become the final mix.** The clean-up step of Ch 6 later gives 116 shots of noisy archive material (8.5% of the frames) a full QTGMC rebuild, and it draws those frames from all three groups. The *final* shares are therefore 74.5% own pair and 4.8% re-paired, which together make the 79.2% exact disc picture (74.46 + 4.78 = 79.24 in the decision files), plus 12.2% QTGMC with the disc's own lines kept and 8.5% clean rebuild. Both sets of numbers are correct; they describe the film before and after Ch 6.

4.4 The impact

4.4.1 Shredded becomes solid, without softening

- **Figure F6 (top of this chapter).** Fence posts, a car roof and belt edges go from shredded to solid, with nothing sharpened, smoothed or reconstructed. Re-paired film is concentrated where archive film dominates; the rolling end credits are almost entirely re-paired film (see the decision barcode in Ch 1).

4.4.2 Process 1 · Gold: lifted, not made

- **Paris street, 3:43 (Figure F9).** 48.4% of the frame's pixels are gold. Every other line was lifted pixel for pixel from the correct neighbouring disc frame.
  - The pixels are still 100% original. They were moved, not made.
  - Gold is just under 50% because some lifted pixels happen to equal the disc's pixel at that spot already. Those show as dark specks.
- **Across the whole film**, lifted pixels are 2.4% of the 112.5 billion output pixels (2,689,459,200). A re-paired frame is half untouched, with the field in its own place, and half lifted.
- **One method per frame.** Whether the other half was captured at the same instant has one answer for the whole picture. So a gold frame never contains blue or green.

Figure F9. Process 1, LIFTED (gold). Paris street, 3:43. Top row: whole frame with a red box; bottom row: the box zoomed. Left to right: 1 "Original DVD", with the fence and cars striped. 2 "Enhanced by re-pairing". 3 "What changed (gold = lifted)": every pixel that differs from the disc's pixel at the same spot, here 48.4% of the frame. They arrive as every other line, each lifted pixel for pixel from the correct neighbouring disc frame. 4 "Final shot", for comparison with the map. Gold pixels are still 100% original disc pixels: moved, not made. Key: gold = lifted · blue = invented · green = reworked · no colour = untouched. Maps describe the restored master before H.265 compression; zooms are unstretched, as in Figure F6.

What to look for: this is the same fence as on the cover and in Figure F6; the change map now proves that no pixel in it was invented. In panel 3's zoom, the regular gold line pattern is exactly the half of the picture that was fetched from the neighbouring frame. The few dark specks are pixels that happened to be identical anyway. Compare the combed fence in panel 1's zoom with panel 2.

4.4.3 Verification in both directions

- **Chosen vs reference vs rejected (Figure F10).** For each decision branch, random samples were rendered side by side: the chosen pairing, QTGMC's from-scratch version of the same moment, and the rejected pairing.
  - The chosen pairing matches QTGMC's shapes. The rejected pairing is heavy combing. The rule was right in both directions: where it kept the disc's own pair, and where it overrode it.
- **Pixel-exact pipeline.** The rendering pipeline was verified per decision type: woven frames are bit-identical to the intended fields.

4.4.4 The same rule on material it was never tuned on

- **Copyright card:** 97.6% exact.
- **Oded Gur-Arie interview:** 99.8% exact. It is progressive, and the rule recognised it.
- **Director interview:** true interlaced video, recorded bottom-field-first (in each stored frame the bottom half-picture was captured first, and it must be processed in that order or motion jitters back and forth), so about 94% goes to QTGMC, as expected. The rule does not force "exact" where the disc does not hold it.

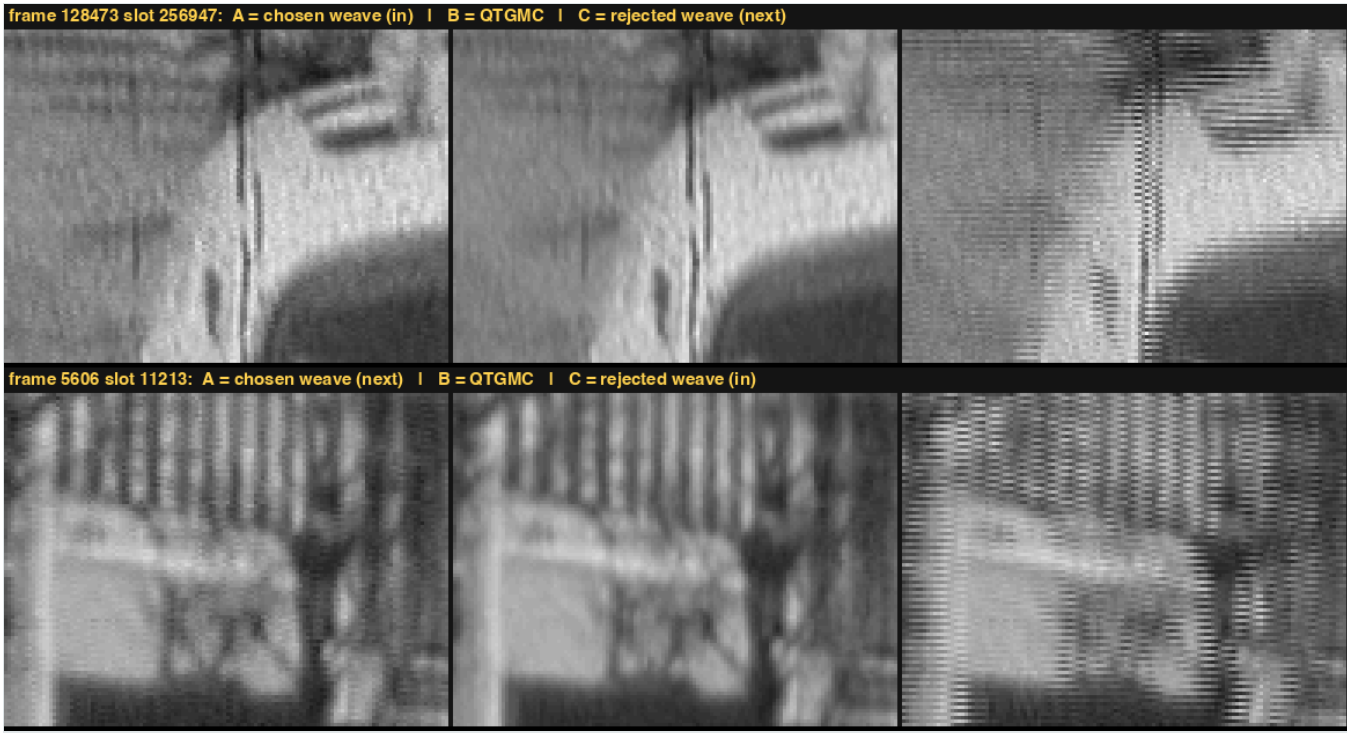


Figure F10. Verification in both directions (3× pixel zooms; each row read left to right). Left (A): the pairing the rule chose, the disc's own lines with no interpolation. Middle (B): QTGMC rebuilding the same moment from scratch, shown only as a reference. Right (C): the pairing the rule rejected. Top row (archive car, 85:39): the field's own partner is correct, and joining it with the next frame (C) shreds the car body. Bottom row (Paris street, 3:44): the true partner is in the next disc frame, and the disc's own pairing (C) is the combed one. In both rows A matches B's shapes while keeping the disc's original pixels. (Brightness channel only, so the colour Paris street appears grey.)

What to look for: C is full of comb teeth; A and B agree on shape. Header terms: "frame" = disc frame index; "slot" = output frame index; "weave in" = joined with its own frame's partner; "weave next" = joined with the next frame's field. B was rendered with QTGMC's earlier setting (reference only).

> **In plain English:** the striped archive film now looks like the film it was shot as, and none of it was repainted. Every repair was checked against an independent method and against the alternative that was rejected. The same rule then gave the right answer on extras it had never seen.

EFFORT LEDGER	CHAPTER 4
BUILT AND DISCARDED	a complete per-scene bwdif master of the film (55.8 GB lossless, about 5 min to render) plus lossless bwdif masters of all 8 extras; two detectors rejected as sole judges (the comb detector and the blurred field measure); a first version of the rule without the stillness evidence.
MEASURED	8 whole-film interlace passes over 135,587 frames, including 271,174 test weaves; more than 100 diagnostic crops inspected by eye; side-by-side verification samples for every decision branch.
TIME	first comb pass about 2 min; core forensics about 30 min (2–6 min per pass); stillness pass 8 min.
KEPT	one per-field rule. 271,174 individual decisions, and mis-paired film repaired with zero invented pixels.

CHAPTER 5

Process 2, Blue: True Video Motion, Reconstructed by QTGMC and Tuned for Fidelity

Chapter 4 handled every field that has a partner captured at the same instant: those frames are joined exactly, with no calculation at all. This chapter covers the fields that have no such partner anywhere on the disc. For them, half of the picture was never recorded, and it has to be calculated. That makes this the first process in the restoration that creates new pixels, so it was tuned with one aim: never alter a line the disc did record.

Thesis. When pixels must be invented, invent only the missing ones, and touch nothing the disc got right.

12.2%
of the film's 271,174 output frames built by QTGMC in exact-lines mode

0
disc lines altered in those frames: the half QTGMC keeps is bit-exact

2,626
QTGMC frames measured across 5 scenes to choose the mode

≈ 73
output frames per second, QTGMC Placebo on one laptop

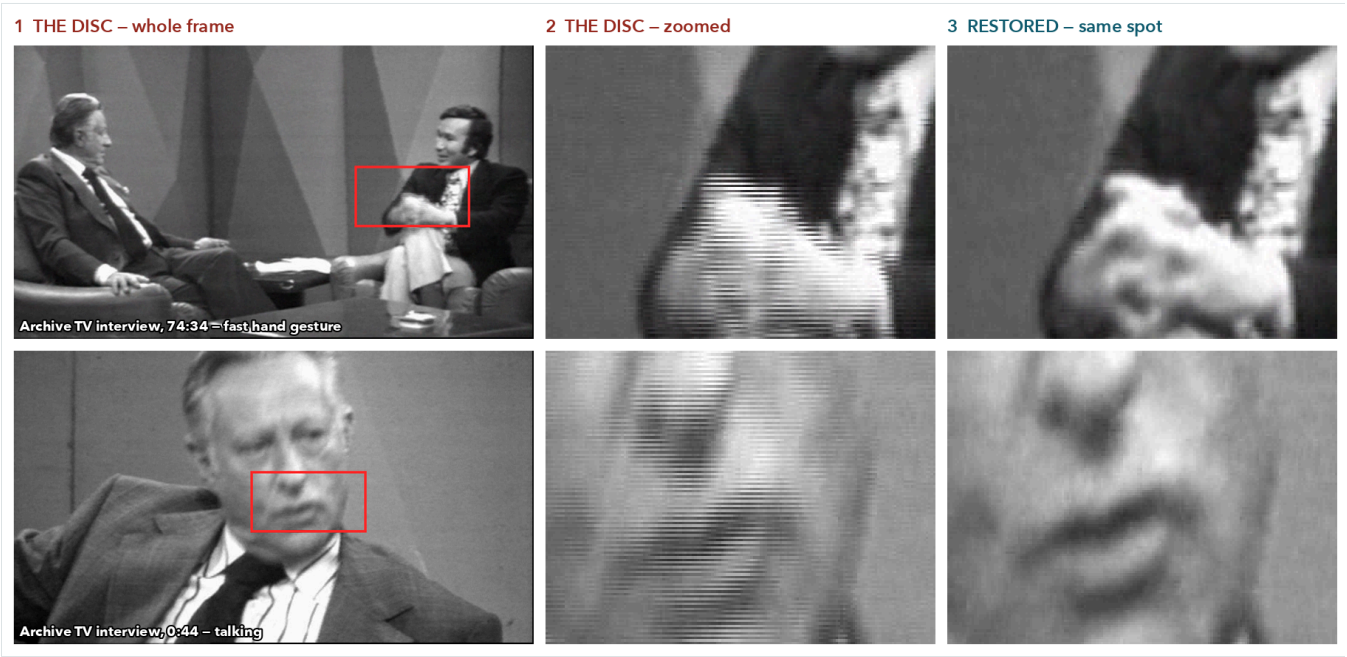


Figure F11. True video motion, reconstructed (archive TV footage). Each row reads left to right. Left: the disc's whole frame with a red box (a fast hand gesture at 74:34; a man talking at 0:44). Middle: the box zoomed from the disc; horizontal stripes where two moments 20 ms apart are woven together. Right: the same spot in the restored 50 fps frame. QTGMC kept the field's own disc lines bit-exact and calculated only the missing ones, giving a solid hand and a clean mouth. These are the same two moments as in Figure F3. Zooms show the disc's 720×576 pixels unstretched (not widescreen-corrected), so they look narrower than the boxed area.

What to look for: the comb-toothed fingers and lips in the Middle column become continuous shapes in the Right column, with no smearing.

5.1 The problem

5.1.1 True video has no partner field

- **A video camera records half-pictures.** PAL video carries 50 fields per second. One field holds the odd lines at one instant; the next holds the even lines 20 ms later. The DVD stores them woven together in pairs, as 25 interlaced frames per second.
- **Film usually has a partner; true video never does.** A film frame is exposed all at once, so its two fields share an instant and can be joined exactly (Chapter 4). Archive TV footage and the director interview were shot on video, so every field is its own moment.
  - When anything moves, no other field anywhere on the disc shows the scene at that instant. Half the lines of every moving frame simply do not exist.
  - Weave the two fields anyway and every moving edge tears into horizontal stripes ("combing"): the striped hand and mouth in Figure F11, Middle column.
  - Scale: per-field matching found no same-instant partner for 13.2% of the film's fields, so they went to QTGMC. The director interview (14:16, bottom-field-first video) is about 94% such fields, as expected for true interlaced video.

5.1.2 The reference tool, and why it would not run

- **QTGMC, in plain words.** QTGMC is a motion-tracking deinterlacer. It follows how things move across neighbouring frames to calculate the half-picture a field is missing, instead of blurring the picture or doubling lines.
  - **Lineage.** It is a long-standing, community-built video restoration script, descended from Didée's TempGaussMC (per the QTGMC v3.33 documentation). This project uses the VapourSynth port in havsfunc r33.
  - Chapter 3 compared it with a simpler deinterlacer (bwdif) in a four-way demo; QTGMC at 50 fps was chosen.
- **An x86-only stack.** QTGMC depends on several compiled plugins that are normally built only for Intel/AMD (x86) processors. Apple Silicon uses a different processor design (ARM), so as distributed QTGMC does not run on it.

5.1.3 A "lossless" mode that is not lossless

- **The promise.** QTGMC's `Lossless` option puts the field's exact source lines back into the output, so only the missing lines are new.
- **The catch.** The commonly used `Lossless=2` mode turned out to alter the very lines it claims to keep. QTGMC's own documentation concedes mode 2 is "not exactly lossless", but does not say by how much. Section 5.2.3 measures it.

> **In plain English:** where the camera recorded only half a picture, something has to draw the other half. The best tool for that job was built for a different kind of computer, and its most popular "lossless" setting quietly changes the half of the picture that was already perfect.

5.2 The analysis

5.2.1 Porting the toolchain to Apple Silicon

- **Three plugins compiled from source for ARM.**
  - *znedi3*: the neural-network line interpolator that gives QTGMC its first estimate of each missing line.
  - *fmtconv*: format and bit-depth conversion.
  - *RemoveGrain*: the small spatial repair filters QTGMC calls throughout.
- **Two QTGMC components re-pointed to ARM equivalents.**
  - The temporal-soften prefilter (an x86-only step that averages neighbouring frames before the motion search; see 5.2.5) now uses VapourSynth's built-in *AverageFrames*.
  - The EEDI3 edge-directed interpolator now uses *vszip*'s EEDI3 port.
- **Speed.** On the laptop, QTGMC at its slowest, most thorough preset (Placebo) runs at about 73 output frames per second, so even that preset was never the bottleneck. The H.265 encoder was: the film's final encode took 7 h 59 min (Chapter 9).

5.2.2 Settings chosen for fidelity, not prettiness

- **Placebo preset.** The most exhaustive motion search and interpolation QTGMC offers.
- **SourceMatch 3 (twice-refined source matching).** After its first reconstruction, QTGMC compares the result with the actual source field and corrects toward it, in two refinement rounds. The calculated lines are pulled toward what the disc really contains.
- **NoiseProcess 0 (no denoising).** Grain is part of the film. Removing it would remove real detail with it.
- **No sharpening added.** No sharpening filter anywhere in the chain; QTGMC's built-in resharpening stays at the reduced level it uses with source matching.

5.2.3 The Lossless modes on trial

- **Round 1: 4 configurations** compared on sample frames. Two finalists remained: the common `Lossless=2`, and `Lossless=1 + TR2=1`.
- **Round 2: 2 configurations measured on 2,626 QTGMC frames across 5 scenes.** Every "kept" line of the output was compared with the same line on the disc.
  - `Lossless=2` changed the kept lines by a mean of 0.7–4.6 brightness levels, depending on the scene (one level is one step of the disc's 256-step scale). At the 99th percentile the change reached 25 levels. It also softened fine detail.
  - `Lossless=1 + TR2=1` kept every disc line bit-exact.
- **Why the extra TR2=1.** The documentation warns that mode 1 can shimmer or show minor combing. One step of final temporal smoothing (`TR2=1`, lighter than the Placebo default) steadies the calculated lines. Mode 1 re-inserts the disc's lines *after* that smoothing, so the kept lines still come out bit-exact. The visual check found no added combing.

5.2.4 The worst cases, reviewed by eye

- **A detector alarm, investigated.** On some frames an automatic comb detector rated the new mode *worse* than the common one. Those frames were pulled and inspected (Figure F12): on the Eiffel Tower the new

mode is the sharper one, and on an epilogue card only the common mode leaves a ghost under the lettering.

- **Verdict.** The alarm was fine detail being kept, not combing: the same false alarm the comb detector raised on clean skin, hair and fabric texture (up to 125 of 256 pixels, Chapter 4). The "lossless" claim was one of several early assumptions that were tested and corrected.

Figure F12. The QTGMC mode check at its worst cases: frames where an automatic detector rated the chosen mode worse than the common one. Each row: Left, QTGMC "Lossless 2" (the common mode); Middle, "Lossless 1 + TR2=1" (chosen; TR2=1 is the light final temporal smoothing of 5.2.3); Right, the whole frame with a red box. In the image headers, "right" means the middle panel (the chosen mode); the far-right panel is the whole frame. Top (Eiffel Tower, 3:45): only the chosen mode resolves the iron lattice openings under the first platform; the common mode blurs them. Bottom (epilogue card, 85:52): the common mode leaves a faint grey ghost under "RIVKA GU", while the chosen mode is clean. "detector +109" = 109 more of 256 pixels flagged; "slot" = output frame index. Crops show the disc's pixels unstretched (not widescreen-corrected).

What to look for: the rows of small lattice windows in the Middle panel against a blur on the Left, and the grey smudge beneath the letters in the bottom-left panel only.

5.2.5 The scene-cut guard

- **The flaw.** QTGMC first smooths neighbouring frames together to build the clip it searches for motion on. The original x86 prefilter refuses to average across a scene change. The ARM replacement, as first patched, did not.
- **Tested at a real cut.** Frames within ±3 of the cut changed by up to 45 brightness levels: the neighbouring shot was leaking into the motion analysis.
- **The fix.** The guard was restored, with cut positions taken from the project's own 908-shot map, so motion analysis never blends two different shots. Both final encodes use it.

- **The cost, accepted.** The extras encode was already running. It was stopped and restarted to include the fix, throwing away about 40 minutes of encoding.

5.2.6 The full toolchain

All of it ran on one Apple Silicon laptop (M5 Pro).

Tool	Role in this project	On Apple Silicon
dd	Bit-exact disc image (ISO)	native
VapourSynth R81	Frame-accurate processing scripts: every per-field decision is applied here	native
BestSource	Frame-exact decoding of the disc's MPEG-2	native
ffmpeg 9	Decode cross-check (matched BestSource pixel for pixel) and the idet interlace detector	native
vszip	Comb mask for the detectors; EEDI3 port for QTGMC	native
MVTools	Motion analysis inside QTGMC	native
QTGMC (havsfunc r33)	Motion-tracking deinterlacing	2 components re-pointed to ARM equivalents
znedi3, fmtconv, RemoveGrain	QTGMC's interpolation, conversion and repair plugins	compiled from source for ARM
x265 4.3	H.265 Main 10 encoding	native
VMAF, SSIMULACRA2	Frame-by-frame quality metrics for encoder calibration	native
SoX	Sample-accurate speed correction of the German audio	native
Tesseract OCR	Subtitle conversion; audit of on-screen text	native
MKVToolNix	Final assembly of the media-server-ready files	native
Python / NumPy	Every custom measurement script	native

> **In plain English:** the best deinterlacing tool did not run on this computer, so its missing parts were rebuilt for it. Its settings were then put on trial with measurements, not taken on reputation. The popular "lossless" setting failed the test, a detector's alarm turned out to be a false one, and a hidden flaw in the rebuilt part was found and fixed even though that meant restarting an encode.

5.3 The decision

Setting	Value	What it means
Preset	Placebo	The most thorough motion search and interpolation
Source matching	SourceMatch 3	Calculated lines refined twice against the disc's own field
Lossless	1, with TR2 = 1	Disc lines re-inserted bit-exact after light smoothing
Denoising	NoiseProcess 0	None: grain is part of the film
Sharpening	none added	Rebuild what is missing; do not "beautify" what is there
Scene cuts	guard restored	Motion never blended across two shots
Scope	12.2% of output frames	Only where no same-instant partner exists

- **Only where it is needed.** The 12.2% share was 13.2% before the clean-up of Chapter 6 took over some of these shots.

- **The rejected mode has one legitimate use.** Chapter 6 explains why the clean-up of archive line noise deliberately runs `Lossless=2`: there, reworking the disc's lines is the goal.

> **In plain English:** for about one frame in eight, the computer draws the missing half of the picture, and it is forbidden to touch the half the disc already has.

5.4 The impact

5.4.1 Striped motion becomes solid

- **In the 12.2% of frames built this way,** a moving hand, mouth or edge is now a clean picture of its own moment instead of two moments woven together, with no smearing (Figure F11).

5.4.2 Process 2 · blue: exactly the missing lines, nothing else

- **42.4% of this frame's pixels are blue** (Figure F13). They are exactly the missing lines of the moving frame. The uncoloured lines between them are the disc's own, bit-exact.
- **Some calculated pixels match the disc anyway.** Measured on 120 sampled frames, 23.2% of QTGMC-reconstructed lines equal the disc's value, mostly in still areas.
- **Across the whole film,** this process calculates 6.9 billion pixels (6,883,522,560), 6.1% of the 112.5 billion output pixels. The other half of each of these frames is counted among the 85.4% of output pixels copied unchanged from the disc. (The blue share in the pixel-origin figure, 10.4%, also includes the missing lines QTGMC calculates inside the clean-up shots of Chapter 6.)
- Maps and counts describe the restored master before compression (caveat in 1.3.2).

Figure F13. Process 2, INVENTED (blue). Archive TV footage, 74:34. Top row: whole frame with a red box; bottom row: the box zoomed. Left to right: 1 Original DVD, with the moving hand striped. 2 Enhanced by deinterlacing. 3 What changed (blue = invented): 42.4% of the pixels, exactly the missing lines of this moving frame, calculated by QTGMC; every other line is the disc's own, kept bit-exact. 4 Final shot, for comparison with the map. Key: gold = lifted · blue = invented · green = reworked · no colour = untouched. At whole-frame size the coloured lines are finer than print can show, so blue on alternate lines reads as a solid blue tint; the zoom row shows the true line-by-line pattern. Zooms show the disc's pixels unstretched (not widescreen-corrected). Maps describe the restored master before H.265 compression.

What to look for: in panel 3's zoom, blue arrives on alternate lines only, never on the kept disc lines. The grey specks inside the blue are calculated pixels that came out identical to the disc anyway.

> **In plain English:** the colour map proves the promise. Blue appears only on the lines the disc never had. Every line the disc did have is still the disc's own.

EFFORT LEDGER	CHAPTER 5
BUILT AND DISCARDED	4 QTGMC "Lossless" configurations reduced to 1; the common `Lossless=2` mode rejected for these frames; the first ARM prefilter (it blended across cuts) fixed; about 40 min of extras encoding thrown away to include the fix.
MEASURED	2 configurations on 2,626 QTGMC frames across 5 scenes; kept lines altered by 0.7–4.6 levels on average (up to 25 at the 99th percentile) under the common mode; up to 45 levels of cross-cut blending before the fix; the detector's worst-case frames reviewed by eye.
TIME	QTGMC Placebo at about 73 output frames/s, never the bottleneck.
KEPT	one QTGMC configuration that never alters a disc line it keeps, applied to 12.2% of frames.

CHAPTER 6

Process 3, Green: Where Does QTGMC Add Value? Cleaning Archive Line Noise

After Chapters 4 and 5, every frame had its most faithful treatment: the exact disc picture, or QTGMC, the motion-tracking deinterlacer, calculating only the missing lines. A fair question then arose: QTGMC's full rebuild often looks cleanest, so should it be used more? Answering it is the project's clearest example of analysis done mostly to prove where *not* to act. The whole film was rendered a second time and every frame was measured. In the end, a full rebuild earns its place in only 116 of 908 shots.

Thesis. A full rebuild is a trade. It removes a real defect, but it costs about 15% of fine detail wherever it is used. So it has to earn its place shot by shot.

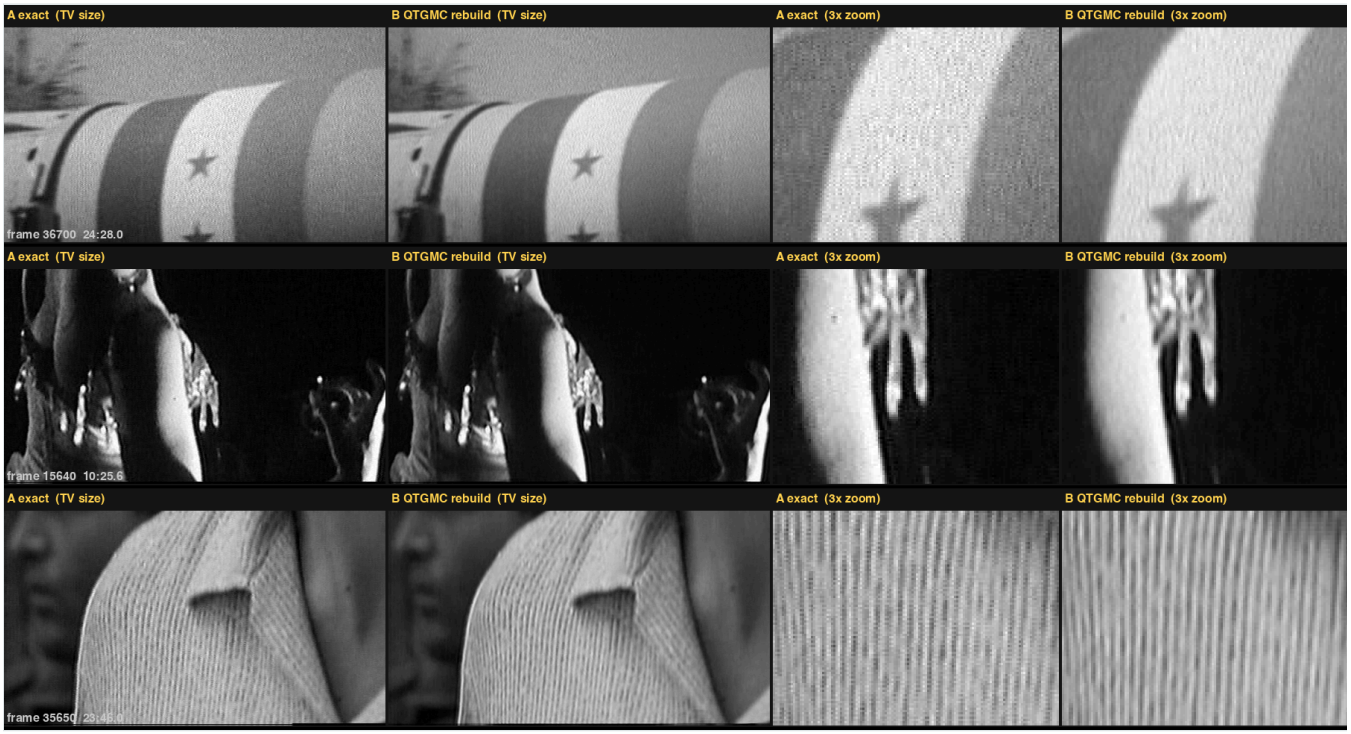


Figure F14. The defect the clean-up targets: three of the noisiest shots in the film. Each row, left to right: A, the exact disc picture at TV size; B, the QTGMC clean rebuild at TV size; A zoomed 3×; B zoomed 3×. Top (24:28): fine horizontal stripes across a painted cylinder and its stars; line-pattern share 0.743 → 0.001. Middle (10:25.6): a dancer's jewelled costume; 0.683 → 0.000. Bottom (23:46): pin-striped fabric; 0.483 → 0.002. Share values are shot averages from the per-frame analysis. The fields were already correctly paired: this is noise baked into the old video transfer. The small label at bottom left gives the disc frame index and film time.

What to look for: the zoom panels on the right. In the bottom row the fabric's real vertical pinstripes survive in B, while the horizontal line noise disappears. The rebuild removes the defect, not the texture.

6.1 The problem

6.1.1 Line noise baked into old transfers

- **A correct pairing is not always a clean picture.** Some archive transfers carry interlaced-video *line noise* baked into the picture itself, even with every field correctly paired. It shows as fine horizontal scan-line stripes and stair-stepped edges.

  - No choice of pairing removes it, because the stripes are in the picture the disc stores. Keeping that picture "exact" therefore preserves a defect (Figure F14).
  - In the worst shots the pattern covers most of the picture: averaged over the painted-cylinder shot at 24:28, it appears in 74% of the textured image blocks (85% in the frame shown in Figure F14).
- **The noise is in the picture, not in the pairing.** Of the 23,086 output frames the clean-up finally rebuilt, 20,562 (89%) had already been correctly joined or re-paired as exact disc pictures in Chapter 4. The pairing logic had done its job. The defect came with the archive transfer.

6.1.2 The cure has a price

- **A full QTGMC "clean" rebuild removes the noise.** It recomputes every pixel from the surrounding fields and smooths over time, and the stripes do not survive that.
- **It also removes fine detail everywhere.** Across the film the rebuild keeps about 0.85 of the fine detail and grain, so it loses about 15%. Applied blindly, it would soften the whole film to fix a minority of shots.

> **In plain English:** some archive clips arrived with fine stripes "printed" into the picture by an old video transfer. A heavier QTGMC pass can scrub them off, but the scrubbing also wipes away a little real texture wherever it is used. So the real question was never "does it look better?" It was "where does the gain beat the loss?"

6.2 The analysis

6.2.1 Render the whole film twice, measure every frame

- **Two complete versions of the film.** Every one of the 271,174 output frames was rendered twice. Version A is the exact plan from Chapters 4–5 (joined, re-paired, or exact-lines QTGMC). Version B is a full QTGMC clean rebuild. That is 542,348 frame renders, and rendering and measuring them took 2 h 17 min.
- **Four kinds of measurement on every frame of both versions.** They were taken on brightness only, in 16×16-pixel blocks, ignoring an 8-pixel border and flat, blank blocks. Twelve numbers were recorded per output frame, about 3.25 million in all.
  - **Alternating-line pattern share:** the share of textured blocks where the difference between adjacent lines is more than 1.15× the difference between lines two apart. Adjacent lines come from different fields; lines two apart come from the same field. In a real picture, neighbouring lines are the more alike pair. In random grain the two differences are about equal. Only interlaced line noise makes neighbours clearly *more* different.
  - **Horizontal and vertical detail.** Horizontal detail is the cost meter. Interlacing never touches it, so any drop in version B is pure smoothing.
  - **Temporal change:** how much each frame differs from the previous one, which indicates noise and flicker.
  - **The difference between A and B:** the average and the worst 16×16 block, which shows how much the rebuild changes the picture.

6.2.2 Finding: the rebuild removes the defect, not the texture

- **Where line noise exists, it goes almost completely.** In the noisiest shots the share falls from 0.48–0.74 to 0.002 or less (Figure F14).

- **Real texture survives.** The fabric's vertical pinstripes are horizontal detail, a real feature of the scene, and they stay standing in version B. Only the horizontal stripes, the interlacing artefact, disappear (Figure F14, bottom row).

6.2.3 The cutoff, calibrated by eye

- **Five review sheets of four shots each (about 20 shots) at rising line-noise levels,** each shown as A and B at TV size, then zoomed 3×. The numbers proposed the bands; the eye set the cutoff at 0.08.

Line-noise band	Shots in the film	Median detail kept by the rebuild	Verdict by eye	Shown in
below 0.04	717	0.87	no benefit: interview close-ups only lose skin texture	F15
0.04–0.08	65	0.84	mixed: the rebuild smeared airport-board lettering	F16
0.08 and above	126	0.81	visibly cleaner: line noise and stair-stepped edges gone (pyramids, a street sign, a suit)	F17, F18

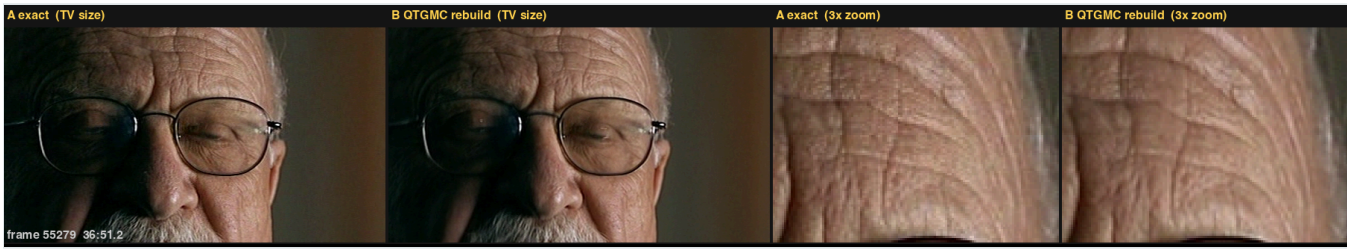


Figure F15. Band 1, below about 0.04: no benefit (36:51, line-noise share 0.037). Left to right: A exact at TV size · B rebuild at TV size · A zoomed 3× · B zoomed 3×. With almost no line noise to remove, the rebuild only smooths real skin texture. Shots like this keep the exact disc picture.

What to look for: the forehead wrinkles and pores in the zoomed A panel, against the slightly waxier zoomed B panel.

Figure F16. Band 2, 0.04–0.08: mixed (1:28, share 0.070). Left to right: A exact at TV size · B rebuild at TV size · A zoomed · B zoomed. The departure-board letters are crisp in A. In B they soften and trail a faint smeared halo. With no clear win and a real risk to lettering, the cutoff went above this band.

What to look for: the edges of the yellow letters in the two zoom panels, crisp on the left and haloed on the right.

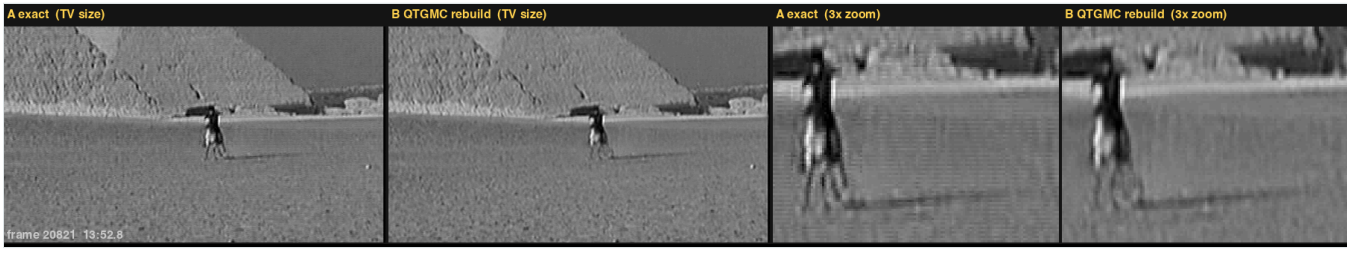


Figure F17. Band 3, at or above 0.08: visibly cleaner (13:53, share 0.125 → 0.000). Left to right: A exact at TV size · B rebuild at TV size · A zoomed · B zoomed. In A the sand is covered in horizontal line noise and the horse and rider have stair-stepped edges. B is clean and smooth, and the pyramids keep their stonework at TV size.

What to look for: the striped sand and jagged rider outline in the third panel, against the fourth.

Figure F18. Above the cutoff, the clean-up clearly wins. Each row, left to right: A exact at TV size · B rebuild at TV size · A zoomed 3× · B zoomed 3×. Top (street corner, 53:02, share 0.209 → 0.000): in A the triangular road sign and striped poles carry comb-like stripes and stair-stepped edges; in B the sign is a clean triangle. Bottom (dark suit, 51:02, 0.200 → 0.000): in A the fabric crawls with fine horizontal line noise and the lapel edge is jagged; in B it is smooth.

What to look for: the road sign in the top row's zoom panels and the lapel edge in the bottom row's zoom panels.

6.2.4 Guard rules

- **Four rules decide which qualifying shots are rebuilt.** Each one is there to stop a specific failure.
  - **Whole shots only.** The look never changes mid-shot.
  - **Only where the rebuild removes at least 70% of the noise.** A partial cure is not worth the detail it costs.
  - **Never on mostly-still shots.** Their frames are already kept exact by the stillness rule of Chapter 4 (Evidence 4, 4.2.5), and this step does not override it.
  - **Never on credits, epilogue or text sequences.** The rebuild ghosts scrolling lettering.
- **Outcome.** The text rule vetoed 10 of the 126 qualifying shots, all inside the opening-credit sequence (0:46–2:21). The other guards vetoed none: above the cutoff, the rebuild worked wherever it was allowed.

6.2.5 Why the clean-up uses the mode Chapter 5 rejected

- **Same tool, opposite job.** The clean-up deliberately runs QTGMC in its `Lossless=2` mode with default temporal smoothing. The exact-lines mode used for true video motion is `Lossless=1` + `TR2=1`.
  - In Chapter 5 (5.2.3), `Lossless=2` was disqualified because it alters the disc lines it claims to keep and softens detail.
  - That property, reworking the disc's own lines, is exactly what the clean-up needs, because here those lines carry the noise. This is why the kept lines of a clean-up frame show as **green** (reworked) in the change maps rather than as untouched.

6.2.6 Every shot on one chart

- **Chart C1 places all 908 shots,** so the decision can be checked at a glance instead of taken on trust.

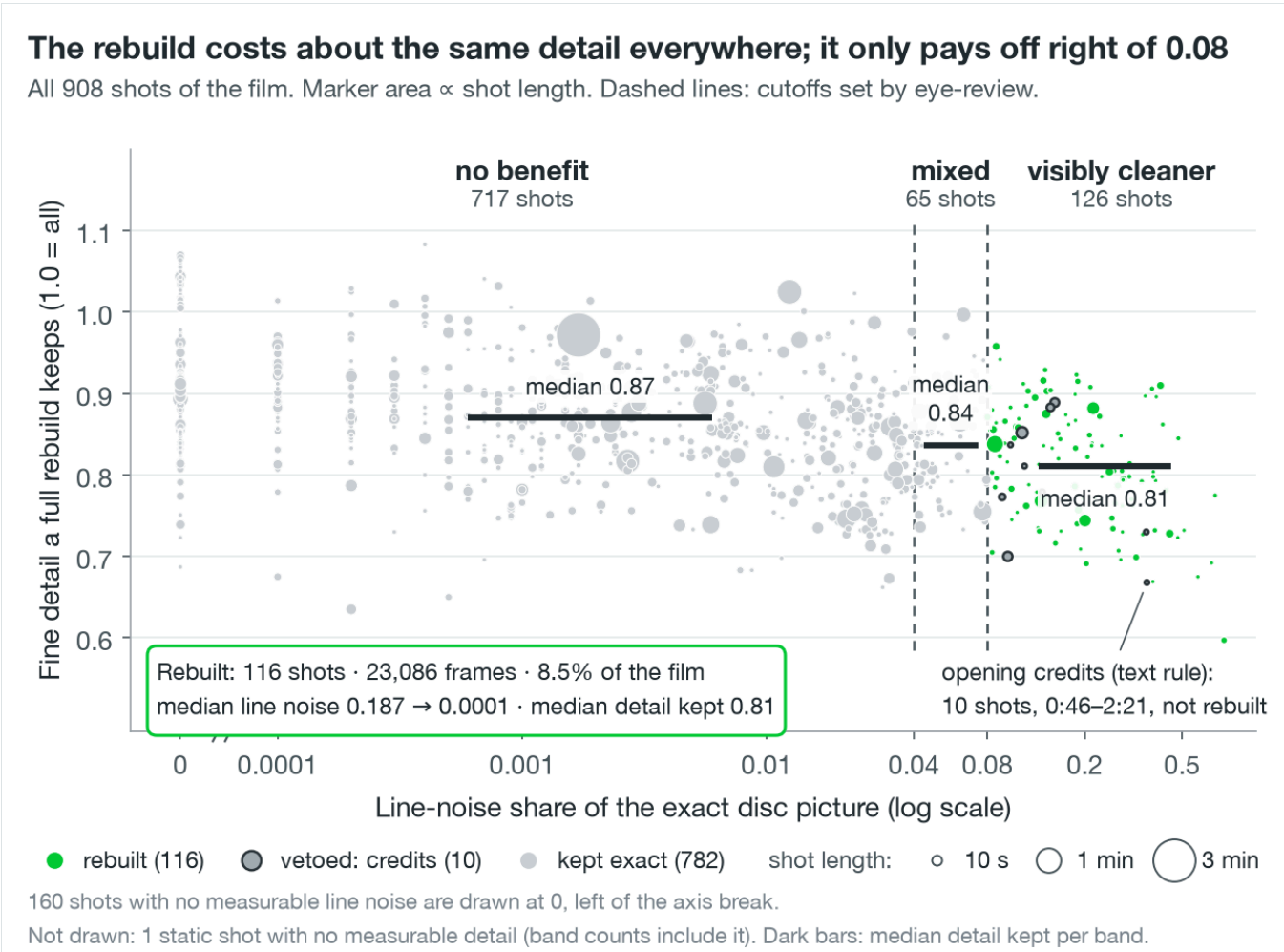


Chart C1. Every one of the film's 908 shots, placed by how much interlaced line noise its exact picture carries (across) and how much fine detail a full QTGMC rebuild would keep (up). The rebuild costs about the same detail everywhere, but it only pays off to the right of the 0.08 cutoff set by eye. 116 shots (green) were rebuilt. 10 more to the right of the 0.08 cutoff (dark-ringed dots) were vetoed because they are opening-credit text. The 160 shots with no measurable line noise are drawn at 0, left of the axis break.

What to look for: the grey cloud and the green cluster sit at about the same height (similar detail cost), but only the shots right of the 0.08 line carry line noise worth removing.

> **In plain English:** a stripe detector that grain cannot fool and a detail meter that only smoothing can move were run on every frame of two complete versions of the film. A person then judged about 20 shots to decide where the stripes start to matter. The resulting rule: rebuild only whole, moving, text-free shots whose stripes are clearly visible and actually go away.

6.3 The decision

116 / 908	**8.5%**	**0.187 → 0.0001**	**0.81**
shots given the clean rebuild	of the film: 23,086 output frames, 7.7 minutes	median line-noise share in those shots	median fine detail kept: the honest cost, accepted only where it removes a visible defect

- **The film's final mix of 271,174 output frames:**

  - 79.2% exact disc picture (74.5% own-frame join, untouched + 4.8% re-paired, lifted; no pixel calculated);
  - 12.2% QTGMC exact-lines mode (true video motion, with the disc's own lines kept bit-exact);
  - 8.5% QTGMC clean rebuild (archive line noise).
  - Before this step the split was 80.3% own-frame, 6.5% re-paired and 13.2% QTGMC.
- **Checked again after the decision.** Random final picks were re-checked side by side, including hand-painted lettering, the risk that set the cutoff (Figure F19).

Figure F19. Random final picks, re-checked after the rule was set. The rule selected all three of these shots for the clean-up. Each row, left to right: A, the exact disc picture at TV size · B, the rebuild at TV size · A zoomed 3× · B zoomed 3×. Top (47:30): a face in dark glasses. Middle (20:54): men in robes inside a rounded home-movie frame. Bottom (23:49): a hand-painted protest placard. B removes faint horizontal line noise without harming faces or brush lettering.

What to look for: the brush strokes of the lettering in the bottom row's zooms have the same shapes in A and B, minus the fine horizontal texture. Faces keep their features.

> **In plain English:** about one shot in eight got the heavier treatment, and only because a measurement and a human eye agreed it was needed. Everywhere else the film keeps the disc's own picture.

6.4 The impact

6.4.1 Process 3 · green, with blue

- **In a clean-up frame, alternate lines are treated differently** (Figure F20, archive parade at 24:16): the missing lines are invented (blue) and the disc's own lines are reworked to remove the line noise (green). The remaining ~9% of pixels were recomputed but came out identical to the disc anyway.
- **Across the whole film,** green is 4.3% of the 112.5 billion output pixels, all inside these 116 shots.
- Maps and counts describe the restored master before compression (caveat in 1.3.2).

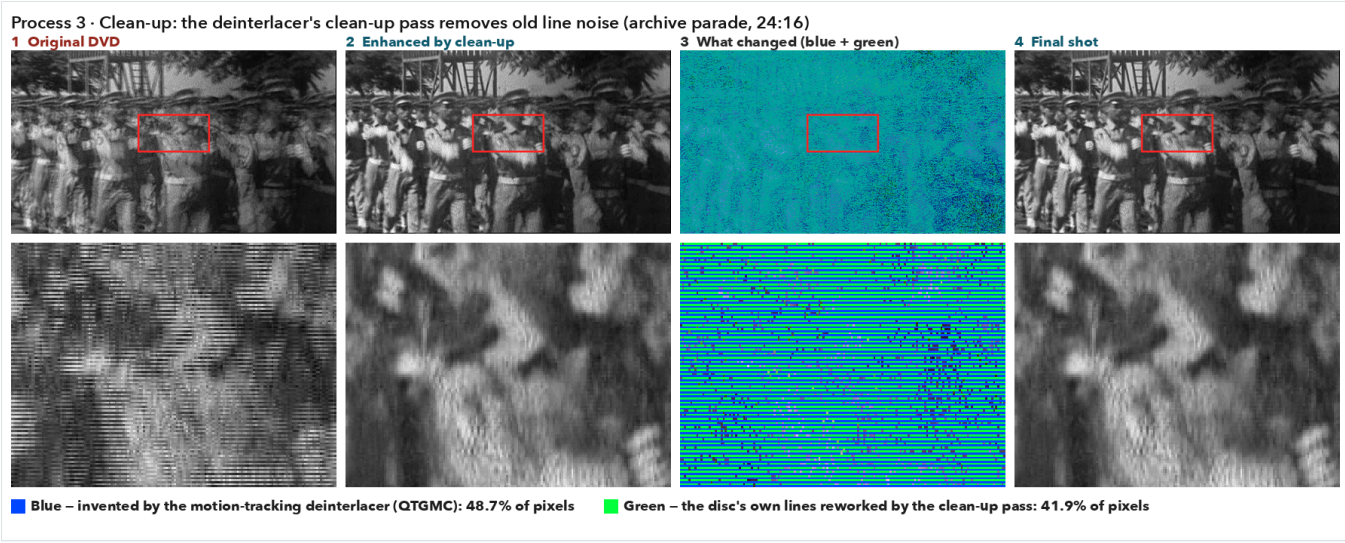


Figure F20. Process 3, clean-up (green, with blue). Archive parade, 24:16, a clean-rebuild shot. Top row: the whole frame with a red box. Bottom row: the box zoomed. Left to right: 1 Original DVD, with heavy horizontal line noise. 2 Enhanced by clean-up. 3 What changed (blue + green): alternate lines are treated differently. Blue = missing lines invented by QTGMC (48.7% of pixels); green = the disc's own lines reworked to remove the line noise (41.9%). 4 Final shot, for comparison with the map. Key: gold = lifted · blue = invented · green = reworked · no colour = untouched. At whole-frame size the coloured lines are finer than print can show, so alternating blue and green read as teal; the zoom row shows the true line-by-line pattern. Zooms show the disc's pixels unstretched (not widescreen-corrected). The maps describe the restored master before H.265 compression.

What to look for: the striped zoom in panel 1, against the calm zooms in panels 2 and 4, and blue and green alternating line by line in panel 3's zoom.

6.4.2 What it means for the viewer

- **91.5% of the film keeps its original detail and grain:** the measurements showed no clear benefit there, or (for 10 opening-credit shots) the rebuild would ghost the lettering.
- **The 8.5% that does need it loses its stripes.** Painted surfaces, costumes, sand, signs and suits that crawled with scan-line noise on the disc are now calm (Figures F14, F17, F18).
- **No shot changes its look partway through,** and the credit, epilogue and text sequences are never rebuilt.

> **In plain English:** the striped archive shots now look clean, and nothing else paid for it.

EFFORT LEDGER	CHAPTER 6
BUILT AND DISCARDED	a complete second rendering of the entire film (271,174 frames) that is used for only 8.5% of it. 91.5% of that analysis exists to prove where *not* to use the rebuild. The 0.04–0.08 "grey zone" was reviewed and rejected.
MEASURED	542,348 frame renders × 4 kinds of measurement (12 numbers per output frame, about 3.25 million in all). Five review sheets (about 20 shots) were compared side by side by eye, and the final picks were re-checked.
TIME	2 h 17 min of whole-film rendering and measurement.
KEPT	the clean rebuild on 116 of 908 shots (23,086 frames, 7.7 min).

CHAPTER 7

All Together: One Method per Frame, 50 Times a Second

Chapters 4–6 showed each process on its own. Here they meet.

Thesis. The method is chosen per output frame, 50 times a second, never per film or per scene, and the result can be audited down to the pixel.

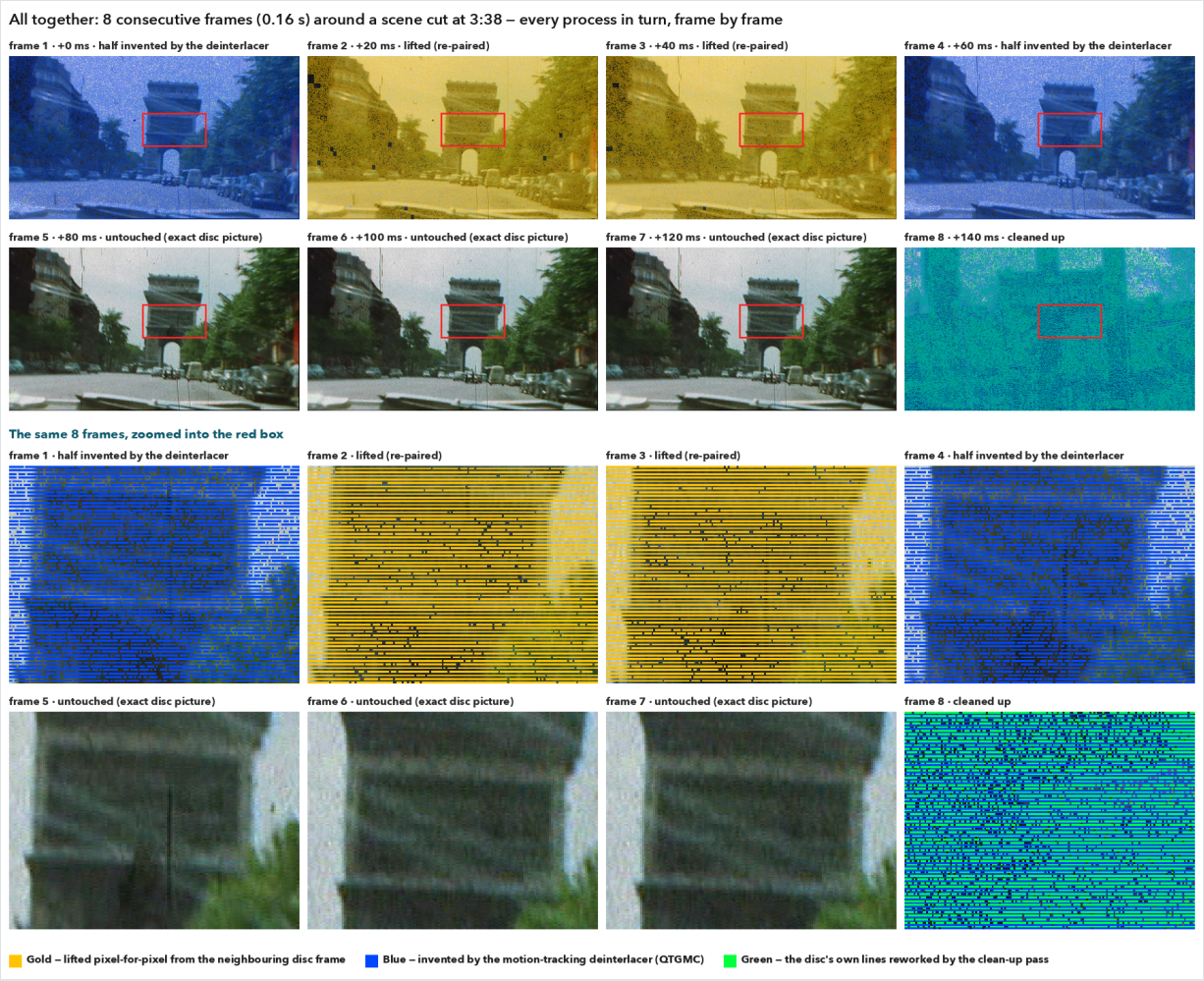


Figure F21. All together: 8 consecutive output frames, 20 ms apart, around the cut at 3:38 (Arc de Triomphe shot → next shot), each coloured by the process that built it. Top two rows: whole frames 1–4 and 5–8, left to right; bottom two rows: the same frames zoomed into the red box. Frame 1 invented (blue) · 2–3 lifted (gold) · 4 invented · 5–7 untouched (no colour) · 8, the first frame of the next shot, cleaned up (blue + green), so its zoom shows the same screen area of a different picture. At whole-frame size the coloured lines are finer than print can show: frame 8's alternating blue and green read as teal; the zooms show the true pattern. Zooms are the disc's pixels unstretched (not widescreen-corrected). Restored master, before compression (7.4.3).

What to look for: the colour changes every 20 ms inside one shot, and no frame mixes gold with blue or green.

50 / s
method decisions per second of film, one per output frame

79.2%
of frames are the exact disc picture (74.5% untouched + 4.8% lifted)

85.4%
of output pixels are the disc's own (96.0 billion)

≈ 88.4%
of output pixels bit-identical to the DVD, measured

7.1 The problem: four methods in one film

- **Four ways to make a frame.** Every one of the 271,174 output frames is built in one of four ways:
  - **Untouched (no colour):** the field woven back together with its own frame's partner, which is exactly the disc's picture.
  - **Lifted (gold):** the field joined with the same-instant field of the neighbouring disc frame. This exactly repairs mis-paired film.
  - **Invented (blue):** QTGMC, the motion-tracking deinterlacer, keeps the field's own disc lines bit-exact and calculates only the missing half.
  - **Clean-up (blue + green):** a QTGMC rebuild of whole archive shots. One half of each frame is invented, and the disc's own lines in the other half are reworked.
- **Mixing four methods in one film raises three risks:**
  - **Risk 1, seams.** If two methods met inside one picture, the boundary between them could show as an edge.
  - **Risk 2, flicker.** If the look changed from one frame to the next inside a shot, the shot could shimmer as it switched.
  - **Risk 3, provenance.** After all this processing, how much of what the viewer sees still comes from the disc? A restoration that cannot answer that question cannot claim to be faithful.

> In plain English: the film is now a mixture of untouched, repaired and calculated pictures. Two things had to be proved: that the joins between them are invisible, and exactly how much of the result is still the original DVD.

7.2 The analysis

7.2.1 Eight frames, 0.16 seconds, every process in turn

Figure F21 follows eight consecutive output frames, 20 ms apart. They are made from the fields of five consecutive disc frames. The per-frame decision files record which method built each one:

- **Frame 1 (+0 ms): invented (blue).** This field had no same-instant partner. QTGMC kept the field's own lines and calculated the other half.
- **Frames 2–3 (+20 and +40 ms): lifted (gold).** Frame 2's field found its same-instant partner in the *previous* disc frame, and frame 3's in the *next* one. This is mis-paired film, repaired exactly with no calculation.
- **Frame 4 (+60 ms): invented (blue)** again.
- **Frames 5–7 (+80 to +120 ms): untouched.** The disc's own pairs, woven back exactly, so there is nothing to colour.
- **Frame 8 (+140 ms): clean-up (blue + green).** This is the first frame of the next shot, one of the 116 shots selected for the clean-up. Every line is coloured, alternating between invented and reworked. The scene-cut guard restored on Apple Silicon (5.2.5) keeps QTGMC's motion analysis from borrowing anything from the Arc de Triomphe shot before the cut.

In the zoom rows, gold and blue fill *alternate* lines only, because each of those frames lifted or invented exactly one field while the other field kept the disc's own lines in their own place. The uncoloured specks inside coloured lines are pixels whose new value happens to equal the disc's value at that spot.

7.2.2 Why a single frame is always built by one method

- **(1) The core question is about time, and it has one answer for the whole picture.** Was the other field captured at the same instant as this one? A field is exposed all at once, so the time gap between two fields is the same in every corner of the frame.
  - Same instant: every line can be joined exactly (untouched or lifted).
  - Different instants: the missing half has to be calculated everywhere (invented).
  - No frame in this film is "same instant" on one half of the screen and not on the other.
- **(2) Inside a deinterlaced frame, the work already adapts region by region.** Where nothing moves, the calculated lines converge on the true picture. Measured: 23.2% of the lines QTGMC invents come out exactly equal to the disc's value anyway, mostly in still areas.
- **(3) Mixing methods would cost more than it gains.** Splitting one frame into regions would create seams where the methods meet. Switching looks from frame to frame inside a shot would flicker. That is also why the clean-up is applied only to whole shots.
- **Consequence.** Gold never shares a frame with blue or green. Blue and green appear together only because the clean-up is a single method that computes both halves. That is why the "all together" view has to be a filmstrip: the processes meet in time, never inside one picture.

> **In plain English:** each picture asks one yes-or-no question: were its two halves photographed at the same moment? The answer is the same for the whole picture, so each picture gets one method. The method can change from one picture to the next, never inside one.

7.2.3 Checks: exact where claimed, harmless where calculated

- **Pixel-exact per decision type.** The pipeline was verified one decision type at a time. Woven frames are bit-identical to the fields they were meant to contain, and the kept half of every exact-lines QTGMC frame is bit-identical to the disc.
- **The methods agree on the picture.** For each decision type, random samples were rendered side by side with QTGMC's version of the same frame and with the rejected pairing. The chosen pairing matched QTGMC's picture, and the rejected one showed heavy combing. So, in the samples checked, a switch of method (blue, gold, untouched) was not a switch of picture.
- **Clean-up picks re-checked.** Random final picks were re-checked after the rule was set (Chapter 6, Figure F19).

7.2.4 From decisions to pixels

Frame shares and pixel shares give different numbers. Each step below follows from how the frames are built, and together they reproduce the per-pixel count across all 112.5 billion output pixels (271,174 frames × 720×576). Chart C2 draws each step.

- **The clean-up took 8.5 percentage points of frames from the earlier split.** 5.9 came from own-pair frames, 1.7 from re-paired frames and 0.9 from QTGMC frames, as counted in the decision files. That moved own pair from 80.3% to 74.5%, re-paired from 6.5% to 4.8%, and QTGMC exact-lines from 13.2% to 12.2% (shares rounded separately).
- **A re-paired frame is half untouched, half lifted.** 4.8% of frames → 2.4% gold pixels (plus 2.4% untouched).

- **An exact-lines frame is half kept, half invented.** 12.2% of frames → 6.1% kept + 6.1% blue.
- **A clean-up frame is half invented, half reworked.** 8.5% of frames → 4.3% blue + 4.3% green.
- **Totals.** Untouched 74.5 + 2.4 + 6.1 = 83.0% · gold 2.4% · blue 6.1 + 4.3 = 10.4% · green 4.3%. This is why 79.2% exact frames become 85.4% original pixels.

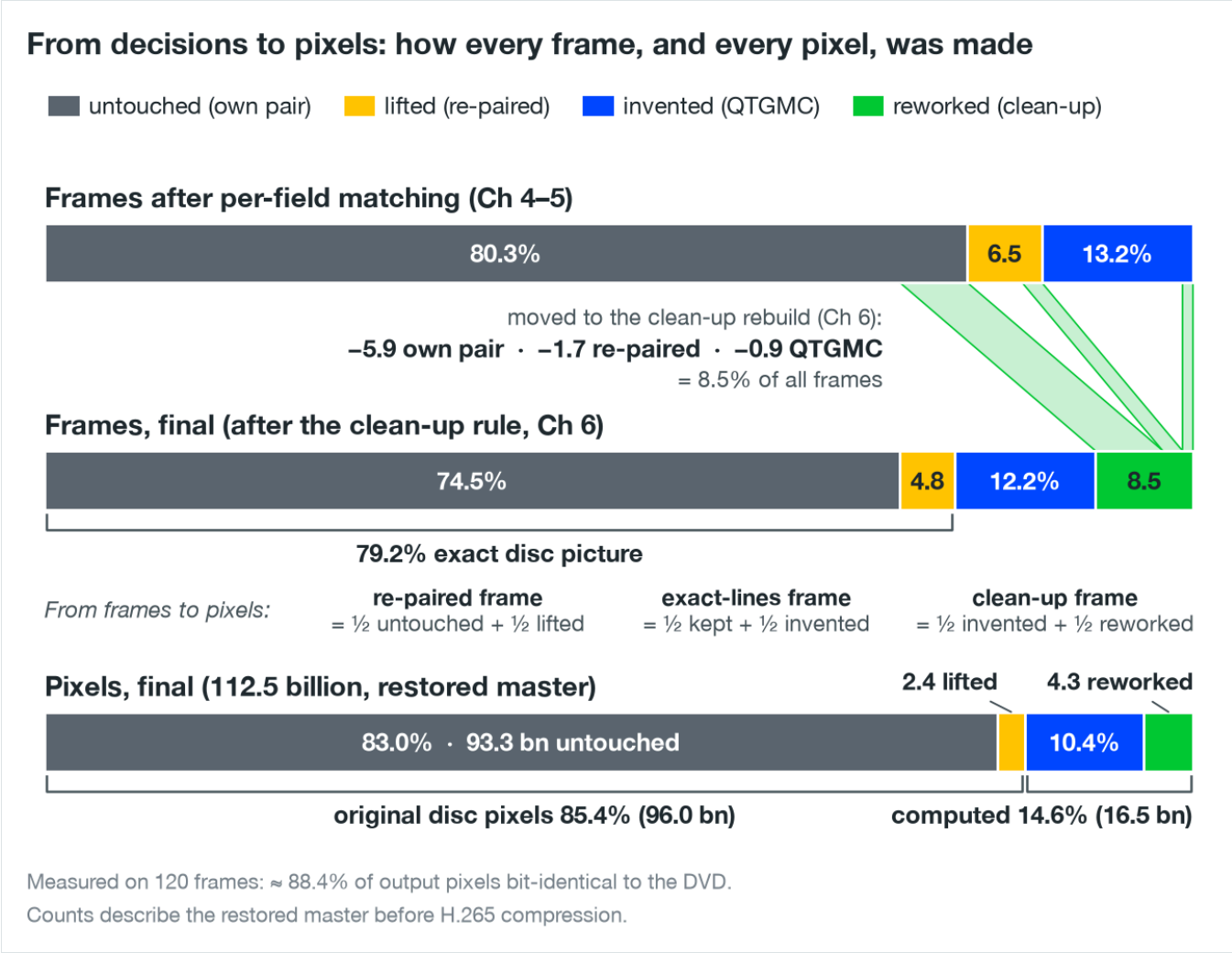


Chart C2. How the final mix came about, and why frames and pixels give different numbers. Per-field matching left 86.8% of frames as exact disc pictures. The clean-up rule then moved 8.5% of the film to a full rebuild, all of it noisy archive shots. Counted in pixels, the 79.2% exact frames become 85.4% original pixels, and only 14.6% of the pixels were computed.

What to look for: the green 8.5% in the middle bar is taken from all three segments above it (the funnel lines); in the bottom bar, slate plus gold make the 85.4% of original pixels.

7.2.5 Measured, not only constructed

- **120 sampled frames compared pixel by pixel with the disc.** Of the lines QTGMC calculated in exact-lines frames, 23.2% came out exactly equal to the disc's value anyway. Of all pixels in clean-up frames, 18.6% did. Both happen mostly in still areas and flat backgrounds.
- **Result.** 85.4% + (23.2% of 6.1%) + (18.6% of 8.5%) ≈ **88.4% of all output pixels are bit-identical to the DVD.**

7.3 The decision: the final mix

One decision per output frame, recorded for all 271,174 frames:

Colour	Method	Share of frames	Share of pixels
no colour	Untouched own pair	74.5%	74.5% untouched
gold	Lifted: re-paired film, repaired exactly	4.8%	2.4% untouched + 2.4% lifted
	Exact disc picture (rows rounded separately)	**79.2%**	
blue	QTGMC exact-lines: true video motion	12.2%	6.1% kept + 6.1% invented
blue + green	QTGMC clean-up: archive line noise (116 shots, 7.7 min)	8.5%	4.3% invented + 4.3% reworked
	Total	**100%**	**83.0 untouched · 2.4 gold · 10.4 blue · 4.3 green**

> **In plain English:** four out of five frames are the DVD's own picture. About one in eight is a moving video frame whose missing half had to be calculated. About one in twelve belongs to a noisy archive shot that was rebuilt to remove old transfer stripes.

7.4 The impact

7.4.1 The method follows the picture, every 20 ms

- **Within one shot, the method changes from frame to frame** (Figure F21: invented, lifted, lifted, invented, untouched), with no seams inside any frame (7.2.2).
- **Frames 5–7 need no colour at all.** That is what about three of every four frames in the film look like (74.5%); the lifted frames that make up the rest of the 79.2% exact picture show gold.

7.4.2 Pixel origin, accounted for completely

- **Chart C2's bottom bar is the same pixel-origin split shown in Figure F2 (Chapter 1).** Of the 112.5 billion output pixels, 85.4% (96.0 billion) are the disc's own (83.0% untouched + 2.4% lifted) and 14.6% (16.5 billion) were computed (10.4% invented + 4.3% reworked).

7.4.3 What these counts measure

> **Honesty note.** These counts and colour maps describe the *restored master* that feeds the encoder. The final H.265 file uses lossy compression, as the DVD's own MPEG-2 does, so the "bit-identical" figures refer to the master, not to the compressed file. The finished film was then scored frame by frame against that master: all 271,174 frames averaged VMAF 96.08 (worst 1%: 91.10), with a mean PSNR (a simple pixel-difference score) of 46.23 dB and no frame below 30 dB (Chapter 16).

EFFORT LEDGER	CHAPTER 7
BUILT AND DISCARDED	nothing new. This chapter audits the three before it, whose ledgers carry the discards: a 55.8 GB bwdif master, QTGMC's Lossless=2 mode for motion frames, and a full-film rebuild that ended up used for only 8.5% of the film.
MEASURED	pixel-exact checks per decision type; 120 frames compared pixel by pixel against the disc; the origin of all 112.5 billion output pixels, with one decision recorded for each of 271,174 frames.
TIME	covered by the chapters above, with no separate pass. The audit reuses the per-field decisions and the 2 h 17 min double render.
KEPT	one decision per output frame, 50 per second.

CHAPTER 8

Framing: Proving the Picture Was Already Right

Thesis. Sometimes the best restoration decision is a proven "leave it alone". Most chapters in this paper end with something changed; this one ends with nothing changed, and that was the rigorous result. One of the project's goals was correct handling of black bars and stretched footage. An automatic detector raised 100 possible problems. Each was checked by eye at true display shape, and the 16:9 shape of the picture was read back from the stream's own flags. The finding: every black edge in the film is either the filmmakers' design or plain darkness, and nothing is stretched.

749	100	5	0
scenes scanned by the black-border detector	scenes flagged as possibly boxed in	contact sheets, every flag checked by eye	scenes cropped or resized

8.1 The problem

8.1.1 Goal 6: black bars and stretched footage

- **The goal.** The brief asked for correct handling of interlacing, black bars and any stretched footage (goal 6). Interlacing is covered in Chapters 3 to 7. This chapter covers the other two.
- **The detector.** An automatic black-border detector measured the non-black picture area of each of the film's 749 merged scenes. It flagged 100 of them as possibly boxed in by black:
  - **55 "pillarbox":** black down the sides, as when a narrow picture sits inside a wide frame.
  - **28 "windowbox":** black all round.
  - **17 "letterbox":** black above and below, as when a wide picture sits inside a narrower frame.

  Together the flagged scenes run about 16.6 minutes, roughly 18% of the film. 647 scenes read as full 16:9, and 2 as black or blank.
- **Why a flag matters.** Genuine bars are wasted screen: a player shows the picture small, inside a black frame of its own. The usual cure is to crop the bars and scale the picture up to fill the screen. Done right, that is a real improvement. Chapter 15 does exactly that for two trailers.

8.1.2 The risk runs both ways

- **A wrong aspect flag would stretch every face.** The disc stores the picture *anamorphically*: 720×576 pixels that are each wider than tall (pixel aspect ratio 64:45). The player widens them to 1024×576 for a 16:9 display. The shape comes from a flag in the stream, not from the pixels themselves.
  - Shown as 4:3 instead, the picture would be 25% too narrow, and every interview subject would look gaunt.
  - "Correcting" a picture that was never stretched would distort it the other way.
- **Cropping designed black would destroy the design.** Many of the flagged scenes use black on purpose: a strip of picture, a rounded inset, a band that carries a caption. Cropping and scaling such a scene would cut off the design, blow a thin strip up to full height, or delete a name caption entirely.
- **Any crop or resize recomputes every pixel.** Scaling resamples the whole picture. That goes against this restoration's principle (Chapter 1) of computing pixels only where the disc's picture is missing or

damaged.

> **In plain English:** a computer flagged 100 scenes that "might" have black bars. Fixing a real black bar makes the picture bigger and better. "Fixing" one that the filmmakers put there on purpose ruins the shot. And getting the overall shape wrong would make everybody look squashed or stretched. So every flag had to be checked before anything was touched.

8.2 The analysis

8.2.1 All 100 flags reviewed by eye, at true display shape

- **Contact sheets.** One frame from each flagged scene was laid out on 5 contact sheets of 20 tiles each (4 across, 5 down). Each tile carries the scene's start time, the detector's guess, and the measured shape of the non-black area.
- **Drawn at true 16:9 shape.** Each tile is 384×216 pixels, a true 16:9 rectangle. So each tile shows the picture as a viewer sees it, not in the squeezed 720×576 shape the disc stores. Judging proportions on the stored shape would have made every face look wrong.

8.2.2 The verdict: design or darkness, every time

- **Deliberate design.** Four kinds of designed black recur:
  - **Split-screen panels and strips:** only part of the frame carries picture. Examples: a thin strip of airport footage at 1:10 (the non-black area measures 5.32:1), a two-panel restaurant scene at 17:37, and a jet in flight at 62:05.
  - **Home movies inside rounded "old TV" frames:** colour home movies at 20:32 and 20:52, presented as if on an old television set.
  - **Black bands that hold name captions:** for example the »AVIGDOR« caption at 8:35. A crop would have removed the caption itself. That same caption is one of the German titles that Chapter 14 translates.
  - **Epilogue cards and credits on black:** the closing cards (85:32), the end credits (86:37) and the final logo (89:54).
- **Plain darkness.** Everything else is a low-key shot whose edges simply fall into shadow:
  - 25 of the 100 flags come from one recurring night-time interview set-up: a speaker against a dark background, with city lights out of focus behind him.
  - Extreme close-ups lit on a sliver of a face read as shapes as narrow as 0.73:1.
  - A telegraph key in deep shadow at 21:22 reads as "pillarbox 1.74".
- **What was not found.** No scene had bars left over from a format conversion, and none showed stretched or squeezed geometry.

8.2.3 The 16:9 flag is correct, checked two ways

- **The flags.** The film stream is 720×576 with a sample aspect ratio of 64:45 and a display aspect ratio of 16:9. It displays as 1024×576.
- **Natural proportions.** On the true-shape contact sheets, interview subjects look natural, neither squeezed nor stretched.

Figure F22. Four scenes flagged by the black-border detector and kept exactly as authored, from one of the five review sheets, drawn at true 16:9 shape. Top left: a split-screen effect at 17:37 in which the right third of the frame is black. Top right: a colour home movie at 20:32, presented inside a rounded "old TV" frame. Bottom left: another home movie at 20:52, in the same rounded frame. Bottom right: a dark, low-key shot of a telegraph key at 21:22, whose black edges are shadow. The black is design or darkness, not bars to remove; none of the 100 flagged scenes was cropped or resized.

What to look for: the rounded frames around the home movies, which are deliberate and kept. The tile labels are the detector's guesses: "letterbox 1.98", for example, is the shape of the non-black area, not the real frame shape, which stays 16:9.

> **In plain English:** every one of the 100 suspect scenes was looked at, drawn at the exact shape a TV shows. Every time, the black was either part of the filmmakers' design (a split screen, an old-TV frame, a strip for a name caption, a credits card) or just a dark room, and the people in the interviews look naturally proportioned at the disc's 16:9 setting.

8.3 The decision

- **Keep the film's framing exactly as authored:** no cropping, no resizing, no padding.
  - The encode writes the same 720×576 frame with the same 64:45 pixel shape and 16:9 flag as the disc (Chapter 9).
  - Every player widens it to 16:9 on screen, exactly as the DVD player did.
- **Why this is the right call.**
  - The filmmakers' designs survive intact.
  - No face can end up stretched.
  - No caption is cut off.
  - No pixel in the film is resampled for framing.

> **In plain English:** the picture was already the right shape, and the black edges belong there. So the film's frame was left exactly as the disc has it, and the TV does the widening, just as a DVD player does.

8.4 The impact

- **The designed look is preserved.** The split screens, the old-TV home movies, the caption bands and the cards on black all appear exactly as the filmmakers composed them.
- **Nothing is stretched.** The stream's flags read back as 16:9, and proportions were checked by eye on all 100 flagged scenes at true display shape.
- **Zero computed pixels from framing.** Of the 14.6% of output pixels the restoration computes (Chapter 1), none comes from framing. This step added no change at all.
- **Real bars were fixed where they exist.** The only genuinely letterboxed pictures on the disc are two trailers among the extras, and they were cropped to fill 16:9 (Chapter 15). The rule was never "do not crop". It was "crop only what is really a bar, and prove it first".

EFFORT LEDGER	CHAPTER 8
BUILT AND DISCARDED	every crop the detector proposed. 100 scenes were flagged (55 pillarbox, 28 windowbox, 17 letterbox); 0 were cropped and 0 resized.
MEASURED	749 scenes scanned for black borders. All 100 flagged scenes reviewed by eye on 5 true-16:9 contact sheets. The stream's aspect flags read back (720×576, 64:45, 16:9). Proportions checked on the interviews.
TIME	not separately logged; part of the Phase 1 whole-film analysis. Leaving the framing alone adds no processing time to the final encode.
KEPT	the film's original framing and 16:9 flag, verified.

CHAPTER 9

Encoding: One Quality Target, Chosen from 60 Test Encodes

Thesis. Compression is chosen by measurement and verified by eye, at the size people actually watch. The restored picture had to be compressed into a file a television can play. No setting was taken from a forum or a default: every candidate was encoded and scored frame by frame against the restored master, the finalists were compared side by side at TV size, and the finished encode was scored again, every one of its frames.

60	10 / 5 / 0	7 h 59 min	96.08
test encodes (30,000 frames scored); 1 setting shipped	wins / ties / losses for standard settings vs grain tuning	to encode the film on one laptop, extras alongside	mean VMAF over all 271,174 film frames, measured against the restored master

9.1 The problem

9.1.1 The master is far too large, and grain resists compression

- **A lossless master is enormous.** The film now has 271,174 frames, twice the disc's 135,587, because every field became its own frame. An earlier lossless (FFV1) master at 25 frames per second measured 55.8 GB, so a lossless 50 fps master would be about 110 GB.
- **Archive grain is nearly incompressible.** Compression works by predicting each frame from the frames around it and storing only the difference. Film grain is new random noise in every frame, so it has nothing to predict from.
  - Much of this film is grainy archive footage: street film, newspapers, crowds and Super-8 home movies.
  - Every encoder setting therefore trades file size against how much of that grain survives.

9.1.2 The devices set hard limits

- **Goal 9:** the film has to play well on Apple TV, Fire TV, Roku and phones. Those devices decide which codecs are usable, whatever their efficiency on paper.

> **In plain English:** the restored film is far too big to store or stream as it is, and its grain is the hardest material there is to shrink. The aim was to shrink it without visibly losing that grain, in a format every target device can play.

9.2 The analysis

9.2.1 Codec study: three candidates, two rejected

- **AV1** is about 26% more efficient than H.265, so it gives the same quality in roughly a quarter less space. *Rejected:* Apple TV has no AV1 hardware decoding. (In the frame-rate test, doubling to 50 fps also cost +63% in size with AV1 against +21% with H.265.)
- **H.266 (VVC)**, the newest standard. *Rejected:* none of the target devices support it.

- **H.265 / HEVC, Main 10 profile (10-bit).** *Kept:* it plays natively on all four device families. Ten-bit precision keeps smooth dark gradients free of banding, which matters in the film's dark reconstruction scenes.

9.2.2 Preset benchmark: how hard the encoder searches

- **Measured on the same 800 frames:** *slower* 55 s · *veryslow* 104 s · *placebo* 170 s.
- **Chosen: veryslow.** It spends nearly twice the analysis of *slower* on every frame. *Placebo* would have added another 63% of encoding time for exhaustive searches that typically give only marginal further gains.

9.2.3 Calibration: 60 test encodes, 30,000 scored frames

- **Ten deliberately hard, representative test scenes** of 500 output frames each: interview, repaired archive street, horse-race credits, dark reconstruction, crowd, Super-8 epilogue, rolling credits, grainy newspaper, motorcade, and crowd close-up.
- **× 2 tunings:** x265 standard settings with SAO off (`--no-sao`) against the purpose-built `--tune grain`. SAO (Sample Adaptive Offset) is a smoothing filter built into the encoder that is commonly switched off for grainy sources because it tends to flatten fine texture.
- **× 3 quality levels:** CRF 14, 16 and 18. CRF (Constant Rate Factor) is a quality target, not a size target, and a lower number means higher quality.
- **Scoring:** every frame of every encode was compared with the restored master by two independent full-reference metrics.
  - **VMAF**, Netflix's perceptual quality model: scores run 0–100, higher is better, and about 95 is widely treated as a rule of thumb for "very hard to tell from the source", which is why 95 is used as the bar here.
  - **SSIMULACRA2**, a newer perceptual metric designed for the high-quality range (up to 100; higher is better).
  - Five encodes were then re-scored with VMAF after scaling to 1080p, the size a TV actually shows.
- **Time:** the whole calibration took about 50 minutes.

> **In plain English:** instead of guessing a setting, every combination was tried on the ten hardest kinds of footage in the film. Two independent "how close to the original?" meters read every one of the 30,000 resulting frames.

9.2.4 Findings: standard settings beat grain tuning

- **Standard settings won at matched file size:** 10 wins, 5 ties, 0 losses against `--tune grain`.
  - At every quality level, grain tuning spent 18–24% more bits (11.5 vs 9.7 Mb/s at CRF 14) without a better SSIMULACRA2 score (84.9 for grain tuning vs 85.1 for standard settings; higher is better).
  - The setting built "for grain" was not the best one for this grain. Only measurement could show that.
- **Average bitrates on these hard scenes:** CRF 14 / 16 / 18 = 9.7 / 7.4 / 5.4 Mb/s, against the DVD's 8.5 Mb/s MPEG-2 average. Averaged native VMAF was 96.3 / 95.8 / 95.0.

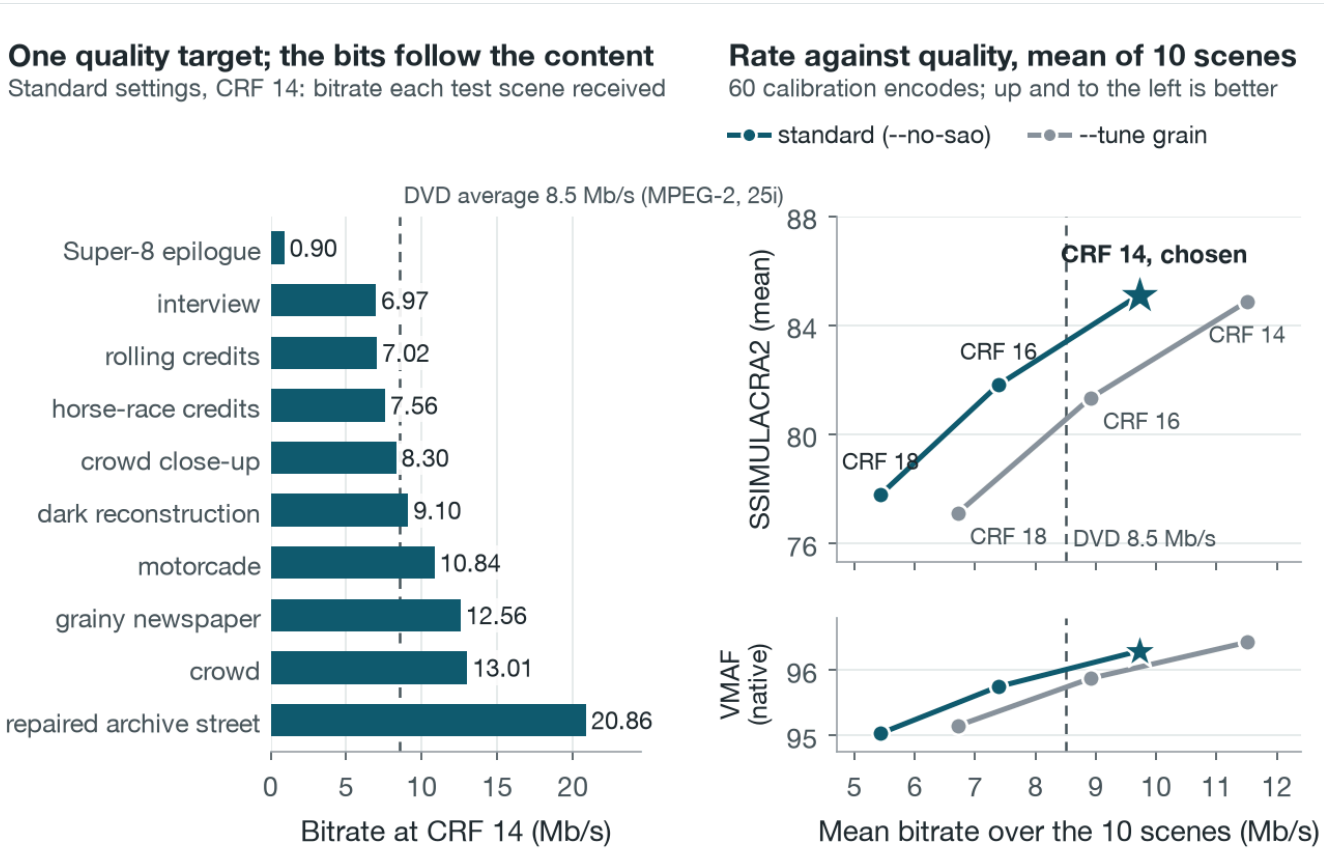


Chart C3. Left: one setting, CRF 14, yet each test scene received a very different number of bits, from under 1 Mb/s for a calm Super-8 clip to about 21 Mb/s for grainy archive street film. That is constant-quality encoding at work. Right: quality against bitrate from the 60 calibration encodes; top right is SSIMULACRA2, bottom right is VMAF at native size, and both tell the same story. "--no-sao" means standard x265 settings with its SAO smoothing filter switched off. The grain-tuned curve needs more bits for the same score, which is why standard settings won (10 wins, 5 ties, 0 losses at matched size). The chosen point is starred.

What to look for: in both right-hand panels the teal (standard) line sits above and left of the grey (grain-tuned) line; on the left, the same setting gives scenes more than twentyfold different bitrates.

9.2.5 The honest finding: grain defeats the meters, so the eye decides

- **Even CRF 14 does not reach VMAF 95 on the very worst frames at TV size.**
  - One example is the repaired archive-street scene at CRF 14. At native size it averaged VMAF 95.8, and its worst frame scored 93.2.
  - Scaled to 1080p, the same encode averaged 90.5, and its worst frame scored 84.0. In plain terms: by the meter, those worst frames fall short of the 95 bar even at the best setting tested, so the meter alone could not settle the choice and the decision went to the eye.
- **Why.** Grain is essentially incompressible. A full-reference metric compares pixel by pixel, so grain that the encoder re-draws slightly differently counts as error even when it looks the same. Enlarging the picture to TV size enlarges that error too.
- **So the finalists were judged by eye, at TV size, on the encoder's hardest frames** (Figure F23). CRF 14 matches the master. CRF 18 keeps every edge but slightly flattens the grain. For the whole film, CRF 18 would have been about 35% smaller.
- **CRF 14 was chosen: quality first.**

Figure F23. The encoder's hardest frames, enlarged to TV size. Left: the restored master, the uncompressed input to the encoder. Middle: H.265 at CRF 14 (chosen). Right: CRF 18 (tested and rejected). Top: the Eiffel Tower's lattice against a grainy sky, from the repaired archive-street test scene. Bottom: grainy Super-8 home-movie footage of a lawn from inside the interview test scene. CRF 14 is indistinguishable from the master; CRF 18 keeps the edges but slightly flattens the grain in the sky and grass.

What to look for: the grain in the grey sky between the tower's legs is alive on the Left and in the Middle, and a touch smoother on the Right. The difference is deliberately subtle; that is the honest result. "Exact master" in the labels means the restored reference before compression. The bottom row's "interview" label names the test scene, not the footage shown.

9.2.6 Constant quality in practice: the bits follow the content

- **One target, very different budgets.** At the same CRF 14 the test scenes received anywhere from 0.9 Mb/s (the calm Super-8 epilogue) to 20.9 Mb/s (the grainy repaired archive street), as Chart C3 (left) shows.
- **These are not per-scene settings.** x265 adjusts its compression frame by frame to hold one quality target, and it spends bits where the picture is hard and saves them where it is easy.

> **In plain English:** scenes were not hand-tuned. One quality bar was set for the whole film, and the encoder automatically gives a quiet shot a trickle of data and a grainy street scene a flood.

9.2.7 Distributed encoding: evaluated and rejected

- **Throughput on the laptop:** one x265 *veryslow* instance runs at about 7.7 frames/s. Five side by side reach about 11 frames/s in total.
- **A second, older desktop computer** (a six-core processor from 2013) was benchmarked at 0.9 frames/s, so it would have added less than a tenth to the total. **A second laptop** was also considered.
- **Rejected:** neither was worth the cost of moving frames between machines and keeping the encoder build and settings identical on each. A single machine also guarantees one consistent encode.

9.3 The decision

9.3.1 One setting for the whole film

Setting	Value	What it means
Encoder	x265 4.3, preset *veryslow*	the widely used open-source H.265 encoder, at its most thorough practical search
Codec / profile	H.265 / HEVC, Main 10, 10-bit, 4:2:0 colour (colour detail stored at half resolution, exactly as on the DVD)	plays natively on Apple TV, Fire TV, Roku and phones
Quality	CRF 14, SAO off, otherwise x265 defaults	one quality target; the bits follow the content
Keyframes	at least every 250 frames (5 s)	a keyframe is a complete picture a player can jump straight to, so seeking is fast and accurate
Frame rate	50 frames/s constant	one frame per field of the disc
Picture	720×576, 16:9, pixel aspect (SAR) 64:45	exactly the disc's raster, shown as 1024×576; no resizing, denoising or sharpening
Colour	BT.601 / PAL flags (stored as BT.470BG and SMPTE 170M), limited range	the TV reads the colours exactly as the DVD intended

9.3.2 Chunked, parallel, verified encoding

- **Split at shot boundaries into 32 chunks** of about 4,000–5,750 source frames each (8,000–11,500 output frames), so no chunk boundary falls inside a shot. With the three extras files, that makes 35 pieces.
- **Up to 8 pieces encode at once** on the one laptop, the extras alongside the film.
- **Each piece is verified before it is accepted.** Its encoded frame count must equal the master's. A piece that fails is simply run again, and an interrupted run resumes without repeating finished pieces.
- **Nothing huge is ever stored.** Frames stream from the restoration pipeline (decode → per-field decisions → QTGMC) straight into the encoder, so the roughly 110 GB lossless 50 fps master never has to exist on disk.

9.4 The impact

- **The encode finished, every piece verified.** All 35 pieces, 362,954 output frames in total, passed the frame-count check. The film ran from 22:17 to 06:16, 7 h 59 min of wall-clock time; the extras ran alongside it from 22:03 to 03:39 (5 h 36 min). The per-piece times add up to 51.0 encoder-hours, an average of 1.98 frames/s per piece while sharing the processor.
- **Every frame scored against the restored master.** After assembly, each delivered frame was compared with the master frame it was encoded from, at native resolution, using VMAF and PSNR (peak signal-to-noise ratio, a simple pixel-difference score in decibels; higher is better). For the film's 271,174 frames: VMAF mean 96.08, worst 1% of frames 91.10, single worst frame 83.72; PSNR on brightness mean 46.23 dB, minimum 36.54 dB, and 0 frames below 30 dB. The extras score as well or better (table below).
- **The same story as calibration.** The average clears the 95 bar. The frames that fall below it are the grainy ones that calibration predicted (9.2.5), where the meter counts re-drawn grain as error, the same kind of frame that was judged by eye at TV size in Figure F23. The "bit-identical" pixel counts of earlier chapters describe the restored master; these scores measure how closely the lossy H.265 film stays to it.
- **Size, by design.** About the disc's size, not a drastic cut; the full accounting is Chapter 10.

File	Frames compared	VMAF mean	VMAF, worst 1%	VMAF, worst frame
The Champagne Spy (2007)	271,174 / 271,174	96.08	91.10	83.72
Featurettes / Intros	25,282 / 25,282	96.27	92.12	85.45
Interviews / Oded Gur-Arie	23,650 / 23,650	96.52	95.68	92.30
Interviews / Nadav Schirman (Director)	42,848 / 42,848	97.33	96.01	94.07

In plain English: the finished film was checked against the restored master one frame at a time, all 271,174 of them. On Netflix's 0–100 meter it averages 96, above the "very hard to tell apart" mark of 95. The frames that score lower are the grainiest ones, where the meter counts re-drawn grain as an error; in the test encodes at the same setting, those frames looked the same as the master on a TV.

EFFORT LEDGER CHAPTER 9

BUILT AND DISCARDED	59 of 60 calibration settings; AV1 and H.266; the *slower* and *placebo* presets; distributed encoding on a second computer; CRF 18 (about 35% smaller, slightly flatter grain).
MEASURED	30,000 encoded frames, each scored with VMAF and SSIMULACRA2, plus 5 TV-size re-scores; preset benchmark of 55 / 104 / 170 s on 800 frames; second computer at 0.9 frames/s vs about 11 frames/s on the laptop; all 362,954 delivered frames scored with VMAF and PSNR against the master.
TIME	calibration about 50 min; final encode 7 h 59 min wall clock for the film with the extras alongside (51.0 encoder-hours in all).
KEPT	one setting (standard tuning, SAO off, CRF 14), encoded in 35 resumable, frame-count-verified pieces (the film's 32 chunks plus 3 extras), up to 8 at once.

CHAPTER 10

The Honest Size Story

Thesis. The restored film carries twice the pictures, more bits per sample and a lossless German soundtrack in the same space the disc gives its own film: 6.20 GB, against about 6.2 GB for the disc's main-film video and audio. The whole delivered set is 7.41 GB, 6.4% smaller than the 7.92 GB disc image. That is *not* a drastic size cut, and under a quality-first brief it should not be; this chapter explains why. Every size below is labelled **MEASURED** (read from a real file) or, in the two cases where no file exists, **ESTIMATE**.

2×
frames: 135,587 on the disc → 271,174 restored

6.20 GB
film file, against ≈ 6.2 GB for the disc's own film video and audio

7.92 → 7.41 GB
disc image → whole restored set, 6.4% smaller

≈ 18×
smaller than the ≈ 110 GB lossless 50 fps master it was encoded from

10.1 The problem

- **The usual expectation.** A DVD re-encoded with a modern codec is expected to come out dramatically smaller. H.265 is far more efficient than the DVD's MPEG-2, so a much smaller file is always available, but only by spending that efficiency on size instead of quality.
- **This brief pointed the other way.** The project's goal was near-zero quality loss first and size savings second, with no limit on processing time.
  - Three of the upgrades add data by design. The film has twice as many frames. It is encoded at 10 bits per sample instead of 8. The German track is lossless (Chapter 11).
- **The risk for this paper.** File size is where a write-up most easily oversells. So every row below says whether it was read from a real file or is an estimate.

10.2 The analysis

Every relevant size, side by side. Disc rows are the yardstick, master rows are the uncompressed reference, and the last rows are the delivered files.

Item	Format	Size	Status
Disc image (whole DVD, bit-exact ISO)	7,922,694,144 bytes: film, audio, subtitles, menus, extras	7.92 GB	**MEASURED**
Disc: main-film video	MPEG-2, 25 interlaced frames/s, 8-bit, ≈ 8.5 Mb/s	5.76 GB	**MEASURED**
Disc: German dub	AC-3 2.0, 320 kb/s (the Hebrew track is the same format)	≈ 217 MB	**MEASURED**
Disc: main-film streams together	video + German and Hebrew AC-3	≈ 6.2 GB	**MEASURED**
Lossless master, 25 frames/s (built, then discarded)	FFV1 lossless	55.8 GB	**MEASURED**
Lossless master, 50 frames/s (never stored)	FFV1 lossless, twice the frames	≈ 110 GB	**ESTIMATE**

Item	Format	Size	Status
Restored film file	H.265 Main 10, 50 progressive frames/s, CRF 14 + German FLAC + Hebrew AC-3 + 2 subtitle tracks + 24 chapters	6,196,136,052 bytes = 6.20 GB	**MEASURED**
… of which the German track	FLAC 16-bit, lossless, re-timed	≈ 445 MB	**MEASURED**
… of which the Hebrew original	AC-3 2.0, 320 kb/s, copied bit-for-bit	same as on the disc	**MEASURED** (bit-identical)
Same film at CRF 18 (rejected)	H.265 Main 10, 50 frames/s, CRF 18	≈ 35% smaller video	**ESTIMATE** (projected from the calibration encodes)
Featurettes / Intros	H.265 Main 10, 50 frames/s, CRF 14; original AC-3	247,207,771 bytes = 0.25 GB	**MEASURED**
Interviews / Oded Gur-Arie	same; English AC-3 + German subtitles	262,906,465 bytes = 0.26 GB	**MEASURED**
Interviews / Nadav Schirman (Director)	same; English AC-3 + German subtitles	705,758,918 bytes = 0.71 GB	**MEASURED**
Whole restored set	the four files above	7,412,009,206 bytes = 7.41 GB	**MEASURED**

10.2.1 The yardstick: the disc's own film

- **5.76 GB of MPEG-2 video.** 135,587 interlaced frames (1:30:23.5) at 25 frames/s, 8 bits per sample, about 8.5 Mb/s on average. Each frame holds two half-pictures (fields) taken 20 ms apart.
- **With its two soundtracks, about 6.2 GB.** The German and Hebrew AC-3 tracks (320 kb/s each) bring the disc's film streams to about 6.2 GB. That is the fair comparison for the restored film file, which also carries its soundtracks. The 7.92 GB disc image adds menus and the extras.

10.2.2 Lossless masters (measured and estimated)

- **A lossless master is enormous.** It stores every pixel exactly, including the film's grain, and random detail cannot be packed tightly.
  - The first master, a per-scene bwdif deinterlace of the whole film at 25 frames/s stored as lossless FFV1, measured 55.8 GB. That is almost ten times the disc's video. It took about 5 minutes to render and was discarded once per-field matching proved better (Chapter 4).
  - A 50 frames/s master would be about 110 GB.
- **It is never written to disk;** frames stream straight into the encoder in checked chunks (Chapter 9.3.2).

10.2.3 The restored film file (measured)

- **6.20 GB for the complete film:** H.265 Main 10 (10-bit) video at 50 progressive frames/s (CRF 14, preset veryslow, SAO off), the lossless German FLAC, the Hebrew original, both subtitle tracks and 24 chapters.
  - That is the same space as the disc's own film streams (about 6.2 GB) for twice the frames: half the bytes per displayed frame, with the soundtracks counted on both sides.
- **Why twice the frames do not cost twice the bytes.**
  - **1. A better codec.** H.265 predicts each picture from its neighbours far more efficiently than MPEG-2 does.

  - **2. Neighbouring frames are nearly identical.** Two frames 20 ms apart differ only where something moved. When this was measured on a worst-case motion clip, going from 25 to 50 frames/s added 21% to the H.265 file, not 100%.
  - **3. Constant quality, not constant size.** CRF gives every scene only the bits it needs, so calm shots cost little (Chapter 9.2.6).
- **Against the master.** The film file is about 18× smaller than the ≈ 110 GB lossless master it was encoded from. How closely it stays to that master is measured frame by frame in Chapter 9.4.

> **In plain English:** the new file holds twice as many pictures in about the same space for two reasons. Pictures 1/50 of a second apart are almost the same, and a modern codec only records what changed. Film grain is the expensive part, because it changes randomly in every picture. Keeping the grain is what keeps the file from shrinking further.

10.2.4 The road not taken (each option measured, each rejected)

- **CRF 18 would have been about 35% smaller.** At TV size it kept every edge but slightly flattened the grain (Figure F23). *Rejected for quality.*
- **AV1 would have saved about a quarter of the video size.** *Rejected because* Apple TV has no AV1 hardware decoding (Chapter 9.2.1).
- **25 frames/s would have avoided the +21%** that 50 frames/s costs on worst-case motion. The price would have been half of the motion. *Rejected* in the four-way demo (Chapter 3).
- **Each of these would have made a smaller file.** Each was measured, then turned down for picture quality, motion or playback.

10.2.5 German audio (measured)

- **FLAC about 445 MB, against about 217 MB for the disc's AC-3.** That is roughly twice the size, about 228 MB more.
  - **Why it grew.** The German dub drifted 22.16 ppm against the picture and had to be re-timed. An AC-3 track cannot be re-timed without being decoded, and encoding it back to AC-3 would add a second generation of loss. FLAC stores the corrected track losslessly (16-bit, triangular dither; Chapter 11).
  - **In proportion.** The extra 228 MB is about 3% of the whole set.
  - **Measured, not guessed.** An early size estimate for this track was corrected along the way; these figures come from the actual files.
- **The Hebrew original adds nothing new.** It is the disc's AC-3 track (320 kb/s), copied bit-for-bit, at the same size as on the disc.

10.2.6 The whole set (measured)

- **7.41 GB in total:** the 6.20 GB film plus 1.22 GB for the three extras files (the intros and the two interviews, also at 50 frames/s and CRF 14).
  - Against the 7.92 GB disc image, the restored set is 6.4% smaller: about the same size, not a drastic cut.
  - The bit-exact disc image stays as the archive. The MKV set is the everyday, media-server-ready copy that sits beside it.

10.3 The decision

- **Quality over size: CRF 14.** It was chosen after seeing the restored master, CRF 14 and CRF 18 side by side at TV size.
- **The codec's efficiency went into the content.** It bought twice the frames, 10-bit encoding and a lossless German track, not a smaller file.
- **Honest reporting.** The paper claims no drastic size cut. Every size is read from a real file; the two that cannot be (a master that was never stored, a setting that was never run on the whole film) are labelled as estimates.

10.4 The impact

- **Twice the motion samples.** 271,174 frames instead of 135,587. Every field instant becomes its own full picture.
- **More bits per sample.** The video is encoded at 10 bits per sample (Main 10) instead of the disc's 8.
- **A lossless German track,** re-timed to within 0.66 ms worst case (Chapter 11). The Hebrew original is untouched.
- **In disc-sized storage:** a 6.20 GB film file against about 6.2 GB of film streams on the disc, and 7.41 GB in all against the 7.92 GB disc image.

> **In plain English:** the choice was a better film at about the disc's size, rather than one about a third smaller with slightly flatter grain. For about the same storage as the DVD, the viewer gets smoother motion, finer picture data and lossless German sound in sync with the lips, and the disc itself is kept too, as the untouched original.

EFFORT LEDGER	CHAPTER 10
BUILT AND DISCARDED	the 55.8 GB lossless 25 frames/s master of the film, plus lossless bwdif masters of all 8 extras; the CRF 18 and CRF 16 options; AV1; a 25 frames/s output.
MEASURED	the 7.92 GB disc image and its 5.76 GB of film video; the 55.8 GB master; 60 calibration encodes (CRF 14 / 16 / 18 averaging 9.7 / 7.4 / 5.4 Mb/s on hard scenes, from which CRF 18 is projected at about 35% smaller); +21% file size for 50 frames/s on worst-case motion; FLAC about 445 MB against AC-3 about 217 MB; the four delivered files, 7.41 GB in all.
TIME	no separate step. The size questions were answered by measurements made for other chapters. The discarded master took about 5 minutes to render, and the calibration encodes that fixed the size took about 50 minutes (Chapter 9).
KEPT	a streaming pipeline with no 110 GB intermediate, and a 6.20 GB quality-first film in a 7.41 GB set, stored beside the bit-exact disc image.

CHAPTER 11

The German Dub, Re-timed to Sub-Millisecond Accuracy

Thesis. Prove what the error is before correcting it. The requirement was simple: voices must not run fast in some scenes and slow in others. The German dub on the disc is out of sync with the picture by an amount that changes over the film. Before anything was changed, the error was measured at 87 points and shown to be one constant speed error. It was then corrected in a single exact step, written losslessly and checked again at 37 independent points.

87	22.16 ppm	83 → 0.66 ms	37
sync measurements across the film before correction	the German dub's measured speed error (≈ 0.002% fast)	worst sync error, disc vs restored	independent verification windows after correction

11.1 The problem

11.1.1 The two audio tracks disagree, and the disagreement grows

- **The Hebrew original is the reference.** The disc's original Hebrew track (AC-3, the Dolby Digital format DVDs use) is locked to the picture from the first frame to the last.
- **The German dub drifts in a straight line.** It starts about 32 ms *late* (+32.2 ms at the first measurement, 0:30) and ends 83 ms *early* (−82.5 ms at 86:30).
  - That is a speed error of 22.16 parts per million: the dub runs about 0.002% too fast.
  - Over the full 90:23.5 running time this adds up to about 120 ms, which is how far the German voices move against the lips from the opening to the last scene.

11.1.2 Why 83 ms matters

- **Viewers start to notice lip-sync errors at around 40–45 ms.** The disc's German track is out by about twice that in the final reels. Viewers see mouths and words come apart.
- **One frame of the restored 50 fps video lasts 20 ms.** At its worst the disc's dub is more than four video frames off.

11.1.3 Context: two different mixes

- **The tracks are separate mixes, not copies.** Measured integrated loudness: German −21.2 LUFS, Hebrew −27.4 LUFS. (LUFS measures loudness the way a listener hears it.) The dub was mixed about 6 LU (loudness units; 1 LU = 1 dB) louder. Any comparison between the two tracks therefore has to look at the *shape* of the sound, not its level.

> **In plain English:** the German soundtrack plays very slightly too fast. The speed difference is inaudible, but over an hour and a half it builds up. By the end, the voices arrive noticeably before the lips move.

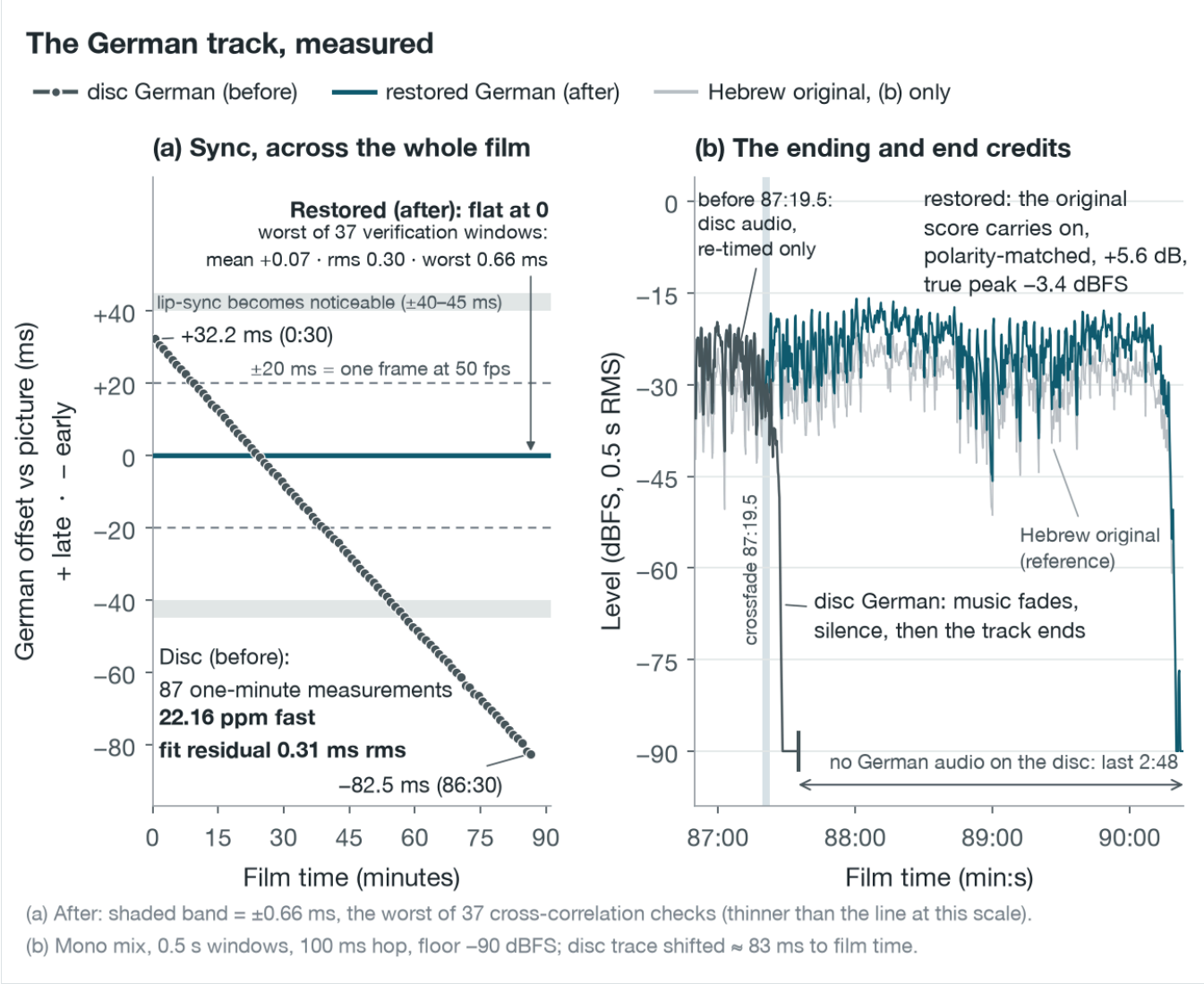


Chart C4. The German track, measured. Left, (a) Sync: on the disc the dub runs 22.16 parts per million fast. All 87 one-minute measurements fall on one straight line, drifting from about 32 ms late to 83 ms early, so one exact speed correction fixes the whole film. After re-timing, 37 independent checks found a worst-case error of 0.66 ms, a thirtieth of one video frame. Right, (b) the ending (covered in Chapter 12): the disc's German track fades to silence and then stops, leaving the 2:48 of end credits with no sound. The restored track hands over to the original score at 87:19.5 and carries the music to the last frame.

What to look for: in the left panel the dots form a straight sloping line, not a scatter, and the restored line sits flat on zero, far inside the dashed one-frame lines. In the right panel the disc trace drops to nothing while the restored trace stays at music level to the end.

11.2 The analysis

11.2.1 87 measurements, one per minute of film

- **Method: cross-correlation against the Hebrew track.** The two mixes share the film's music and much of its sound. One minute of the German track can therefore be slid against the same minute of the Hebrew track until the waveforms line up best. The slide needed to line them up is the German offset at that point in the film.
- **Coverage.** One measurement every 60 seconds, from 0:30 to 86:30: 87 points. Every minute in that range gave a measurement, each with a correlation peak of strength 0.18 to 0.80.

11.2.2 The decisive test: what shape is the error?

- **Three possible explanations, each with its own signature in the data.**
  - *A constant offset* (the dub simply starts late) would give a flat row of dots. Ruled out: the offset changes by 114.7 ms across the film.
  - *Scene-by-scene wandering*, the original concern, would give jumps at edits and scattered dots. Ruled out: there are no jumps.
  - *One constant speed error* would give a perfectly straight sloping line. Confirmed.
- **The fit.** offset (ms) = 32.64 − 0.0221629 · t (s). Residual: only 0.31 ms rms (root-mean-square: a typical size of the leftover errors). The slope, 0.0221629 ms per second, is the 22.16 ppm speed error.
- **What this proves.** The dub is uniformly fast everywhere. It is not fast in some scenes and slow in others. One exact, uniform correction therefore fixes every scene at once and cannot introduce scene-to-scene differences of its own.

11.2.3 Alternatives weighed and rejected

- **A single constant delay (the usual fix).** A container delay (a delay value stored in the file that shifts the whole track) needs no decoding at all. The measurements rule it out: the best possible constant delay would still leave both ends of the film about 57 ms off, which is above the visibility threshold. A delay can correct a late start but not a speed error.
- **Re-timing, then re-encoding to AC-3.** AC-3 is a lossy, compressed format, so it cannot be re-timed without first decoding it. Encoding the re-timed audio back to AC-3 would add a second generation of loss on top of the disc's own compression. Rejected.
- **Correcting scene by scene.** This was unnecessary because the error is one straight line. It would also have created the very thing the requirement ruled out: timing steps at every cut.

11.2.4 A discovery: the German mix is upside down

- **Every correlation peak is negative.** This was noticed during verification, and it holds for all 87 drift measurements as well (−0.18 to −0.80). The German mix is polarity-inverted relative to the Hebrew mix: its waveform is the mirror image of the original's.
- **Harmless on its own, critical in a blend.** Played alone, the German track sounds normal and was left exactly as the disc has it. But if the two tracks are ever mixed together, the inverted waveform partly cancels the original. This becomes decisive for the seamless ending in Chapter 12.

> **In plain English:** the German voices were checked against the original-language soundtrack once a minute for the whole film. All 87 checks fell on one straight line. That means the dub is not wandering; it is simply running at a very slightly wrong speed. One exact speed adjustment therefore fixes all of it. A simple delay could not do that, and saving the result in the disc's lossy format would have cost quality, so both were ruled out.

11.3 The decision: one exact, lossless correction chain

Step	Operation	Why
1	Decode the disc's AC-3 to 32-bit floating point	Gives full precision and headroom, so the intermediate steps add no meaningful rounding error.

2	Apply the measured 22.16 ppm speed correction: SoX `speed 0.9999778371` with a very-high-quality resampler	Slows the dub by exactly its measured speed error. Over the film this lengthens the track by about 120 ms. Pitch drops by 0.04 cent (a cent is 1/100 of a semitone), far too little for anyone to hear.
3	Shift by 1,567 samples (32.65 ms at 48 kHz)	Removes the start offset. 32.64 ms is the fitted value at the start of the film, rounded to the nearest whole sample.
4	Write 16-bit FLAC with triangular (TPDF) dither	FLAC is lossless: it decodes to exactly these samples. Dither turns the final rounding to 16 bits into a tiny steady noise floor instead of distortion.

- **Timing only.** The chain contains no level, equalisation or compression step. The German mix keeps its own sound and its own loudness. The ending is the one exception, covered in Chapter 12.

> **In plain English:** the German soundtrack was unpacked, slowed by 22 millionths and moved 33 thousandths of a second earlier. It was then saved in a format that keeps every sample exactly. Nothing else about the sound was touched.

11.4 The impact

- **Verified, not assumed.** The corrected track was cross-correlated against the Hebrew track at the full sample rate over 37 windows across the film. Results: mean error +0.07 ms, rms 0.30 ms, worst 0.66 ms.
  - The remaining scatter (0.30 ms rms) is the same size as the measurement points' scatter around the line before correction (0.31 ms). The correction removed all of the systematic drift. What is left is the measurement's own small scatter, not drift.
- **From visible to imperceptible.** The worst error went from 83 ms to under 1 ms, more than 100 times smaller. That is about a thirtieth of one 20 ms video frame and far below the 40–45 ms at which lip-sync errors become noticeable.
- **Identical everywhere.** Because the correction is one uniform speed change, the first scene and the last scene are in sync to the same sub-millisecond accuracy. The requirement is met by construction and confirmed by measurement.
- **The cost, stated plainly.** The lossless German FLAC is about 445 MB, against about 217 MB for the disc's AC-3 (Chapter 10). This was a deliberate choice: lossless, with no second generation of loss.

> **In plain English:** the German voices now match the lips to within less than a thousandth of a second, everywhere in the film. Before, they were up to a twelfth of a second off. The fix was not assumed to work: it was measured again at 37 more places. The price is a larger audio file, chosen deliberately so that no quality is lost.

EFFORT LEDGER	CHAPTER 11
BUILT AND DISCARDED	re-encoding the re-timed track to AC-3 (rejected: a second generation of loss). Two common shortcuts were also ruled out on the numbers: a single constant delay (would leave about 57 ms at both ends) and a scene-by-scene correction (unnecessary: one straight line fits all 87 points to 0.31 ms rms).
MEASURED	87 drift points before correction (one per minute, 0:30–86:30); 37 verification windows after; the polarity of every correlation peak (all negative); the loudness of both tracks.
TIME	about 2 min to re-time the track and write the FLAC; the real work was the 87 + 37 measurements that decide and confirm it.
KEPT	one lossless German track, changed in timing only, accurate to 0.66 ms in the worst case (it was up to 83 ms off).

CHAPTER 12

Audio Integrity: A Seamless Ending, an Untouched Original, Frame-Exact Intros

Thesis: change audio only where it is broken, and prove that every other sample is untouched. Chapter 11 re-timed the German dub to the picture. This chapter covers the three other audio jobs, and each one is about restraint. The German track had one real defect: its music runs out before the end credits. That defect was repaired only from 87:19.5 onward: a 3-second crossfade hands over to the original score, which then carries the last three minutes; nothing before 87:19.5 changed. The original Hebrew track had no defect, so it was not touched at all. The intro audio was only at risk of being damaged by the conversion itself, so its original frames were kept exactly and pinned to their original times.

3 s	+5.6 dB	−3.4 dBFS	0 bits	0.01 ms
crossfade at 87:19.5 into the original score; every German sample before it bit-identical to the re-timed track (Ch 11)	measured level difference of the same music in the two mixes	true peak after the fix, so nothing clips	of the original Hebrew track changed (AC-3 copied bit-for-bit)	accuracy to which every intro audio frame's timestamp was verified

12.1 The problem

12.1.1 The German track ends before the film does

- **The music fades out early.** On the disc's German track the score fades out at 87:23 and the track is silent until 87:32.
- **Then the track stops.** Measured on the decoded track, the German audio ends at 87:35 (5,255.07 s), while the picture runs to 90:23.5. That leaves the 2 min 48 s of end credits with no sound at all.
- **The original track keeps playing.** Over the same pictures, the Hebrew track's music plays straight through into the credits with no gap (Chart C4, panel b, Chapter 11). The German dub simply lost its last minutes of score, and anyone watching in German would have sat through the credits in silence.

12.1.2 The intros are six pieces with fragile timing

- **Six separate pieces; five carry their own AC-3 audio,** at bit rates from 192 to 320 kb/s. The intros file joins the copyright notice, the distributor's logo and four trailers.
- **The offsets matter.** The logo's sound starts 80 ms before its picture, exactly as authored on the disc. The copyright card has no audio.
- **Careless muxing (combining the streams into one file) would damage it.** A typical conversion either re-encodes mixed-bit-rate audio into one uniform stream, which adds a second generation of lossy compression, or appends the pieces end to end, which re-times them.

> **In plain English:** the German soundtrack goes quiet and then stops almost three minutes before the film ends, while the original-language version carries its music right through the credits. The short intro clips were never broken, but a routine conversion would quietly degrade or shift their sound.

12.2 The analysis

12.2.1 First ending: built, listened to, discarded

- **Version 1.** The first fix was built along with the re-timed German track. It left the German audio as it was and brought the original credits music back with a fade-in after a 5-second gap.
- **Why it was rejected.** On listening, the hole was still there. The score faded, stopped, and returned, where the original film plays one continuous piece of music. A fix that leaves an audible seam is not a fix, so version 1 was replaced.

12.2.2 Polarity: the two mixes are mirror images

- **Already known from the sync work.** Every correlation peak in Chapter 11 came out negative (11.2.4): the German mix is polarity-inverted relative to the Hebrew mix.
- **Why it matters here.** On its own an inverted mix sounds identical. But a crossfade briefly plays both signals together, and if one is inverted the two partly subtract. The music would thin out and dip in the middle of the handover.
- **The fix.** The original track's music is polarity-flipped so that it matches the German mix before the two are blended.

12.2.3 Level: measure the music, not the average

- **The averages would mislead.** Whole-track loudness is −21.2 LUFS for the German track and −27.4 LUFS for the Hebrew, a 6.2 LU gap. That figure averages dialogue and music together, so it is the wrong number for matching music.
- **The measurement used.** The level of the same music was measured in both mixes over the minute before the handover. It was 5.6 dB louder in the German mix, so the original track's music is raised by +5.6 dB.
- **Headroom checked.** True peak after the gain is −3.4 dBFS, which leaves room to spare: nothing clips, even between samples.

12.2.4 Intros: copy the frames, pin the clock

- **The simple plan, replaced.** The first plan on record was to append each piece's audio end to end. That would butt every piece against the one before it. An AC-3 frame lasts 32 ms (1,536 samples at 48 kHz) and a video frame lasts 20 ms at 50 fps, so audio and picture lengths almost never end together. The small mismatches would add up across six pieces, and the logo's 80 ms lead would be lost.
- **The adopted plan: copy every frame and pin it to the clock.**
  - **Frames copied, not decoded.** Each piece's original AC-3 frames are copied unchanged, and this was verified by an MD5 checksum of every frame. AC-3 frames are self-contained: each header carries its own bit rate, so 192 kb/s and 320 kb/s frames can sit in one track without re-encoding.
  - **Pinned with a timestamp map.** A Matroska (MKV) timestamp file gives every 32 ms frame its exact time. Each piece starts at its picture's start plus the disc's own audio/video offset, so the logo's sound still leads its picture by 80 ms.
  - **Silence only where the disc has none.** Silence sits under the copyright card and after the logo music. It is made of newly encoded AC-3 frames of digital silence, and these are the only new audio frames in the intros.

– **Lacing disabled.** Matroska normally "laces" several audio frames into one block with one timestamp. With lacing off, every frame keeps its own exact timestamp, verified to 0.01 ms.

> **In plain English:** the first ending was discarded because the gap could still be heard. Before building the second one, two things a casual edit would miss were checked. The two soundtracks are mirror images of each other, so blended naively they would partly cancel out. And the German mix is louder, so the difference was measured on the music itself. For the intros the original sound data was copied unchanged and every 32-millisecond slice of it was given its original place in time.

12.3 The decision

12.3.1 German: one repair, from 87:19.5 to the end

- **The handover.** At 87:19.5 a 3-second crossfade hands over to the original track's music, polarity-flipped to match the German mix and raised +5.6 dB. The music then continues under the full 2:48 of credits.
- **Everything before 87:19.5 is untouched.** That audio is bit-identical to the re-timed German track from Chapter 11; up to that point the re-timing is the only processing the German track received. It ships as 16-bit FLAC, the film's default audio track.

12.3.2 Hebrew original: copied bit-for-bit

- **Not one bit changed.** The disc's AC-3 at 320 kb/s is copied bit-for-bit as the "Hebrew (Original)" track. It was already locked to the picture, which is why it served as the reference clock for the German re-timing.

Track	What was changed	What was left alone	Proof
German (default, FLAC)	Re-timing (Ch 11); from 87:19.5 a 3 s crossfade into the polarity-flipped, +5.6 dB original score	Every sample before 87:19.5	Bit-identical to the re-timed track before the handover; true peak −3.4 dBFS; approved by ear
Hebrew (Original, AC-3 320 kb/s)	Nothing	Everything	Bit-for-bit copy
Intros (AC-3, 192–320 kb/s)	Silence frames added where the disc has no audio	Every original audio frame and its timing	Per-frame MD5; timestamps verified to 0.01 ms

> **In plain English:** only the last three minutes of the German track were repaired. Everything before them is exactly the re-timed track from Chapter 11. The Hebrew track was copied exactly. The intro clips sound exactly as they do on the disc.

12.4 The impact

- **A continuous score through the credits.** In German, the music now carries through the end credits without a seam. The ending clip was approved by ear.
- **An untouched original.** The Hebrew track is the disc's own data, bit for bit, so viewers who choose the original language hear exactly what the disc holds.
- **Intros that are faithful to the frame.** The intros sound exactly as they do on the disc, down to the 80 ms by which the logo's sound leads its picture.

- **Proved again on the finished files.** After assembly, the German FLAC in the film file decoded bit-identical to the prepared seamless track. Every other audio track was compared packet by packet and is bit-identical to its source: against the disc, the Hebrew track's 169,483 AC-3 packets and the interviews' English tracks, 14,781 (Oded Gur-Arie) and 26,779 (Nadav Schirman); against the prepared frame-pinned intro track, all 15,798 intro packets.

EFFORT LEDGER	CHAPTER 12
BUILT AND DISCARDED	ending version 1 (a fade-in after a 5-second gap), listened to and replaced. An end-to-end append of the intro audio was planned and replaced by a timestamp map.
MEASURED	polarity (all correlation peaks negative, so the German mix is inverted); level difference of the same music, +5.6 dB; true peak after the fix, −3.4 dBFS; the disc's German track ending at 87:35; a per-frame MD5 of the intro audio; intro timestamps to 0.01 ms; after assembly, every audio packet of the finished files against its source and the decoded German FLAC against the prepared track (all bit-identical).
TIME	the seamless ending was built and verified in about 4 minutes, quick because the polarity finding had already come out of the sync verification.
KEPT	the 3-second polarity- and level-matched crossfade; the Hebrew original bit-for-bit; every original intro AC-3 frame unchanged and frame-pinned.

CHAPTER 13

Subtitles I: From Disc Bitmaps to Correct Text

Thesis: a subtitle that is stored as a picture cannot be corrected, restyled or searched. So the work came in a fixed order: first make it text, then make it right. The disc carries four subtitle streams: English and German for the film, and German for each of the two interviews. All four are pictures of words, not words. This chapter covers how those pictures were turned back into text, how the reading machine's mistakes were found and counted, and what the four finished text tracks now are. Chapter 14 builds on this. It adds the 15 new English lines that translate the film's German on-screen text and sets their timing and placement.

1,155	204	194 of 788	26
subtitle lines converted from bitmaps to text (788 + 35 + 110 + 222)	capital "I"s that the raw OCR read as "|", corrected to 0	English lines (about 1 in 4) needed at least one correction	mismatched quotation marks corrected

13.1 The problem

- **The disc's subtitles are pictures.** DVD-Video stores subtitles as a "subpicture" stream, which ripping software usually calls VobSub. Each subtitle is a small bitmap drawn with only four colour slots. The real colours for those slots come from a 16-entry palette stored in the disc's navigation (IFO) files.
  - The lettering is soft grey with anti-aliased edges. It is drawn on the 720×576 PAL grid and gets enlarged along with the picture on a modern TV, so it softens further (Figure F24).
- **A picture of a word cannot be edited.** That rules out four things the project needed:
  - *Correction.* A misspelling inside a bitmap is part of the image.
  - *Restyling or moving.* Font, size and position are baked in. One translated caption in Chapter 14 has to move to the top of the screen, and a bitmap cannot do that.
  - *Searching and copying.* No player or tool can search the dialogue.
  - *Extension in a matching style.* The 15 new English lines of Chapter 14 could never look like the disc's own lines.
- **Raw OCR does not solve it on its own.** Optical character recognition (OCR) reads letters out of images. On soft, low-resolution lettering, though, it makes *systematic* errors: the same wrong guess comes back every time the same letter shape appears. In the raw English output, the capital "I" came back as a vertical bar, "|", 204 times.

> **In plain English:** The DVD's subtitles are tiny pictures of words. A typo in a picture cannot be fixed, and the picture cannot be moved out of the way or made sharper. The first job was to turn the pictures back into real text. The software that does that makes its own predictable mistakes, and those then had to be found and fixed.

13.2 The analysis

13.2.1 Method: render, prepare, read, re-time

- **Render each subtitle with the disc's own palette.** ffmpeg drew each subtitle stream onto a black 720×576 canvas at the disc's 25 frames/s. It used the palette read from the disc's IFO file, so every letter has the shades the disc specifies. A generic palette can put the letter body and its outline in similar shades, and that blurs the letter shapes the OCR depends on.
- **Keep one image per change.** A duplicate-frame filter (mpdecimate) kept only the frames where the subtitle picture changes, which means every appearance and every disappearance. Each kept image carries its exact timestamp.
- **Prepare the image for the reader.** Each image was turned into
  - greyscale;
  - inverted to dark letters on white, the polarity that current Tesseract versions are documented to read best;
  - enlarged 2× with Lanczos resampling (a sharp, high-quality enlargement method), so each letter shape has more pixels to recognise.
- **Read with Tesseract.** Tesseract OCR ran with its English or German language model in page-segmentation mode 6 ("one uniform block of text"), on six images at a time in parallel.
- **Rebuild the timing from the images.**
  - An image with text starts a subtitle, and the next change ends it. Cue times therefore fall on the disc's own 40 ms frame grid.
  - Slivers shorter than 0.2 s were dropped as fade or redraw artefacts. Identical text redrawn back-to-back was merged into one line.
  - Any image with visible marks but no readable text was set to be tagged "[unreadable]" for manual review. None was: every subtitle on the disc produced text.

13.2.2 The corrections, counted

The raw OCR output was kept untouched next to the corrected track, so every fix can be checked by comparing the two line by line. Of the 788 English lines, **194 (about one in four) needed at least one correction**. The correction pass changed text only. No line's start or end time moved.

Error class	OCR read	Corrected to	Track	Count	What went wrong
Capital I read as a bar	| remember it well.	I remember it well.	English	204	In the subtitle font the capital I is a plain vertical stroke, the same shape as "|"
Small letters read as capitals	Know, Knew, Knowing, Suitcase (mid-sentence)	know, knew, knowing, suitcase	English	–	Lower-case k and s returned as their capital forms
Mismatched quotation marks	"because my life depends on it."	"because my life depends on it."	English	26	A curly opening quote closed by a straight one. The closing mark was matched to the opening mark
Lost dash in a date range	Son of Hans and Helena 19211993	… 1921–1993	English	1	The thin dash between the years was dropped, fusing two years into one number
Lower-case l for capital I	Died on 23 lyyar 5752	Died on 23 Iyyar 5752	English	1	l and I are near-identical strokes. Iyyar is a month of the Hebrew calendar

Error class	OCR read	Corrected to	Track	Count	What went wrong
Umlaut missed	Agypten	Ägypten	German (Forced)	–	The two dots of Ä were not recognised
Umlaut invented	Ähnung, Äbenteuer	Ahnung, Abenteuer	Interview German	–	Umlaut dots added where the disc has none

— = corrected, but the occurrences were not counted separately.

> **In plain English:** The OCR's mistakes were not random. The same misreading came back wherever the same letter shape appeared. That meant the mistakes could be found by pattern: search the whole track for the tell-tale shape, check each hit, fix it, and go on to the next kind of mistake. Keeping the raw output alongside the corrected text means every one of these fixes can be checked afterwards.

13.2.3 The alternatives, and why they lost

- **Copy the bitmap tracks unchanged.** This was the zero-effort option. It was rejected on all four counts in 13.1: the subtitles would stay soft and stay uncorrectable, and they could be neither moved nor extended in a matching style.
- **Ship the raw OCR as it came out.** Built, measured and rejected. One English line in four had a fault, and the most common one, "|" for "I", is exactly the kind of error a viewer notices in every scene with first-person narration. A film told largely through the spy's son, in the first person ("I remember the first time…"), has a great many of those lines.
- **Correct by error class, keeping an audit trail.** Chosen. Each class of misreading was searched for across the whole track and fixed, and the untouched raw output is kept so every change can be checked against it.

13.3 The decision

- **Four text subtitle tracks in SRT, the standard plain-text subtitle format every player reads, replace the four bitmap tracks.**

Track	File	Lines	Default	Content
English	Film	788 dialogue lines (803 with the 15 new lines of Chapter 14)	On	Full English translation of the dialogue
German (Forced)	Film	35	Forced flag set	Place-and-date cards and German captions for on-screen headlines and messages, e.g. "Cairo 1961", "Briefbomben erschüttern Kairo"
German	Interview: Oded Gur-Arie	110	Off	Subtitles for the English-language interview
German	Interview: Nadav Schirman (Director)	222	Off	Subtitles for the English-language interview

- **Default: German audio with English subtitles on.** This is the intended way to watch the film.
- **The forced flag is kept.** In the container it marks the German track as one that players should show even to viewers who have not switched subtitles on. That suits a track that only captions on-screen text.

- **Timing comes from the disc.** Every line keeps the start and end times read from the disc's own subtitle changes. Corrections touched wording only.

> **In plain English:** Every subtitle the disc had is still there, in the same place in time, but now as real, corrected text. English comes on by default with the German soundtrack. The short German track that labels places, dates and newspaper headlines is set so players treat it as always-relevant.

13.4 The impact

- **Crisp instead of soft.** The player now draws the letters itself, at the full resolution of the screen, in a clean font. Before, it enlarged a 720×576 bitmap of grey lettering (Figure F24).
- **Corrected, class by class.** The 204 bars are now 0, the 26 mismatched quotation pairs match, and 194 English lines read better than the raw OCR (every class in 13.2.2).
- **Editable and searchable.** Any line can be fixed in seconds in a text editor, and the dialogue can be searched. The entire corrected English track is about 65 kB of plain text.
- **Verified in the finished files.** Extracted directly from the final files, all four tracks (English 803 cues with the Chapter 14 lines, German (Forced) 35, Oded Gur-Arie 110, Nadav Schirman 222) have text identical to the prepared files and times identical to the millisecond.
- **Ready for Chapter 14.** The 15 new English lines that translate the film's German on-screen text go into the same track, in the same style. A text cue can also carry a position tag (`{\an8}`, top centre). That tag is how the one caption that would collide with dialogue, "Avigdor" at 8:35, is lifted to the top of the screen. None of this would have been possible with the disc's bitmaps.

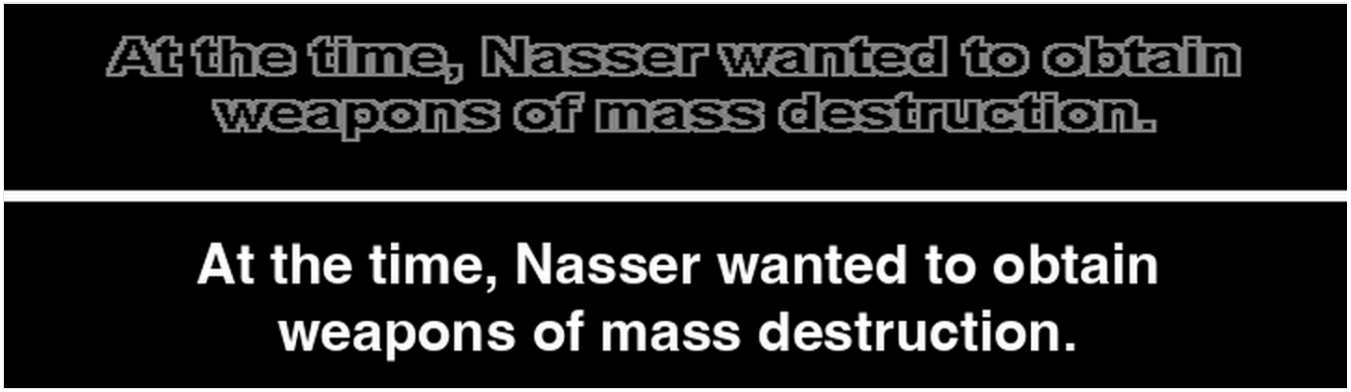


Figure F24. The same dialogue subtitle at 8:37, enlarged 2×. Top: the disc's bitmap subtitle, grey outlined lettering drawn on the 720×576 grid and enlarged along with the picture. Bottom: the corrected text line, drawn by the player at screen resolution (rendered here in a typical TV-player style; the real look depends on the player).

What to look for: the blurred, stepped letter edges on top against the clean edges below.

> **In plain English:** The subtitles now look sharp on any TV, the typos the reading machine introduced are fixed, and because they are ordinary text, the missing translations could be added in the next step so that they look exactly like the rest.

BUILT AND DISCARDED	the raw OCR output of all four tracks. In English alone that meant 204 "|" read for "I" and 194 of 788 lines with a fault. It was kept for audit and never shipped.
MEASURED	788 + 35 + 110 + 222 = 1,155 lines read. Raw and corrected English compared line by line: 194 changed, 0 timings moved, 0 unreadable images.
TIME	about 2 min of OCR for all four tracks.
KEPT	four corrected text tracks.

CHAPTER 14

Subtitles II: Nothing on Screen Left Untranslated

Thesis: the disc translated what people say, but not what the screen shows. The German release has German titles burned into the picture: the opening credits, the title card, the captions that name who is speaking, and the cards that close the film. "Burned in" means the text is part of the image itself, not a separate subtitle track that a player can switch on or off. The disc's English subtitle track translates none of them. Chapter 13 turned the disc's 788 bitmap dialogue lines into corrected text. This chapter covers the 15 new English lines written for the German titles: their wording, their frame-exact timing, where they sit on screen, and the frame-by-frame check behind each one. Every German title burned into the picture (credits, title card, name captions, epilogue cards) now has an English line.

67	0	15	1 of 15
on-screen text events found by a whole-film OCR scan	German burned-in titles translated by the disc's English subtitles	new English lines, each timed to its German title's own frames	lines placed at the top (6 of 15 in the first, rejected placement)

Left: the original DVD. Right: the restoration.

Figure F25. Left (before): the original DVD at 8:37 with its English subtitles on. The spoken line is translated (grey bitmap, bottom), but the German caption just above it, '»AVIGDOR« / Leiter der Einsatzabteilung des Mossad', is not, so an English viewer never learns who is speaking.

What to look for: the untranslated typewriter caption, and the soft grey lettering of the disc's own bitmap subtitle.

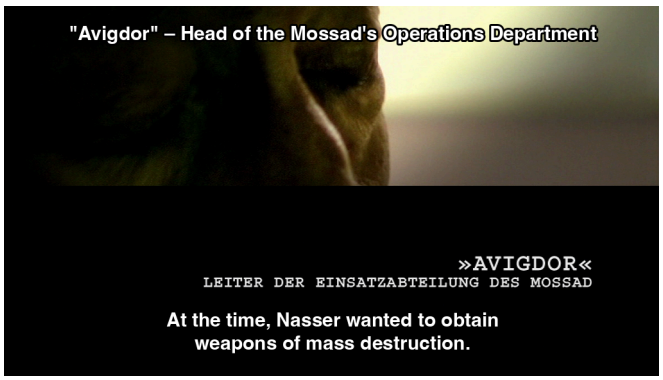


Figure F26. Right (after): the same moment with the new English line '"Avigdor" – Head of the Mossad's Operations Department' at the top, because the dialogue subtitle already occupies the bottom. Three texts share the screen and none overlaps. (Subtitles drawn in a typical TV-player style for review.)

What to look for: the translation at the top, the German caption in the middle, the dialogue at the bottom. Nothing covers anything else.

14.1 The problem

14.1.1 A whole-film text audit

- **OCR of every second of the film.** Tesseract OCR, with its English and German language models in sparse-text mode, read one frame per second of a frame-exact copy of the film at true 16:9 (1024×576). That is 5,423 frames, and the pass took about 3 minutes. Every recognised word was stored with its position and confidence. The words were then grouped into **67 on-screen text events** and compared with the subtitle tracks.
  - Many of the events are already in English, such as newspaper headlines, signs and the film's own English end credits. Those need no translation.

- The gap was the German titles. The disc's English subtitles translate **none** of them:
  - the opening presenter credit and the title card *DER CHAMPAGNER SPION*;
  - the German opening-credit job titles (*Kamera, Schnitt, Ton, Visuelle Effekte, Originalmusik, Produzenten, Buch und Regie*);
  - three name captions in typewriter type;
  - three epilogue cards.

14.1.2 The worst case: who is speaking?

- **8:37, Figures F25 and F26.** A man is shown in extreme close-up. The English viewer gets the spoken line ("At the time, Nasser wanted to obtain weapons of mass destruction."). The viewer never learns that the speaker is "Avigdor", head of the Mossad's operations department. The caption is there to answer exactly that question, and only German viewers could read it.
- **The same is true of the other two people the film introduces by caption,** and of the closing cards, which tell what became of Rivka Gur-Arie, Wolfgang Lotz and Oded Gur-Arie.

> **In plain English:** the DVD's English subtitles only translate speech. Anything printed onto the picture itself in German stayed in German: who made the film, its title, who is speaking, and what became of the people in it. An English viewer missed all of it.

14.2 The analysis

14.2.1 Wording: the filmmakers' own English

- **First drafts were literal.** *Kamera* became "Camera", *Schnitt* "Editing", *Originalmusik* "Original Music" and *Buch und Regie* "Written and Directed by".
- **Then the film's English end credits were read.** The same text audit had already captured them. All four drafts were rewritten to the credits' own words: **Cinematography, Editor, Original Score, Written & Directed by**. The other three (Sound, Visual Effects, Producers) needed no change.
- **Captions and cards keep their two-part form,** a name followed by a description: *LOTZ' KONTAKTMANN IN PARIS* → "Lotz's contact in Paris"; *HEIRATETE NICHT WIEDER / STARB 1990* → "Never remarried. Died 1990." Names are spelled exactly as on screen.
- In plain terms: the subtitles use the filmmakers' own English, so the opening and closing credits now agree.

14.2.2 Timing: measuring when each German title is on screen

- **Method v1, colour and brightness masks: discarded.** Selecting the text's pixels by colour or brightness cannot tell letters apart from a bright or yellow background behind them. The readings came out contaminated, so the method was dropped rather than tuned.
- **Method v2, stroke matching: adopted.**
  - **Template.** In a frame from the middle of each title, the letter strokes are isolated: pixels more than 30 levels brighter than their 15×15-pixel surroundings, and for the yellow credits also yellow in colour. Only strokes that are still present 5 frames earlier and 5 frames later are kept. The templates range from 572 stroke pixels (*Kamera*) to 10,218 (*Produzenten*).

  - **Visibility.** For every frame from 4 s before the title to 4 s after it, visibility is the share of those exact stroke pixels that are strokes again. The background level is subtracted and the result is scaled to the title's peak.
  - **Definition of "on screen".** At least 50% visible. A cue starts on the first frame that passes and ends on the last, so every time falls on the disc's 40 ms frame grid. For titles that fade, the final windows follow the fade itself (14.2.3).
- **Result on the nine yellow opening credits.** For seven of them, the measurement matched the existing timing to the frame.
- **The two disagreements were checked by eye (Figure F27).**
  - For *Kamera* and *Ton*, the measurement put the end 1.48 s and 1.96 s early (93.36 s and 117.76 s).
  - The background changes behind the letters there, and fewer stroke pixels stand out against it. The letters themselves stay legible.
  - The frame-exact end times were kept, 94.84 s and 119.72 s, which end exactly when the German text leaves the screen.

Figure F27. A timing measurement double-checked by eye. Each thumbnail is the quarter of the frame that holds the credit, at true widescreen shape; thumbnails run left to right in time, labelled with the credit and the film time. Top row: 'Kamera / Itai Neeman' at 92.80 s, 93.80 s, 94.80 s (its last frame) and 94.84 s (gone: the next shot has begun). Bottom row: 'Ton / Issachar Vishnia' at 117.20 s, 118.40 s, 119.68 s (its last frame) and 119.72 s (gone). The automatic stroke-matching measurement judged both titles gone 1.5–2 s early, because the background changed behind the letters. The strip shows them readable up to their last frame, so the frame-exact end times were kept: 'Cinematography: Itai Neeman' and 'Sound: Issachar Vishnia' end exactly when the German text leaves.

What to look for: in each row, the yellow name stays legible as the background changes (the picture darkening to black on top, a blurred horse race turning white below), and is gone one disc frame (40 ms) after its last frame.

14.2.3 Duration: up with the German title, never after it

- **The yellow credits cut on and off.** They appear and vanish between one frame and the next, so their existing times were already frame-exact (confirmed in 14.2.2).
- **The captions and epilogue cards fade in and out.** The first-draft times had clipped those fades. In the final film each English line follows the fade of its German title: it appears when the title reaches 10% of full brightness and leaves while the title is still at 25% or more on its fade-out. Against the first drafts, the lines now start **0.32–0.64 s** earlier and end **0.24–0.72 s** later (table below). A translation never outstays its title, so it cannot linger into the next shot.
- **Overlapping credits become a two-line subtitle.** *Originalmusik* appears at 2:05.16 while *Visuelle Effekte* is still on screen (2:03.24–2:08.80). The English is split into two cues: "Visual Effects: Kiril Rosenfeld" alone

until 2:05.16, then "Original Score: Ran Bagno" above "Visual Effects: Kiril Rosenfeld" until both German credits leave.

- **Frame-exact in the 50 fps film too.** The restored film keeps the disc's own timing, and each 40 ms disc frame becomes two 20 ms output frames. So a cue that starts on a disc-frame boundary also starts on an output-frame boundary.

German title	First-draft line	Final line (fade-aware)	Starts earlier by	Ends later by
»AVIGDOR« caption	8:35.84 – 8:43.84	8:35.52 – 8:44.08	0.32 s	0.24 s
ARIE SIVAN caption	14:15.92 – 14:19.28	14:15.60 – 14:19.52	0.32 s	0.24 s
HANNELORE WEIZENEGGER caption	34:32.68 – 34:38.32	34:32.36 – 34:38.56	0.32 s	0.24 s
RIVKA GUR-ARIE card	85:52.00 – 86:00.60	85:51.36 – 86:01.16	0.64 s	0.56 s
WOLFGANG LOTZ card	86:03.20 – 86:11.80	86:02.60 – 86:12.52	0.60 s	0.72 s
ODED GUR-ARIE card	86:14.32 – 86:22.44	86:13.72 – 86:23.04	0.60 s	0.60 s

14.2.4 Placement: tested on a television, then replaced

- **Placement v1: top whenever the German text sat low.** This put **6 of the 15 lines** at the top of the screen (Editor, Visual Effects, Original Score, Avigdor, Arie Sivan, Hannelore Weizenegger). It used the common top-centre position tag (`{\an8}`) inside the text subtitle.
- **Device test.** A 2-minute test clip was played on a living-room television through the media server. From that viewing, two rules were set:
  - **Placement v2:** always at the bottom, and at the top only when a dialogue subtitle is on screen at the same moment.
  - **Duration:** each translation stays up as long as its German title and never longer (implemented in 14.2.3).
- **Result:** only **1 of 15** lines is now at the top, "Avigdor" at 8:35. The test clip on the media server was replaced by the v2 version.
- **A side finding from device testing.** A stutter on the Apple TV was traced to its "Match Frame Rate" setting being off, which made it play 50 fps material on a 60 Hz output. The cause was the setting, not the file.

14.2.5 Collision check: no line covers German text

- **Every added line was drawn onto frames of the restored master at its measured time and checked frame by frame.** The aim was to confirm that it covers no German on-screen text. The 15 checks are collected on three review sheets; Figures F28 and F29 come from them.
- **The tight case is 8:35.** The German caption sits low in the frame and the dialogue subtitle sits below it, as on the DVD. The translation therefore goes to the top, and all three texts stay readable (Figure F26).

> In plain English: the translations were not just typed in. Their wording was taken from the film's own English credits. The computer measured, frame by frame, exactly when each German title is visible. The first placement was tried on a television and changed. Then every line was checked against the picture, so no subtitle ever sits on top of the text it translates.

14.3 The decision

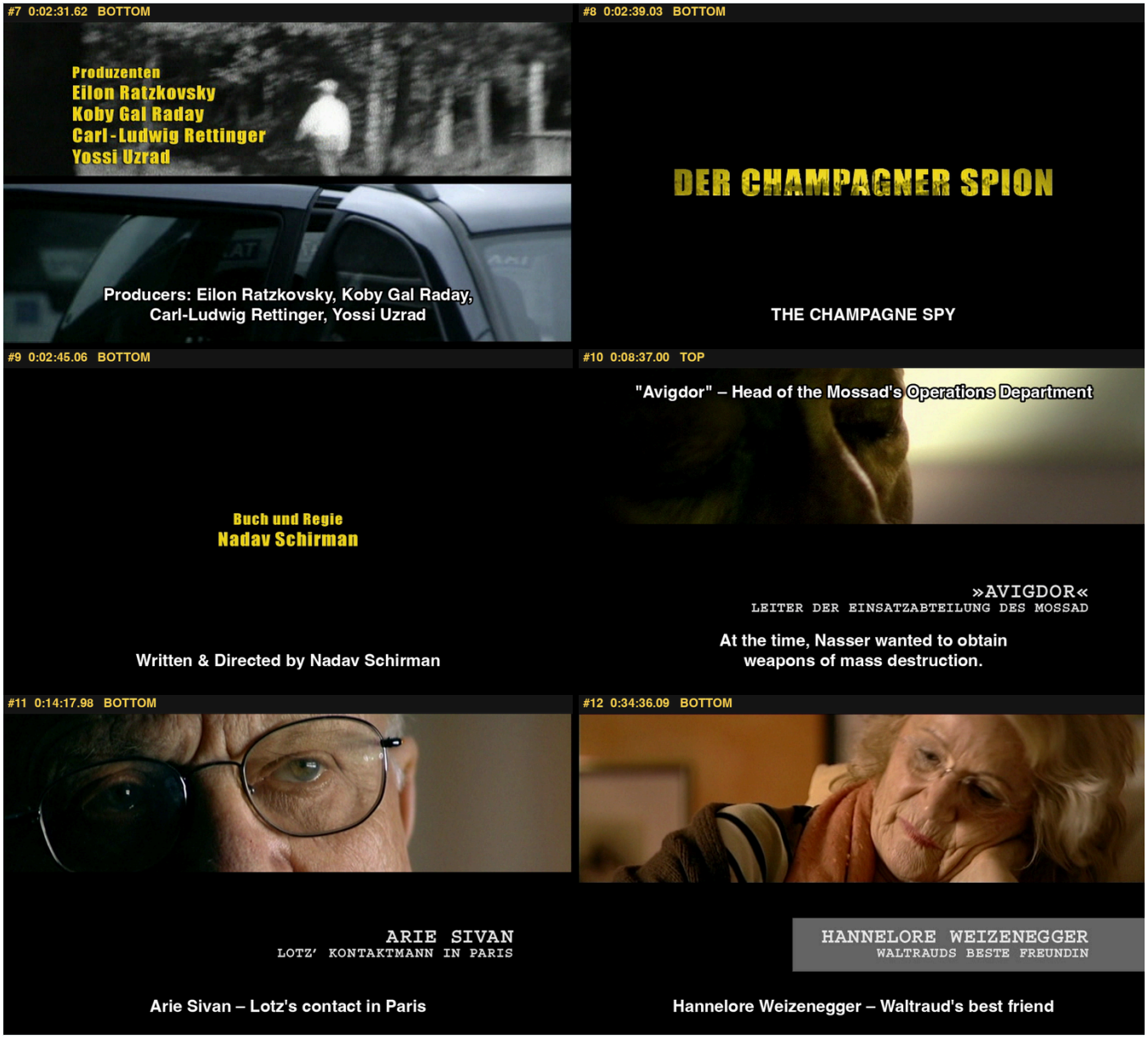


Figure F28. Six of the 15 new English lines, drawn on frames of the restored master in a typical TV-player style (the real look depends on the player). Top row, left: 'Produzenten' → 'Producers: …' (2:31); right: the title card 'DER CHAMPAGNER SPION' → 'THE CHAMPAGNE SPY' (2:39). Middle row, left: 'Buch und Regie' → 'Written & Directed by Nadav Schirman' (2:45); right: the 'Avigdor' caption, translated at the top because dialogue occupies the bottom (8:37, shown full size in Figure F26). Bottom row, left: 'Arie Sivan – Lotz's contact in Paris' (14:18); right: 'Hannelore Weizenegger – Waltraud's best friend' (34:36). Yellow or typewriter text is German burned into the picture; the white outlined line is the new translation.

What to look for: every German title now has its English line, and no line covers on-screen text. The dark strip above each tile is a review label giving the line number (#7–#12), the film time sampled and the placement (BOTTOM or TOP).

- **15 new English lines,** all at the bottom except "Avigdor" at 8:35, which goes to the top:
  - **Opening presenter credit** (1:22): "July August Productions and Lichtblick Film present".
  - **Opening-credit job titles** (1:31–2:46), in the film's own English: Cinematography, Editor, Sound, Visual Effects, Original Score (sharing the screen with Visual Effects), Producers, Written & Directed by.

  - **The title card** (2:35): "THE CHAMPAGNE SPY".
  - **Three name captions:** "'Avigdor" – Head of the Mossad's Operations Department' (8:35, top); "Arie Sivan – Lotz's contact in Paris" (14:15); "Hannelore Weizenegger – Waltraud's best friend" (34:32).
  - **Three epilogue cards** (85:51–86:23): "Rivka Gur-Arie / Never remarried. Died 1990."; "Wolfgang Lotz / Ze'ev Gur-Arie / Died 1993. Buried in Israel."; "Oded Gur-Arie / Lives in the USA. Happily married, three children."
- **Delivered inside the English track.** The film's English subtitle track holds 803 lines, the 788 dialogue lines plus these 15. It is stored as text (SRT) and is on by default with the German audio, so the translations appear without the viewer doing anything.

14.4 The impact

- **An English viewer now reads every title a German viewer reads:** who made the film, what it is called, who each witness is, and how the story ended for its people (Figures F28 and F29).

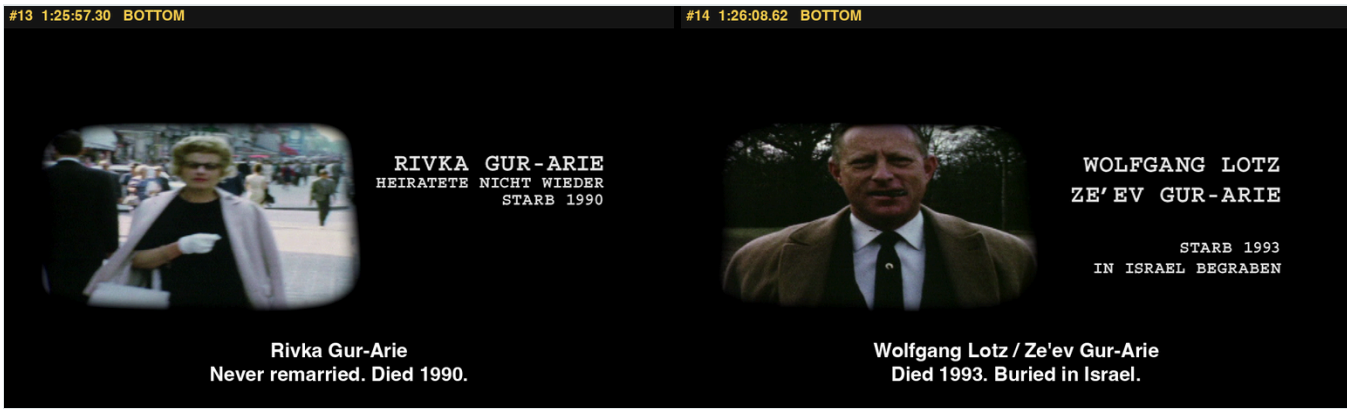


Figure F29. The film's closing cards, now in English. Left: 'RIVKA GUR-ARIE / HEIRATETE NICHT WIEDER / STARB 1990' → 'Rivka Gur-Arie / Never remarried. Died 1990.' Right: 'WOLFGANG LOTZ / ZE'EV GUR-ARIE / STARB 1993 / IN ISRAEL BEGRABEN' → 'Wolfgang Lotz / Ze'ev Gur-Arie / Died 1993. Buried in Israel.' The right card shows the spy the film is about. These cards fade in and out, so each line appears as its card reaches 10% of full brightness and leaves while the card is still at 25% or more.

What to look for: the German text on the right of each card and its English line at the bottom. The label strip gives the line number (#13, #14), the film time sampled and the placement.

> **In plain English:** a viewer watching the film in English now gets every title a German viewer gets: every credit, every name caption, every closing card. Each one appears and disappears with the German text it translates, and none of them covers anything.

EFFORT LEDGER	CHAPTER 14
BUILT AND DISCARDED	timing method v1 (colour and brightness masks); placement v1, which put 6 of 15 lines at the top, and its 2-minute television test clip, replaced on the media server by v2; 4 literal job-title translations, rewritten to the end-credit wording; the first-draft timings of 6 fading captions and cards, replaced by measured, fade-aware windows.
MEASURED	5,423 frames read by OCR, yielding 67 on-screen text events; the on-screen span of all 15 German titles, frame by frame (at least 50% of each title's letter strokes visible), and the fades of the 6 fading titles (in at 10% brightness, out while still at 25% or more); the 2 credits where the measurement disagreed, checked by eye; all 15 added lines checked against on-screen text.
TIME	the whole-film OCR text audit took about 3 minutes.
KEPT	15 frame-timed lines (14 at the bottom, 1 at the top), the v2 placement rule, and an 803-line English track on by default.

CHAPTER 15

The Extras: Trailers Reframed Subtitle-Safe, Interviews Treated Like the Film

Thesis. The extras get the film's method, not a shortcut. Besides the film, the disc carries eight short pieces: a copyright notice, the distributor's Neue Visionen logo, four trailers for other films and two interviews, about 30 minutes in all. Every one of them went through the same per-field analysis as the film (Chapter 4), the same QTGMC settings and scene-cut guard (Chapter 5) and the same CRF 14 encode (Chapter 9). Two of the trailers also had a problem the film never had: they are stored as small letterboxed pictures, and one of them has subtitles burned in close to the edge that any crop would remove first.

24 rows
trimmed from the top of trailer 2 only, so its burned-in subtitles stay whole

97.6%
of the copyright card's fields joined exactly with their own partner

99.8%
of the Oded Gur-Arie interview's fields joined exactly with their own partner; none sent to QTGMC

≈ 94%
of the director interview sent to QTGMC: true interlaced video, as expected

Figure F30. Reframing the letterboxed trailers. Left: trailer 1 as the disc presents it on a 16:9 TV, a 1.78:1 picture letterboxed inside a pillarboxed 4:3 frame, with black on all four sides (the thin grey line marks the edge of the screen). Middle: the same frame after cropping the bars (712×428 active picture) and scaling to fill 16:9, with natural proportions. Right: trailer 2 in an early check render with a symmetric crop, where the bottom of the burned-in German subtitle ('und starke Jäger sein.') is clipped. That check caught the problem. The final crop takes all 24 rows from the top instead, so the subtitle stays whole.

What to look for: Left: the small window. Middle: no black bars on any side. Right: the cut-off second subtitle line, which is the reason for the top-only crop. The thin dark edges at the sides of the right-hand panel are part of that trailer's own picture.

15.1 The problem

15.1.1 Two trailers are small pictures inside black bars

- **Letterboxed inside 4:3.** Trailers 1 and 2 are widescreen pictures, but the disc stores them in 4:3 frames (720×576, pixel aspect 16:15) with black bars above and below. The other six extras are 16:9 anamorphic, like the film.
  - A widescreen TV shows a 4:3 frame with black pillars at the left and right. Together with the trailer's own bars above and below, the picture becomes a small window with black on all four sides.

15.1.2 Trailer 2's subtitles sit near the bottom edge of its picture

- **Burned-in, not a track.** Trailer 2 carries German subtitles burned into the image. They are part of the picture, so they cannot be moved or switched off like a subtitle track.
- **Five rows of margin.** The picture occupies rows 60–515 of the 576-row frame, and the lettering reaches row 510, only five pixel rows above the picture's lower edge. Any crop that takes picture from the bottom

removes letters first, and lost letters cannot be recovered.

15.1.3 The extras mix progressive and truly interlaced material

- **Both kinds of picture appear, just as in the film.** The per-shot classification found:
  - trailer 1: 21 progressive, 16 interlaced and 3 mixed shots out of 40;
  - the Oded Gur-Arie interview: progressive (all 15 shots);
  - the director interview with Nadav Schirman: true interlaced video, bottom field first (all 5 shots).
- **One treatment cannot fit all of them.** A single treatment per piece would mishandle one kind of picture or the other, the trap the film's per-field method was built to avoid.

> **In plain English:** two trailers were stored as small pictures framed in black, and one has German subtitles painted into the image right at its lower edge, so the obvious crop would cut words off. The extras also mix film-style and video-style footage, which need different handling, as in the main film.

15.2 The analysis

15.2.1 Trailer 1: an exact 16:9 fit

- **Active picture measured at 712×428.** Inside the 720×576 frame there are 4 black columns on each side and 74 black rows above and below.
- **Already 16:9.** With 4:3 pixel shape, 712×428 displays at 1.78:1, so cropping to exactly those pixels and scaling to the 16:9 frame (720×576, pixel aspect 64:45) enlarges the picture about 1.35× both across and down. The two scale factors agree within 0.2%, so nothing is stretched and nothing is cut.

15.2.2 Trailer 2: the symmetric trim, caught and rejected

- **The geometry.** Rows 60–515 are 456 rows, a nominally 1.66:1 picture (768 × 456 display pixels, about 1.68:1 as stored). To fill 16:9 at this width the picture must be 432 rows tall (720 pixels at 16:15 display as 768 wide, and 768 × 9/16 = 432), so 24 rows of picture have to go.
- **The obvious choice failed its check.** The natural choice is a symmetric trim of 12 rows top and bottom. An early check render made that way showed the second subtitle line cut through its letters (Figure F30, right).
  - Measured: the lettering reaches row 510 of the 515-row picture. A symmetric trim ends the picture at row 503, which removes the bottom seven rows of the letters. The symmetric trim was rejected.
- **The fix.** Take all 24 rows from the top: crop 84 rows above (60 rows of bar plus 24 of picture) and 60 rows below (bar only). That leaves a 720×432 picture, which scales by exactly 4/3 in both directions to fill 16:9. The only cost is a thin strip along the top edge of the picture instead of the subtitles.

15.2.3 Per-field matching on every extra

- **Same evidence, same rule.** Each extra was run through the film's detectors: the cross-weave comb test, the grain-proof local field difference and the stillness check. Every field of every extra was then decided individually: joined with its own frame's partner, re-paired with a neighbouring frame's field, or sent to QTGMC, the motion-tracking deinterlacer. Field order is set per shot, and each extra's own shot list feeds the scene-cut guard, so QTGMC never blends two shots.

- **Same QTGMC settings as the film.** Placebo, SourceMatch 3, Lossless 1 with TR2 1, no denoising and no sharpening. Even in the director interview, the disc's own field lines stay bit-exact in every QTGMC frame; only the missing half is calculated. The whole-shot clean-up of Chapter 6 was calibrated on the film's archive footage and is not applied to the extras.

Extra	Length	Joined with own partner (exact)	Re-paired (exact)	QTGMC (disc lines kept)
Copyright notice	16 s	97.6%	0.1%	2.2%
Neue Visionen logo	24 s	91.1%	0.2%	8.7%
Trailer 1	2:06	63.0%	1.2%	35.9%
Trailer 2	1:51	98.2%	0.4%	1.4%
Trailer 3	2:01	91.5%	0.3%	8.2%
Trailer 4	1:47	93.8%	0.7%	5.5%
Intros file (all six pieces)	25,282 frames	86.6%	0.6%	12.8%
Oded Gur-Arie interview	7:53	99.8%	0.2%	0.0% (no fields)
Director interview (Nadav Schirman)	14:16	4.5%	1.3%	94.3%

Shares are of output frames (one per field, 50 per second), measured from the decision files that drive the encode. They describe the restored master before H.265 compression. Rows may not add to exactly 100% because of rounding.

- **The numbers follow the material.** The still copyright card and the progressive Oded interview come out almost entirely exact; the director interview, true video, goes about 94% to QTGMC; trailer 1's 35.9% QTGMC share tracks its 16 interlaced shots.

15.2.4 Discarded along the way

- **A first complete set of extras masters.** Lossless masters of all 8 extras were built with per-scene bwdif deinterlacing, one simple deinterlacer per shot, the same approach as the film's first, discarded master. In that set, trailers 1–2 were left as 4:3 frames pillarboxed into 16:9. All of it was discarded when the per-field method proved better (Chapter 4), and the bars were cropped so that the trailers fill the screen.
- **The first extras encode.** It was stopped after about 40 minutes and restarted so that it would include the restored scene-cut guard, and those 40 minutes of encoding were thrown away. The guard had been tested at a real cut, where frames within ±3 of the cut changed by up to 45 brightness levels (Chapter 5).
- **The symmetric trailer-2 trim** (15.2.2).

> **In plain English:** the crop for each trailer was worked out from measured pixel rows, not guessed. The obvious "trim a bit off the top and bottom" was tried, caught cutting words in half, and replaced. Every second of every extra then got the same field-by-field decisions as the film, and the numbers come out just as each kind of footage predicts.

15.3 The decision

- **Trailers 1–2: bars cropped, picture scaled to fill 16:9.** Cropping and Spline36 scaling (a high-quality resizing filter) run after the per-field step, on whole progressive frames. Scaling an interlaced frame vertically would blend its two half-pictures together.
  - Trailer 1: 712×428 → 720×576, 16:9.
  - Trailer 2: all 24 rows of its trim taken from the top only, 720×432 → 720×576, 16:9.
  - The other extras are already 16:9 anamorphic and are left exactly as authored.
- **Intros in one file, interviews separate** (library layout in Chapter 16):
  - Intros: copyright notice, Neue Visionen logo and the four trailers, joined into one file with one chapter each (6 chapters).
  - Interviews: one file each, Oded Gur-Arie and Nadav Schirman (4 chapters taken from its own title cards: A Story Finds You · Your Role as a Director · After the Release, Parts 1–2).
- **Sound and text untouched where possible.** Each intro piece's original AC-3 frames are copied bit-exact and pinned to their original times (Chapter 12). The interviews' English AC-3 audio is copied bit-for-bit. Their German subtitles were read by OCR, corrected (e.g. "Ähnung" → "Ahnung", "Äbenteuer" → "Abenteuer") and carried as text (110 and 222 lines, off by default).
- **Same encode as the film.** H.265 Main 10, CRF 14, preset veryslow, 50 frames/s.

15.4 The impact

- **Trailer 1 fills a widescreen TV edge to edge**, with natural proportions (Figure F30, middle), instead of sitting in a small window with black on all four sides.
- **Trailer 2 fills it with every subtitle line intact.** The only cost is a 24-row strip at the top of its picture.
- **Each interview gets exactly the treatment its material needs.**
  - The Oded Gur-Arie interview is the disc's own picture, every field joined or re-paired exactly, at 50 frames/s.
  - The director interview turns each of its fields into its own frame. That keeps the full 50-moments-per-second motion of the original video, with the disc's own lines kept bit-exact.
- **Verified like the film.** Every extra's frame count matched, and every encoded frame was compared with its restored master (VMAF, 0–100):

Final file	Frames	VMAF mean	Worst 1% of frames	Worst frame	Size
Featurettes/Intros.mkv (6 chapters)	25,282	96.27	92.12	85.45	0.25 GB
Interviews/Oded Gur-Arie.mkv	23,650	96.52	95.68	92.30	0.26 GB
Interviews/Nadav Schirman (Director).mkv (4 chapters)	42,848	97.33	96.01	94.07	0.71 GB

Every audio packet of the extras is bit-identical to its source (Chapter 12), and the interviews' German subtitles match the prepared text exactly (Chapter 13).

> **In plain English:** the trailers fill the screen without losing a word, and each interview was handled according to how it was shot, not with a one-size-fits-all filter.

BUILT AND DISCARDED	the symmetric trailer-2 crop, caught on a check render; lossless bwdif masters of all 8 extras, with trailers 1–2 pillarboxed; about 40 minutes of extras encoding, restarted for the scene-cut fix.
MEASURED	the trailers' active areas (712×428; rows 60–515); trailer 2's subtitles reaching row 510 of 515; per-field decisions on 91,780 output frames across all 8 extras (copyright card 97.6% exact, Oded interview 99.8% exact, director interview about 94% QTGMC).
TIME	the analysis was not separately logged. The extras encoded unattended in 5 h 36 min, alongside the film; the one measured loss is about 40 minutes of discarded encoding.
KEPT	an exact-fit crop for trailer 1, a top-only 24-row trim for trailer 2, and per-field processing on every extra.

CHAPTER 16

Delivery: A Media-Server-Ready Package, Verified

Thesis. A restoration is not finished until it plays correctly, by default, on the devices people actually use. Every earlier chapter improved a stream: the picture, the German dub, the subtitles, the extras. This chapter packages those streams so that a media server recognises the film and its extras without manual matching, sets the defaults so that a viewer presses Play and gets the intended German audio with English subtitles, and builds the final checks into the assembly itself. Those checks have run, and every one of them passed.

4 files	German + English	362,954	0 errors
film + 3 extras, in folders a media server recognises	default audio + default subtitles	video frames across the 4 files, every one present and compared with its master	Apple hardware decode of the whole film, all 271,174 frames

16.1 The problem

- **A DVD is menus and title sets; a media server needs named files (goal 2).** On the disc, the film and each extra sit in separate title sets that the disc menu links together. A media server has no menu. It identifies a film from the name of its folder and file, and it identifies extras only by the sub-folder they are in.
  - Without that structure, the two interviews and the intros would show up as unidentified "films", or not at all.
- **The defaults must be right without anyone touching a menu (goal 8).** Both audio tracks and all extras are kept. A viewer who simply presses Play should hear the German dub and see English subtitles, with the Hebrew original one click away.
- **It must play well on Apple TV, Fire TV, Roku and phones (goal 9).** This file is unusual for a DVD conversion: it is 10-bit, at 50 frames per second. Each target device must decode it in hardware and show it at its true rate.

> In plain English: A disc works through its menu. A media library works through file and folder names. So the project's last job was to turn one menu-driven disc into a few clearly named files that a media server files in the right places and that play correctly the moment someone presses Play.

16.2 The analysis

16.2.1 Folder names a media server already understands

- **Film: `Title (Year) {tmdb-ID} [tags]` for both folder and file.** `Title (Year)` is the naming convention common to current media servers, and the TMDB ID (`{tmdb-131748}`, the film's entry in The Movie Database) pins the match to the right title and artwork; the bracketed tags (`[576p] [DVDRip] [x265]`) record resolution, source and codec.
- **Extras: `Featurettes` and `Interviews`.** These are extras sub-folder names that media servers recognise, so the extras appear on the film's own page, under the film.
  - A `Trailers` folder also exists in that convention, but it is meant for the film's own trailer. The four trailers on this disc advertise other films. They therefore stay in their disc order with the copyright notice

and the distributor logo, as one "Intros" featurette with a chapter for each piece.

16.2.2 Device test

- **The 2-minute test clip that set the subtitle rules** (Chapter 14.2.4) doubled as the first device test, on a living-room television.
- **A second clip in the final format,** 1:20–3:00 of the film, was then served through the media server for a direct-play test of the exact delivered format (10-bit H.265 at 50 frames per second, FLAC and AC-3 audio, text subtitles).

16.2.3 A stutter traced to the device, not the file

- **Symptom.** The 50 fps test clip stuttered on an Apple TV.
- **Cause.** The Apple TV's "Match Frame Rate" setting was off, so it was sending 50 fps video to a 60 Hz output.
  - The arithmetic shows why: 60 refreshes per second cannot divide evenly among 50 frames. Ten frames in every second must be held for two refreshes and the other forty for one. That uneven rhythm shows up as judder on any smooth movement.
- **Consequence.** The file stays at a constant 50 fps, which is the disc's true motion rate (Chapter 3). The fix is one setting on the playback device: with frame-rate matching on, the box switches to 50 Hz for this film and the motion plays as smoothly as it was restored.

16.2.4 Colour flags written, not guessed

- **Colour flags written explicitly.** PAL / BT.601 colour, stored as BT.470BG primaries and matrix with the SMPTE 170M transfer, limited range, the same as the DVD. No player has to guess how to turn the numbers back into colour.

> **In plain English:** Three real-world questions were checked before the final build. Will the media server know what these files are? Will the television show them smoothly? Will every device read the colours as the DVD intended? The answers came from the media server's naming rules, from a test clip on an actual television, and from colour flags written into the file. The one stutter found was a playback-device setting, and changing the file would not have fixed it.

16.3 The decision: the library layout

```
The Champagne Spy (2007) {tmdb-131748} [576p] [DVDRip] [x265]/
├─ The Champagne Spy (2007) {tmdb-131748} […].mkv
│                                German (FLAC, default) · Hebrew (Original) · English subs (default) · German (Forced) · 24 chapters
├─ Featurettes/Intros.mkv         copyright notice + Neue Visionen logo + 4 trailers, one chapter each
└─ Interviews/
   ├─ Oded Gur-Arie.mkv
   └─ Nadav Schirman (Director).mkv   4 chapters: A Story Finds You · Your Role as a Director · After the Release (Parts 1–2)
```

File	Video	Audio (flag)	Subtitles (flag)	Chapters
The Champagne Spy (2007)	H.265 Main 10, 50 fps, 720×576 shown as 16:9	German, FLAC 16-bit 48 kHz (default) · Hebrew (Original), AC-3 320 kb/s copied bit-for-bit	English, 803 lines = 788 dialogue + 15 new (default) · German (Forced), 35 lines (forced flag)	24, from the disc, each moved one frame (below); named from content

File	Video	Audio (flag)	Subtitles (flag)	Chapters
Featurettes / Intros	as above; trailers 1–2 cropped to fill 16:9	each piece's original AC-3 frames, unchanged and pinned to their original times	none	6: Copyright Notice · Neue Visionen Logo · Trailer 1–4
Interviews / Oded Gur-Arie	as above	English, AC-3 320 kb/s, copied bit-for-bit	German, 110 lines (off by default)	none
Interviews / Nadav Schirman (Director)	as above	English, AC-3 256 kb/s, copied bit-for-bit	German, 222 lines (off by default)	4

- **Defaults are declared inside the file.** Every track carries a language tag, a readable name and a default or forced flag in the Matroska container. Media servers and players read these flags to pick German audio with English subtitles.
- **All tracks keep the disc's own timing.** On the disc, the film's streams start at 0.04 s. The newly encoded video and the re-timed German FLAC are placed with the same 40 ms start, and the copied tracks keep theirs. The sub-millisecond sync from Chapter 11 is therefore not undone at the final packaging step.
- **Chapter marks moved by one frame, and named.** The disc's 24 chapter times point one frame (40 ms) early. A final touch-up remux moved every mark +40 ms, so on the hard-cut chapters each mark now lands exactly on the first frame of the new shot; the disc's unnamed chapters were also given names from their content. The same remux gave the six fading German captions fade-aware subtitle windows (Chapter 14), and it left every video and audio packet identical to the verified encode.
- **Staged first.** The set was assembled in a staging folder and placed in the media library only after a final visual review; nothing earlier in the chain was deleted, and the bit-exact disc image remains the archival original.

> **In plain English:** One folder, named the way a media server expects, holds the film and its extras in the server's own extras sub-folders. The film file already "knows" its defaults: German speech, English subtitles, with the original Hebrew and a forced German track available. Nothing needs setting up on the viewer's side.

16.4 The impact: verification built into assembly

The checks below ran on their own as the encode finished, without waiting for anyone. Verification ran for 48 minutes immediately after muxing. The last two rows were run after the final touch-up remux.

Check	What it would catch	Result
Encoding and assembly: each of the 35 pieces accepted only if its frame count equals the master's; assembly waits until every piece has passed	a piece cut short by a crash or a full disk; a film quietly assembled with a gap	35 of 35 pieces accepted
Frame count of each file: film 271,174 · Intros 25,282 · Oded 23,650 · Nadav 42,848	a dropped or duplicated frame, which would shift sync from that point on	all four exact: 362,954 frames
Picture: every frame compared with the exact 50 fps master it was encoded from, by VMAF + PSNR (peak signal-to-noise ratio, a simple pixel-difference score) per frame	shifted, swapped or damaged frames	all 362,954 frames compared; film VMAF mean 96.08, 0 frames below 30 dB PSNR (Chapter 9.4)

Check	What it would catch	Result
Audio: decoded-audio checksum for the FLAC, per-packet checksums for the copied AC-3	any change to audio meant to be untouched	German FLAC decodes bit-identical to the prepared seamless track. AC-3: Hebrew 169,483 · Oded Gur-Arie 14,781 · Nadav Schirman 26,779 packets, bit-identical to the disc; Intros 15,798 packets, bit-identical to the prepared frame-pinned intro track (original disc frames MD5-verified, plus the added silence frames)
Subtitles: compared cue by cue, text and times	a lost or altered subtitle line	English 803 · German (Forced) 35 · Oded 110 · Nadav 222 cues; text identical to the prepared files, times identical to the millisecond (checked by direct extraction)
Tracks, flags and chapters read back	a wrong default, a missing chapter, a wrong picture or colour flag	as designed: German (default) · Hebrew (Original) · English (default) · German (Forced, forced flag); 24 chapters, Intros 6, Nadav 4; HEVC Main 10, 720×576 shown as 1024×576, 50 fps, BT.470BG / SMPTE 170M, limited range
Start time of every track read back	a track offset from the disc's own timing	film and both interviews: every track at 0.040 s; Intros: video and audio together at 0
After the remux: Apple VideoToolbox hardware decode of the whole film	a stream that a device's hardware decoder rejects	271,174 frames, 0 errors
After the remux: field-order rhythm of every shot; chapter marks against their cuts; subtitle windows	a shot whose two half-pictures play in the wrong order; a chapter mark off its cut; a translation that outstays its German title	0 reversed shots (the one suspect shot, at 32:15, was tracked: its motion is steady); chapter and subtitle-window checks passed

In plain English: The finished files checked themselves as soon as the encode ended. Every frame is present and was compared with its master, every untouched audio track is truly untouched, every subtitle line arrived intact, and every flag and start time reads back as designed. A problem anywhere would have stopped the chain before the files reached the library.

EFFORT LEDGER	CHAPTER 16
BUILT AND DISCARDED	nothing new; delivery is where earlier choices are checked, not made. (The subtitle test clip's first version was replaced on the media server by the second.)
MEASURED	playback of the test clips on a television; the Apple TV stutter traced to a device setting; frame counts of all 35 pieces and 4 files (362,954 frames); every frame scored against the master; bit-identity of all 5 audio tracks; 1,170 subtitle cues compared; flags, chapters and start times read back; hardware decode of the whole film; a field-order sweep of every shot.
TIME	verification took 48 minutes, immediately after the film encode and muxing.
KEPT	a media-server-ready set of 4 files, 7.41 GB, with German audio + English subtitles by default, verified and placed in the library.

CHAPTER 17

Method and Effort: AI-Assisted Versus Conventional

Thesis: with AI, a non-specialist got specialist-grade restoration in one evening of elapsed time; without it, the same person would realistically spend weeks and still miss several of these results. Chapters 2–16 show what was done to the film and what each choice cost to prove. This closing chapter asks a different question: what would the same job take without AI? The comparison person has professional video tools and real ability but no deep restoration expertise. One side of the comparison is measured. The other is an **estimate**, labelled as one, with its assumptions stated. This is the only place in the paper where total effort is compared.

≈ 6.5 h	≈ 8 h	95–255 h	4
measured: from inserting the disc to starting the final encode, on one laptop	measured: unattended final encoding overnight (machine time, the same with or without AI)	ESTIMATE: hands-on hours for the same job without AI	key results the estimate says would realistically be missed without AI

17.1 What actually happened (measured)

17.1.1 One evening of elapsed time, then the machine

- **About 6.5 hours elapsed** from inserting the disc to starting the final encode, all on one Apple Silicon laptop (M5 Pro). Nearly all the analysis and engineering described in Chapters 2–16 happened inside that window. Only the final encode, the checks on its output and a small touch-up remux of chapter marks and subtitle windows came after it.
  - It includes 16 minutes of disc imaging, eight whole-film interlace forensic passes (about 2–6 minutes each, except the 8-minute stillness pass), 60 calibration encodes with frame-by-frame scoring (about 50 minutes), and the 2 h 17 min "where does QTGMC add value" pass, which made 542,348 frame renders.
  - Jobs ran side by side. The German audio was re-timed and saved as FLAC in about 2 minutes while the forensic passes were running. The calibration encodes, the seamless German ending and the start of the double render also overlapped.
- **Then the unattended final encode, measured.** Film and extras ran as 35 pieces, 362,954 output frames in all, up to 8 pieces at once; every piece's frame count was checked before it was accepted. The film took 7 h 59 min overnight, with the extras alongside (5 h 36 min); the per-piece times add up to 51.0 encoder-hours. Assembly and a 48-minute check of every encoded frame against the restored master followed. This is machine time, and it would be the same with or without AI.

17.1.2 Division of labour

- **The author made the decisions and the reviews, in conversation with the AI.** Every judgement call came back as a short visual or listening check.
  - The author chose QTGMC at 50 fps from a 4-way demo, asked the question behind Chapter 6 ("where does QTGMC add value?") and chose CRF 14 over CRF 18, putting quality first.
  - The author approved the German ending by ear, and set the subtitle placement rule and the caption-timing rule after watching a 2-minute test clip on a living-room television.

- **The AI coding agent did the engineering, scripting, measurement and verification.**
  - It built the Apple Silicon toolchain: three plugins compiled for ARM and two QTGMC components re-pointed to ARM equivalents.
  - It wrote the detectors, ran the forensic passes and rendered more than 100 diagnostic crops to calibrate thresholds. It then built the per-field matcher that decided all 271,174 fields one by one.
  - It ran and scored the 60 test encodes. It measured the German drift at 87 points and checked the correction at 37 more.
  - It corrected the OCR'd subtitles and frame-timed 15 new lines. It also built the assembly checks: frame counts, a full decode of every file, the start time of every track, and a VMAF and PSNR comparison of every encoded frame with the restored master.

17.1.3 Process discipline

- **No long-running job was started without the author's explicit approval,** however quickly the AI had prepared it.
- **The AI's own early assumptions were tested, and several were corrected.** These corrections are the rigour working as intended. They are not errors that slipped through.
  - *The "Lossless=2" claim.* QTGMC's common "lossless" mode was measured altering the original lines it claims to keep, by a mean of 0.7–4.6 brightness levels (99th percentile up to 25). The pipeline switched to Lossless=1 with light smoothing (TR2=1).
  - *The comb detector's false alarms.* On its own it flagged clean skin, hair and fabric at up to 125/256. That finding is why the final rule combines it with a grain-proof field-difference measure.
  - *The AC-3 size estimate.* An early estimate of the audio track sizes was re-measured and corrected.
  - *The scene-cut guard.* The ARM replacement prefilter at first averaged across cuts, changing frames within ±3 of a cut by up to 45 levels. It was fixed, and the extras encode was stopped and restarted. About 40 minutes of encoding were thrown away.

> **In plain English:** In one evening, on one laptop, the author made the choices and the AI did the technical work. That covered the analysis, the custom tools, the measurements and the checks. Then the computer encoded the film overnight on its own. Several of the AI's own first guesses turned out to be wrong, and a measurement caught each one before it reached the finished files.

17.2 The comparison and its assumptions

ESTIMATE, not measured. Everything in the "Without AI" column below is the AI's own assessment, not a measurement. The assumptions are:

- **17.2.1 Who.** A capable person with professional video tools, for example MakeMKV, VapourSynth/AviSynth with QTGMC or a GUI front-end such as Hybrid, DaVinci Resolve or Premiere, a DAW (digital audio workstation, i.e. audio-editing software) such as Audacity or Reaper, Subtitle Edit with OCR, MKVToolNix, ffmpeg and x265. They have *no* deep restoration, signal-processing (DSP) or programming expertise, and *no* AI assistance.
- **17.2.2 What is counted.** Hands-on working hours, given as ranges. The overnight final encoding (about 8 hours) is machine time on both sides and is left out of both.

- **17.2.3 Which way the comparison leans.** The with-AI figure is total *elapsed* time, including machine time for the analysis passes. The without-AI figures count only the person's working hours, so the comparison leans in the without-AI side's favour.

17.3 Task by task

ESTIMATE: the same job by a capable person with professional tools, no deep expertise and no AI (hands-on hours)

Task	Without AI (est.)	Realistic outcome without AI
Image the disc, extract title sets	0.5–1 h	Same result
Learn interlacing and set up the QTGMC toolchain (on Apple Silicon: ARM builds of 3 plugins and 2 code patches)	10–30 h	Often abandoned for a GUI preset
Whole-film interlace forensics (8 passes, custom detectors, threshold calibration with 100+ crops)	40–120 h (needs programming and signal-processing skill)	In practice skipped: one deinterlacer applied to the whole film
Per-field matching: repairing mis-paired film exactly	Not realistically achievable without writing custom software	Mis-paired film stays striped, or a generic deinterlacer softens it
QTGMC mode tuning (Lossless modes, scene-cut guard)	4–10 h	Defaults used (the "lossless" default alters lines)
"Where does QTGMC add value" (every frame rendered twice, measured)	10–20 h + scripting	Not done
Encoder calibration (60 encodes scored with VMAF and SSIMULACRA2)	8–20 h	A CRF guessed from a forum
German audio drift: measure at 87 points, correct to sub-millisecond	6–15 h by hand in a DAW, precision about 5–20 ms at best	Drift likely unnoticed, or fixed roughly
Seamless German ending (polarity and level matched)	1–3 h	Doable; the polarity issue likely missed
Subtitle OCR and proofreading of 788 lines	4–8 h	Doable
Translate and frame-time 15 burned-in titles	2–4 h	Doable, timed by eye
Extras: subtitle-safe trailer crops; intro audio bit-exact and frame-pinned	3–6 h	Crops likely symmetric (clipping subtitles); audio re-encoded
Assemble (mux) the files into the library structure; verification	2–4 h	Doable
Visual verification of every decision	5–15 h	Partial
TOTAL (sum of the rows)	**≈ 95–255 h** (≈ 2.5–6.5 weeks at 40 h/week)	**Several results not achievable at all**

17.3.1 Where the hours go

- **Forensics is the biggest row.** Eight passes over all 135,587 disc frames included 271,174 test weaves. Two of the detectors turned out to be unreliable on their own, and that was only discovered by inspecting real crops. Without programming and signal-processing skill, this row is either very long or skipped.
- **Five of the 14 rows are "doable" or "same result".** These are imaging, the German ending (with its polarity caveat), subtitle OCR, translation and muxing. For these rows AI saves hours. It does not change what is possible.

17.3.2 Where the outcome differs

- **Exact film repair needs custom software.** The film's 79.2% exact disc picture rests on deciding each of 271,174 fields individually, using a best-to-other ratio of at least 2 and a stillness test of ≤ 1.0 levels. Re-paired film alone is 4.8% of the output frames, and it is recovered bit-exact.
  - The generic alternative was measured too: a full QTGMC rebuild removes about 15% of fine detail and grain *everywhere* (detail kept ≈ 0.85). That is what "one deinterlacer for the whole film" costs.
- **Defaults are not neutral, and their damage is invisible to the person using them.** The "lossless" default alters the original lines it claims to keep (mean 0.7–4.6 levels). A guessed CRF is never checked against the master.
- **Audio precision is the starkest gap.** The disc's dub runs from about 32 ms late to 83 ms early, which is enough to see lips out of sync. Hand alignment in a DAW reaches about 5–20 ms at best (estimate). The measured correction here is 0.30 ms rms and 0.66 ms worst case.

> **In plain English:** A capable person with the right software could image the disc, fix the subtitles and package the files without AI. That would be slower, but the result would be similar. The parts that make this restoration special are the frame-by-frame forensics, the exact repair of mis-paired film, the measured use of the deinterlacer and the sub-millisecond audio sync. Those need custom code and signal-processing know-how that most people do not have. Without AI they would most likely be skipped or done roughly, and the defaults that replace them fail silently.

17.4 Side by side

	With AI (measured)	Without AI (estimate)
Time	≈ 6.5 h elapsed	≈ 95–255 h hands-on (≈ 2.5–6.5 weeks)
Final encoding	≈ 8 h unattended, overnight	The same
Mis-paired archive film	Re-paired exactly, bit-exact disc pixels	Striped, or softened
Deinterlacer use	Measured: none on the 79.2% exact disc pictures; exact-lines mode on 12.2% of frames; clean rebuild on 8.5% (116 of 908 shots)	One deinterlacer on default settings for the whole film
Encoding	CRF 14 chosen from 60 scored encodes	A guessed CRF
German lip-sync	0.30 ms rms, 0.66 ms worst	Unnoticed, or about 5–20 ms at best

- **The difference, as estimated.** Even when the person's hands-on hours (95–255 h, estimate) are set against the AI side's total elapsed time (about 6.5 h, measured), the gap is roughly 15 to 40 times, and the without-AI result would still lack the exact film repair, the measured use of QTGMC, calibrated encoding and sub-millisecond sync.

17.5 Summary

- **With AI, a non-specialist got specialist-grade restoration in one evening of elapsed time.** That means custom per-frame forensics, exact repair of mis-paired film, sub-millisecond audio sync and calibrated encoding.

- **Without AI, the same person would realistically spend weeks** and still not reach several of these results (estimate, 17.2–17.3).
- **How to read this chapter.** The table is an estimate made under the assumptions in 17.2. Only the with-AI hours are measured. The measured record of what was built, tested and discarded is in the Effort ledgers that close Chapters 2–16.

> **In plain English:** AI did not replace the work; it made the careful, specialist route affordable. The author could ask for the best version of every step, see the proof and decide, all in one evening instead of weeks of learning and hand work. Without it, by this estimate, the steps that set this restoration apart would most likely have been skipped or done roughly.

APPENDIX A

Glossary

Every term of art in this paper, defined briefly and grouped by topic, so any chapter can be read on its own. Project numbers are given where they help.

A.1 Picture and interlacing

- **Field.** A half-picture: every other line of a frame. PAL records 50 per second, each at its own instant, 20 ms apart.
- **Frame.** A complete picture: 25 per second on the disc (720×576), 50 in the restored film.
- **Interlaced / progressive.** An interlaced frame is two fields that may show different moments; a progressive frame shows one moment, as modern screens expect.
- **Weave.** Joining two fields line by line. If both were captured at the same instant, the result is the original picture exactly.
- **Combing (comb teeth).** Horizontal stripes on moving edges where fields from two moments are woven together.
- **Phase-shifted (mis-paired) film.** Film stored with each half-picture paired with half of the next film frame: it looks combed, yet nothing is lost. Re-pairing restores it exactly.
- **Same-parity fields.** The same half (top or bottom) of neighbouring frames.
- **Bottom-field-first.** Field order in which each frame's bottom half-picture was captured first (the director interview); it must be processed in that order or motion jitters.
- **Inverse telecine (IVTC; TIVTC).** Undoing a film-to-video conversion by re-pairing fields; the principle behind per-field matching.
- **Field-insertion cadence (24→25).** Fitting 24-frame film into PAL's 25 by inserting extra fields; each insertion shifts the pairing after it, combing clean film.
- **PAL.** The disc's European TV standard: 25 frames (50 fields) per second, 576 lines.
- **4:2:0.** Colour stored at half resolution in each direction, as on the DVD.
- **Anamorphic / pixel aspect (SAR 64:45).** A 16:9 picture stored in 720×576 wide pixels, displayed as 1024×576. Verified correct and kept.
- **Letterbox / pillarbox / windowbox.** Black bars above and below, at the sides, or all round. Two letterboxed trailers were cropped to fill 16:9.
- **Lanczos / Spline36.** High-quality resizing filters: Lanczos enlarged the subtitle images for OCR, Spline36 scaled the reframed trailers.
- **Deinterlacing.** Turning fields into progressive frames, by weaving matching fields or calculating missing lines.
- **bwdif.** ffmpeg's fast deinterlacer; it made the first complete master, later discarded.
- **QTGMC.** A motion-tracking deinterlacer: it follows how things move across neighbouring frames to calculate a field's missing half-picture, rather than blurring or doubling lines. Run at its most thorough preset, Placebo.
- **SourceMatch.** QTGMC option that refines its result to agree with the original fields; level 3 used.
- **Lossless mode (1 vs 2).** When QTGMC re-inserts the field's own disc lines. Mode 2 lets later stages alter them (0.7–4.6 levels on average); mode 1 keeps them bit-exact.
- **Exact-lines mode.** This paper's name for QTGMC with Lossless 1: disc lines kept, only missing lines calculated. 12.2% of frames.
- **TR2.** Reach of QTGMC's final smoothing over time; 1 = light, one frame either side.
- **NoiseProcess.** QTGMC's denoiser; set to 0 (off) to keep the grain.
- **EEDI3 / znedi3.** Interpolators that fill a missing line along edge directions, so diagonals stay smooth; both made to run natively on Apple Silicon.
- **MVTools.** Motion compensation for QTGMC: finds where each block moved between frames.
- **Scene-cut guard.** Stops motion analysis blending two shots across a cut. Missing from the Apple Silicon prefilter (frames within ±3 of a cut changed by up to 45 levels); restored.
- **idet.** ffmpeg's interlace detector; one of eight whole-film forensic passes.
- **Comb detector (CombMask, TIVTC-style).** Counts comb pixels per 16×16 block. It flagged clean hair and skin at up to 125/256, so it is never trusted alone.
- **Grain-proof field difference.** Compares blurred fields so grain is not mistaken for motion; finds which neighbour shares a field's instant.
- **Line noise / line-pattern share.** An alternating-line pattern baked into some archive transfers, and the share of picture showing it. Cutoff 0.08; median 0.187 → 0.0001 after clean-up.
- **Clean rebuild (clean-up).** QTGMC recalculating every line, disc lines included (Lossless 2), to remove line noise. Whole shots only: 116 of 908, 8.5% of the film.

A.2 This paper's own terms

- **Change map.** A frame with every pixel that differs from the disc painted in the colour of the process that changed it.
- **Untouched / lifted / invented / reworked.** The four origins of an output pixel (shares of 112.5 billion):
  - **Untouched** (no colour; slate in charts): the disc's pixel as on the DVD, 83.0%.
  - **Lifted** (gold): a disc pixel moved from the neighbouring frame by re-pairing, 2.4%.
  - **Invented** (blue): calculated by QTGMC where the disc has no picture, 10.4%.
  - **Reworked** (green): a disc line recalculated by the clean-up, 4.3%.
- **Restored master.** The full-quality restoration before compression ("exact master" in figures), streamed into the encoder rather than stored (it would be ≈ 110 GB).
- **Bit-identical.** The same value, bit for bit: ≈ 88.4% of master pixels match the disc.
- **Output frame / slot.** One of 271,174 restored frames, one per disc field; "slot" is its index.
- **Shot vs scene.** A shot runs cut to cut (908); a scene groups related shots (749).

> **In plain English:** no colour is the disc's own pixel, untouched; gold is still the disc's pixel, only moved to its correct partner; blue was calculated because the disc never had that picture; green is a disc line cleaned of old transfer noise. All of it describes the master before final compression.

A.3 Tools and formats

- **dd and ISO image.** dd copies a disc block by block into one ISO file: 7.92 GB in 16 minutes, 0 read errors.
- **CSS.** Content Scramble System, the usual DVD copy protection; absent here.
- **VTS (title set).** A self-contained video block on a DVD: the film is set 9, the extras sets 1–8.
- **VobSub.** DVD subtitles stored as pictures, not text.
- **VapourSynth.** Python-scripted video framework (R81) running every frame-level step.
- **BestSource.** Frame-exact reader; matched ffmpeg pixel for pixel at first, middle and last frame.
- **x86 / ARM.** Intel/AMD vs Apple Silicon processor designs; QTGMC's usual plugins are x86-only, so three were compiled for ARM.
- **vszip.** VapourSynth plugin set whose EEDI3 port replaced an x86-only QTGMC part.
- **ffmpeg.** Standard toolkit for decoding, converting and analysing media.
- **x265.** Open-source H.265 encoder (4.3) making the final video.
- **SoX.** Audio processor that applied the German speed correction.
- **Tesseract OCR.** Open-source text-recognition engine that read the subtitle pictures and on-screen text.
- **MKV and MKVToolNix.** A container holding video, several audio and subtitle tracks and chapters; the tools that assemble it.
- **MPEG-2.** DVD video compression: 5.76 GB for the film, ≈ 8.5 Mb/s.
- **H.265/HEVC Main 10.** The final codec, 10-bit, far more efficient than MPEG-2.
- **AV1.** ≈ 26% more efficient, but Apple TV cannot decode it in hardware: rejected.
- **H.266.** Newest standard (VVC); target devices cannot play it.
- **FFV1.** Lossless codec of the discarded 25 fps master (55.8 GB).

A.4 Encoding and quality

- **Lossy / lossless.** Lossy compression (MPEG-2, AC-3, H.265) discards detail the eye is unlikely to notice; lossless (FLAC, FFV1) keeps every bit.
- **Keyframe.** A complete picture a player can jump to.
- **CRF.** Constant rate factor: constant quality, each scene gets the bits it needs. Lower = better; CRF 14 chosen from 60 test encodes.
- **Preset (veryslow).** How hard x265 searches: 104 s per 800 test frames vs 55 s (slower), 170 s (placebo).
- **SAO.** Sample adaptive offset, an H.265 filter that can soften fine detail; off.
- **--tune grain.** x265's grain tuning; at matched size standard settings beat it 10–0, with 5 ties.
- **VMAF.** Netflix's 0–100 perceptual quality score; computed per frame at native and TV size, and on every final frame against the restored master (film mean 96.08).
- **PSNR.** Peak signal-to-noise ratio, a simple pixel-difference score; with VMAF it compared every final frame with the master (film mean 46.23 dB, none below 30 dB).
- **SSIMULACRA2.** Independent perceptual metric, a second opinion beside VMAF.
- **Bitrate (Mb/s).** Megabits per second: disc 8.5; CRF 14/16/18 averaged 9.7/7.4/5.4 on hard test scenes.
- **10-bit.** 1,024 steps per channel instead of 256; avoids banding.
- **BT.601.** Standard-definition PAL colour standard, flagged so colours display as authored.
- **Limited range.** TV levels 16–235 for black–white; flagged so blacks are not crushed or washed out.

- **Chunked / resumable encoding.** The film in 32 chunks cut at shot boundaries; with the extras, 35 pieces encoded up to 8 at a time, each frame-counted before acceptance. An interruption repeats only unfinished pieces.

A.5 Audio

- **AC-3.** Dolby Digital, the disc's lossy audio, stereo: 320 kb/s for the film's two tracks and the Oded interview, 256 kb/s for the director interview, 192–320 kb/s in the intros.
- **FLAC.** Lossless compression, decoding to identical samples; German ≈ 445 MB vs ≈ 217 MB as AC-3.
- **32-bit float.** Working precision: processing adds no rounding damage or clipping.
- **Resampling.** Recalculating samples on a new time grid; here a speed factor of 0.9999778371.
- **ppm.** Parts per million: the German dub ran 22.16 ppm fast, drifting from about 32 ms late to 83 ms early.
- **Triangular dither.** Faint noise added when going to 16-bit, so rounding becomes hiss, not distortion.
- **Cross-correlation.** Sliding two signals to find their best match: 87 points measured, 37 verified, worst 0.66 ms.
- **Polarity.** Whether a waveform is upright or inverted; the German mix is inverted relative to the Hebrew.
- **Crossfade.** One source fading out as another fades in: 3 s at 87:19.5, into the original music (+5.6 dB).
- **LUFS.** Perceived loudness: German −21.2, Hebrew −27.4.
- **LU.** Loudness unit, a difference in LUFS; 1 LU = 1 dB.
- **rms.** Root-mean-square: a typical size of a set of errors; German sync 0.30 ms rms.
- **DAW.** Digital audio workstation (audio-editing software).
- **True peak / dBFS.** Highest level including between-sample peaks, in decibels below digital full scale; −3.4 dBFS = no clipping.
- **Lacing.** Packing several audio frames under one timestamp; off, so each intro frame keeps its exact time (0.01 ms).
- **MD5.** A fingerprint that changes if one bit changes; proves the intro audio frames are unchanged.

A.6 Subtitles and delivery

- **OCR.** Optical character recognition: reading text from pictures. 788 English lines converted and corrected; 67 on-screen text events found.
- **SRT.** Plain-text subtitle format every player reads; all four subtitle tracks ship as SRT.
- **Forced subtitles.** A track shown automatically, carrying only essential lines; the German forced track (35 lines) is kept.
- **Burned-in text.** Text inside the picture itself (title card, credits, captions, epilogue cards); 15 new English lines translate it, each shown while its German title is on screen and never after it.
- **Subtitle placement (top-positioned line).** Bottom by default; the "Avigdor" caption at 8:35 goes to the top because dialogue is on screen.
- **Mux.** Combining video, audio and subtitle streams into one file.
- **Extras folders (Featurettes, Interviews).** Folder names most media servers recognise, so the extras appear under the film automatically.
- **Chapters.** Jump points: 24 in the film, one per intro piece, 4 in the director interview.
- **Match Frame Rate.** Apple TV setting that switches the TV to the video's own frame rate; off, 50 fps on a 60 Hz output stutters. A player setting, not a file fault.

Colophon

The restoration ran entirely on one Apple Silicon laptop (M5 Pro). The disc was imaged with dd and its title sets extracted with ffmpeg 9; frames were read frame-exactly with VapourSynth R81 and BestSource; deinterlacing used QTGMC (havsfunc r33) with MVTools, znedi3, fmtconv and RemoveGrain compiled for Apple Silicon and vszip's EEDI3 port; all forensic measurements, field decisions and change maps are custom Python/NumPy scripts; encodes were made with x265 4.3 and scored with VMAF and SSIMULACRA2; audio was re-timed with SoX; subtitles were read with Tesseract OCR; and the library is assembled with MKVToolNix. All frames reproduced in this report are from *The Champagne Spy* (2007, dir. Nadav Schirman), © its producers and Neue Visionen, shown as small crops for technical commentary; no copy of the film is distributed with this report. This paper was set in Iowan Old Style and Avenir Next with HTML and CSS, its figures and charts prepared with Python (Pillow, NumPy, Matplotlib), and it was rendered to PDF with Google Chrome. Project: Levi Horsley.